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Microscopic Simulation of Solid State Sintering Regarding Irregularly Shaped Powder Particles

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Doktor-Ingenieur (Dr.-Ing.)

vorgelegt von

Dipl.-Ing. Max Weiner

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Dissertation

Microscopic Simulation of Solid State Sintering Regarding Irregularly Shaped Powder Particles

submitted to the

Faculty of Materials Science and Technology
of the Technical University Bergakademie Freiberg

to attain the academic degree of

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Supervisor: Prof. Dr.-Ing. Ulrich Prahl
Date of submission: September 05, 2025

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- Dr.-Ing. habil. Matthias Schmidtchen, Leiter der Arbeitsgruppe (Diskussion während des Arbeitsprozesses, Kommentare und Ratschläge zum Manuskript)
- Prof. Dr.-Ing. habil. Christos G. Aneziris, Leiter der Forschergruppe (Diskussion während des Arbeitsprozesses zum technologischen Hintergrund)
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- Wilhelm Winkler (Literaturrecherche zu Grenzflächenenergien und Diffusionsdaten im Rahmen einer studentischen Arbeit)
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Max Weiner, Freiberg den 30. Januar 2026

Zusammenfassung / Summary

Deutsche Version

Es wurde ein neuartiges Modell der Sinterung zweier bzw. weniger Teilchen unter Verwendung scharfer Grenzflächen entwickelt und implementiert. Die Anwendung des thermodynamischen Extremalprinzips erlaubt eine elegante und erweiterbare Formulierung. Das numerische Verhalten des Modells unter Anwendung von Neuvernetzungsroutinen wurde untersucht und bewertet. Es wurden Parameterstudien zu klassischen dimensionslosen Kenngrößen von Zweiteilchenmodellen, sowie asymmetrischer Material- wie Geometriepaarung durchgeführt, mit Literaturwissen abgeglichen und erklärt. Kontakte mehrerer Teilchen wurden auf den Einfluss der Porenschließung untersucht, auch bei Anwesenheit inerter Teilchen. Die Formcharakteristik von Pulverteilchen wurde unter statistischer Anwendung einer Formfunktion beschrieben und zur Generierung von Pulverteilchen als Eingabe in die Simulation verwendet. Das mittlere Verhalten der zufällig generierten Teilchen wurde zur Beschreibung des Verhaltens des Pulvers genutzt und Unterschiede zu klassischen Verfahren der Mittelung diskutiert.

English Version

A novel model for the sintering of two or a couple of particles using sharp interfaces was developed and implemented. The application of the thermodynamic extremal principle allows a concise and extendable formulation. The numerical behavior of the model was investigated and evaluated under application of remeshing routines. Parameter studies were conducted on classical dimensionless parameters of two-particle models, as well as asymmetric material and geometry pairing, which were compared with literature knowledge and explained. Contacts between multiple particles were investigated for the influence of pore closure, even in the presence of inert particles. The shape characteristics of powder particles were described using a statistical shape function and used to generate powder particles as input for the simulation. The average behavior of the randomly generated particles was used to describe the behavior of the powder, and differences from classical averaging methods were discussed.

Part I.

Introduction

1. Introduction to Sintering Technology

The following elaborations on the basics of sintering are summaries from standard literature on the topic. Some notable textbooks are those of German [1, 2], Geguzin [3], Kang [4], Exner [5], Fang [6], Rahaman [7] and Routschka and Wuthnow [8].

Sintering in general is a process, in which compacted powders are densified and strengthened by the application of a heat treatment. The actual mechanisms acting therein are discussed in section 2.1. Sintering allows near-net shaping of possibly otherwise hard-to-shape materials, as well as creation of composites or alloys that are not producible via the melt route. To achieve required tolerances in geometry and material properties, a precise control and prediction of the density evolution (shrinkage) during the heat treatment is necessary. In industrial practice, this is often done using empirical approaches, due to the complexity of the process (see section 1.2), Physical modeling, however, is required to improve understanding of mechanisms and observed effects and thereby allowing to design novel processing strategies and material concepts. [1]

1.1. Process Steps for Manufacturing of Sintered Products

The principal processing steps involved in manufacturing of sintered products shall be briefly discussed below. Figure 1.1 visualizes the process via a flow chart.

Powder Mixing Powders of differing particle size distribution, particle shape or even substance are mixed to achieve the desired sintering behavior and product properties. Optionally, additives facilitate compaction or achieve particular processing behavior (polymer resins, water, …).

Compaction The loose powder must be compacted to obtain a green body of sufficient density and strength. This can be accomplished by casting with or without vibrational assist, pressurized compaction, injection molding, or others. Usually, this step defines the desired shape of the product. Attention have to be paid to possible segregation of particles of distinct size or substance.

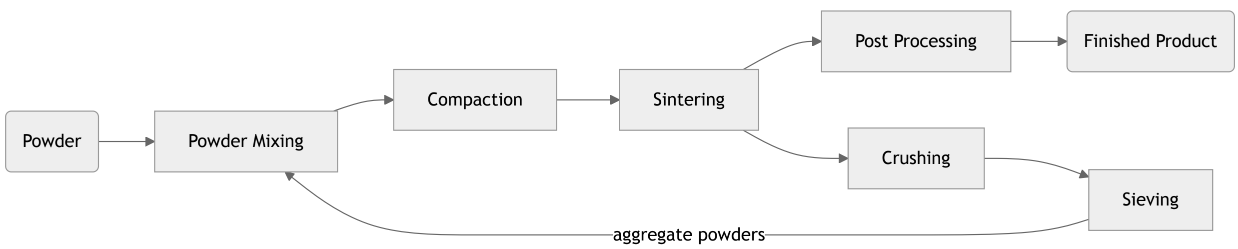


Figure 1.1: Chart of Common Process Steps for Sintered Products

Sintering The green body is heat treated to allow for diffusional processes, which are bonding the distinct powder particles together, and so achieve the desired strength and final density. During sintering the body usually shrinks in its dimensions, pore volume is reduced and the density increases due to internal volume displacement. If resin binders were used in the powder mixture, they have to be removed from the body by a low temperature heat treatment which pyrolyzes and evaporates the binder, otherwise the component will crack during sintering due to fast gas formation which effects high stress. In some advanced processes, compaction and sintering is accomplished in one single step (pressure assisted sintering, field assisted sintering, …).

Post-Processing If required, the sintered body is machined to meet requested tolerances or surface properties. Porous parts for bearings are often infiltrated with lubricant.

In some cases, aggregate particles are formed by sintering simple bodies, crushing them, sieving the fragments and using them in mixing a novel powder for the actual component manufacturing. This is used especially for refractory materials, where coarse grained microstructures are desirable for thermal shock resistance. The aggregate particles are characterized by low internal sintering activity, so fresh active powder fractions have to be added to the mixture. [8]

1.2. Influencing Factors on Sintering Processes

The actual behavior of a sintering process is a complex problem, as there are many factors that influence the process and some of them are hard to describe or even quantify. Most are interrelated. They have been grouped to geometry, material and process related factors by German [2]. Below, the most important of these factors will be described, important interrelations are *emphasized*.

1.2.1. Geometric Factors

These factors involve both the particle geometry and the geometry of the macroscopic sintering body.

Particle Size The particle size is the main geometric factor influencing sintering. Sintering is mainly driven by the reduction of energy bound in interfaces, either surfaces or grain boundaries. The mean particle size gives a rough estimate of the total energy bound in interfaces and thus of the available driving force. It also determines the length of diffusion paths and the required displaced volume to achieve a certain macroscopic response. The particle size determines the maximum achievable *density* before sintering. Inhomogeneous size distributions aid to achieve high densities as pores can be filled by small particles.

Particle Shape While *particle size* affects the overall mean curvature of the particle, the local curvature and thus the driving force of sintering is determined by the actual shape of the particles. Shape deviations from ideal sphere let particles behave locally like larger or smaller ones in certain aspects.

Density and Porosity The current relative density, or its inverse, the porosity, determines the still available volume reduction. With increasing density during sintering *time*, the typical shape characteristic of pores changes from interconnected pore networks, to tube-like networks and finally isolated pores. The latter may be thermodynamically stable, especially when they are filled with gas due to trapped *atmosphere*.

Neck Size The current size of sinter necks (grain boundaries between particles) determine the mechanical strength of the body, the length of grain boundary diffusion paths, as well as the required volume to be displaced for certain shrinkage response. It also determines the available cross-section for inter particle mass flows.

Particle Packing Depending on *particle size* and *particle shape* the packing may be more dense or loose and the coordination number of particle contacts changes. This affects *density* and the mutual hindering of particles in their movement during sintering. Cracks may form due to inconvenient packings around diffusional inert particles.

1.2.2. Material Factors

These factors are related to the substance of the powder particles.

Surface and Interface Energies As the main driving force for sintering is the energy bound in interfaces, the energy per unit area in those interfaces plays a major role. Not only their absolute values, but also the ratio between distinct types of interfaces determines how the *geometry* may evolve, as creation of a low-energy interface in favor of consuming a high-energy one may be thermodynamically beneficial.

Diffusivity The advance of sintering in *time* is heavily determined by the speed of the various diffusional processes acting (see section 2.1). Diffusional speed is dependent on the required activation energy and thus on *temperature*. The ratio of the distinct mechanism's diffusivities affect the evolution and the final state of *geometry*.

Vapor Pressure A seldom encountered mechanism is evaporation/condensation of particle substance and transport via the gas phase (possibly with chemical reactions in reactive *atmospheres*) as discussed in section 2.1. The actual vapor pressure at the respective *temperature* determines the importance of this mechanism. Sometimes, only parts of the powder substance (impurities, specific alloying elements, binder resins) evaporate during the sintering process, which may change chemical composition, create defects in the microstructure or facilitate corrosion of the product (f.e. zinc in brass).

Melting Low-melting phases may be formed, which accelerate sintering and facilitate achievement of near-full density (liquid phase sintering) as the liquid is able to fill pores rapidly by viscous flow.

Flow Stress In processes with high *pressure*, plastic yielding may occur as an additional mechanism for *densification*.

Crystal Structure Depending on the crystal structure, material properties like *surface and interface energies* and *diffusivities* may be significantly anisotropic. This leads to non-spherical equilibrium *particle shape*.

Chemical Interaction Particles of different substance may be soluble or insoluble in each other, or may form new chemical substances through reactions. Concentration gradients within the particles or compound formation affects driving force and *diffusivities*. Chemical interactions with the *atmosphere* are often met.

1.2.3. Processing Factors

These factors describe the process setup.

Temperature The acting mechanisms are activated by elevated temperature to overcome their respective activation energy. Temperature determines the occurrence and speed of these mechanisms, as well as their relative importance. Key factors dependent on temperatures are the *interface energies* and *diffusivities*. In addition, temperature may activate several (maybe unwanted) side processes like *melting*, chemical reactions with *atmosphere* or affect *chemical interaction* of different substances present.

Time With advancing process time, the *geometry* of the system evolves especially by increasing *neck size* and *density*. Time principally affects all conditions in the system as the state of the system evolves towards a lower energy configuration.

Pressure Outer pressure on the compact during sintering acts as an additional driving force for *densification* (pressure assisted sintering). Atmospheric pressure may act counter the *vapor pressure*, thus inhibiting gas phase mechanisms.

Atmosphere The surrounding atmosphere may cause *chemical reactions* with the solid phase (especially oxidation). Gas adsorption affects surface *diffusivities* and gas atoms may also diffuse through grain boundaries, possibly altering the interface properties or form new substances.

2. Basic Theory of Sintering

This chapter provides a brief overview of the basic theory of sintering, namely the mechanisms found in sintering processes, the mathematical treatment of those and the modelling of simple particle systems' evolution during sintering.

2.1. Mechanisms of Sintering

The following elaborations on the basic theory of sintering are based on standard literature for the topic. Some notable textbooks are those of German [1, 2], Geguzin [3], Kang [4] and Exner [5].

This work concentrates on solid state sintering, meaning that there are only solid phases present in the system beside the surrounding atmosphere. Liquid phase sintering, where the material partially melts to assist closing pores, is not regarded here. Therefore, the term sintering shall always refer to solid state sintering from here on.

Sintering in general is a process, where powder material is densified and strengthened by thermal treatment at elevated temperatures. The process is mainly driven by reduction of the energy stored in the system's microstructural features, such as grain boundaries, surfaces and other crystal defects. The energy is dissipated as heat, so it is an irreversible process. The microstructural state of the material is mainly transformed by diffusional flows to reach a state of lower energy. In some systems, other transport mechanisms such as viscose or plastic flow can be observed, but are not regarded in the current work.

In common sintering processes, the following diffusion mechanisms may occur. Their importance depends on powder geometry, process conditions and material properties. The place and usual direction of their occurrence is visualized in figure 2.1.

Volume/Lattice Diffusion is the diffusion of vacancies or interstitial atoms in the bulk material. It is usually one of the slowest mechanisms and often negligible.

Surface Diffusion is the diffusion of mobile atoms on free surfaces (interfaces with surrounding atmosphere or vacuum). It is much faster than volume diffusion.

Grain Boundary Diffusion is the diffusion of mobile atoms along grain boundaries (solid-solid interfaces). It is usually slower than surface diffusion but faster than volume diffusion. The effect of grain boundary diffusion compared to volume diffusion was paraphrased by Fisher [9] with the heat flux in a foil of copper embedded in cork.

Evaporation and Condensation is the evaporation of atoms, their diffusion through the gas phase and finally their condensation at another location. This mechanism features fast transport through the gas phase, but usually occurs only at very high temperatures, with very volatile substances or due to chemical reactions. The rate of evaporation is usually the rate determining mechanism.

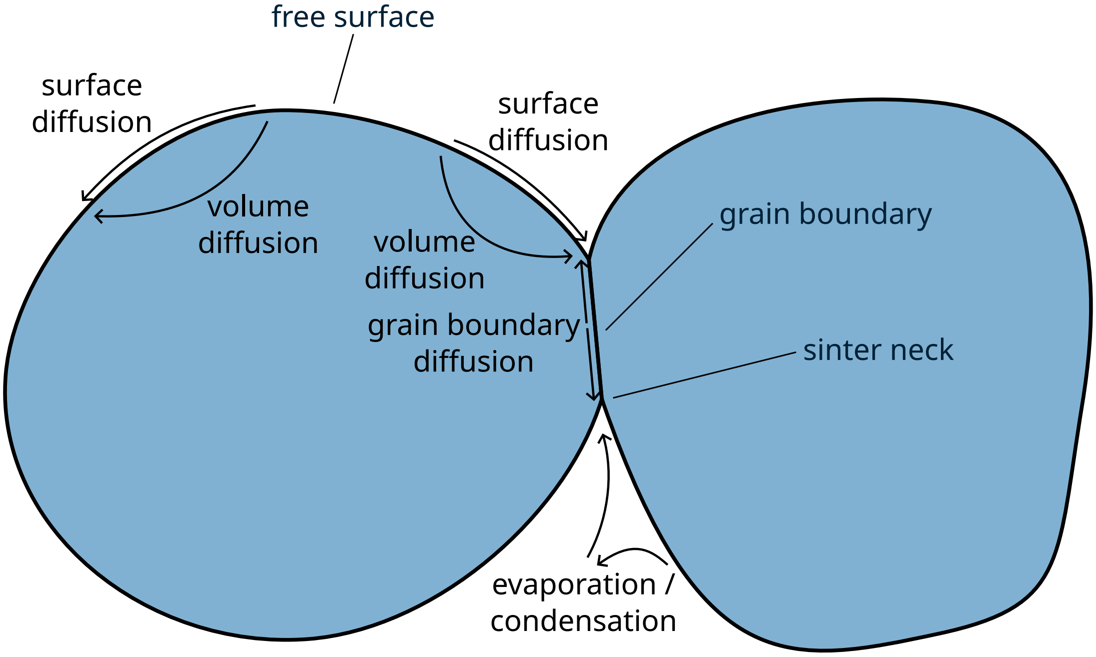


Figure 2.1: Schematic Overview of Diffusion Mechanisms Observed in Sintering Processes

Macroscopically these mechanisms can be observed via properties of the body such a size and shape or of the material such as strength, density, electrical and magnetic properties or chemical surface activity.

The most commonly observed effect is shrinkage, which means that the sintered body is smaller than the green body. This is caused by transport of material from between the particles to the surface of the neck via grain boundary diffusion or volume diffusion. Sole surface diffusion does not cause shrinkage. As the mass of the system is constant, shrinkage must coincide with an increase in density, respectively a decrease of the amount of voids (porosity) in the body. Shrinkage as a linear strain ε is usually quantified normed on the initial (green body) size L_0 as in equation (2.1), where L is the current size.

$$\varepsilon = \frac{L - L_0}{L_0} \tag{2.1}$$

Most material properties are directly correlated with the size of the boundaries between the particles which form during sintering, called sinter necks. The driving force for neck formation is the reduction of surface area in interaction with the energy stored in grain boundaries, where the latter is usually much smaller per area unit. Most notably, the strength varies directly with the neck size, since the largest part of the load is transmitted via the necks, but also electric and magnetic properties. The neck is usually measured in terms of its radius r_N or cross-section area A_N.

By the dominating mechanisms and their effect on neck growth and shrinkage, sintering is usually divided in three stages:

Initial/Early Stage The initial contact is created. Major diffusion occurs along surfaces to the sinter neck (triple points of surfaces and grain boundary). The neck size rapidly increases, but shrinkage is usually low.

Intermediate Stage The porosity is still open and connected. Major diffusion occurs along the grain boundaries to the pores. The removal of substance at grain boundaries leads to large shrinkage.

Final/Late Stage The porosity is mainly closed. The gas pressure in pores may affect sintering. There is still mainly grain boundary, but also volume diffusion. There is slow progress in shrinkage and densification.

2.2. Mathematical Description of Diffusion

The classic description of diffusional processes is accomplished by Fick's laws [10], which have the same mathematical form as the heat conduction laws by Fourier [11]. Equation (2.2) shows Fick's first law for the one-dimensional diffusion along an interface. It proposes a linear relationship between the concentration gradient and the diffusional flux j. For the concentration, here the vacancy concentration c_{Va} is taken, as we assume only one diffusing species. The flux of vacancies equals the negative flux of atoms. The law includes an empirical scaling coefficient, commonly referred to as the diffusion coefficient D, which has the dimension of $[D] = \mathrm{L}^2\mathrm{T}^{-1}$. δ is the thickness of the diffusion layer or the interface, respectively. In some publications, δ is directly included in the diffusion coefficient for interface diffusion as $D' = D\delta$ with the dimension $[D'] = \mathrm{L}^3\mathrm{T}^{-1}$. The interface thickness δ is hard to determine, often it is estimated as the size of one or two atom layers.

$$j = D\delta \frac{\mathrm{d}c_{\mathrm{Va}}}{\mathrm{d}x} \tag{2.2}$$

The diffusion coefficient D heavily varies with temperature. Its temperature dependence is usually described by an Arrhenius-type equation as in equation (2.3) with a pre-exponential factor D_0 and the activation energy Q.

$$D = D_0 \exp\left(-\frac{Q}{RT}\right) \tag{2.3}$$

The local vacancy concentration c_{Va}, however, depends on the thermal equilibrium vacancy concentration c°_{Va} and the local chemical potential μ as in equation (2.4).

$$c_{\mathrm{Va}} = c^{\circ}_{\mathrm{Va}} \left(1 - \frac{\mu - \mu^{\circ}}{RT}\right) \tag{2.4}$$

The main factor influencing the chemical potential in the regarded sintering processes are mechanical stresses, especially stresses originating from curved surfaces or interfaces (also known as surface tensions). The local chemical potential by a stress σ is given as in equation (2.5) with the molar volume V_{m}.

$$\mu = \mu^{\circ} + \sigma V_{\mathrm{m}} \tag{2.5}$$

The stress by a curved surface is given as in equation (2.6) with the surface resp. interface energy γ and the curvature κ.

$$\sigma = \gamma\kappa \tag{2.6}$$

Other stresses present affect the progress of sintering, too. The benefit of pressure assisted sintering is apparent from these equations, since the applied external pressure drives the diffusion from the inner of the grain boundaries to the necks.

In the triple point of three interfaces as shown in figure 2.2, the dihedral angles between the interfaces ψ_i will follow Young's equation (equation (2.7)) in equilibrium with $\psi_1 + \psi_2 + \psi_3 = 2\pi$, where the γ_i are the interface energies of the interface opposed to the angle with the same index.

$$\frac{\gamma_1}{\sin\psi_1} = \frac{\gamma_2}{\sin\psi_2} = \frac{\gamma_3}{\sin\psi_3} \tag{2.7}$$

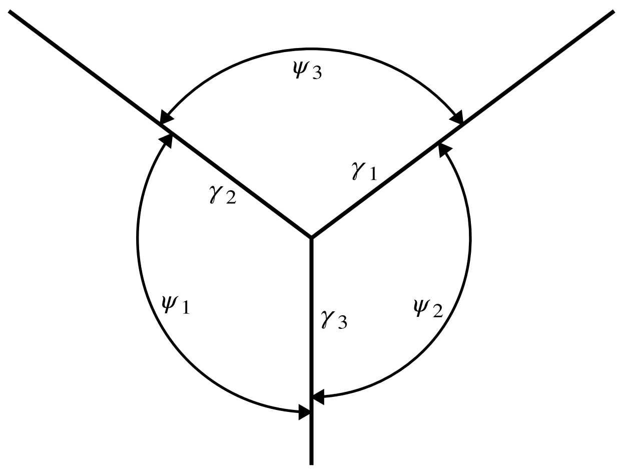


Figure 2.2: Dihedral Angles Between Three Interfaces

This is equivalent to a triangle with side lengths γ_i and the respective opposite angles $\pi - \psi_i$, as Hoffman and Cahn [12] explains. This leads to the condition that the sum of either two surface energies must be larger than the third, or this triangle does not exist and equation (2.7) can not be solved. The non-existence of the triangle implies that such a junction can never be in equilibrium, if no other forces on the junction are present (f.e. pinning forces). It is then always beneficial from energetic point of view to consume the boundary with the largest interface energy while enlarging the others [13].

2.3. Non-Equilibrium Thermodynamics

Sintering processes are by their nature non-equilibrium thermodynamical processes which drive the system state towards an equilibrium state. The major problem of non-equilibrium thermodynamics is that the common thermodynamical quantities temperature and entropy are only defined in equilibrium state. Therefore, the second law of thermodynamics (entropy must always increase) only holds for transitions between equilibrium states and may thus not describe the pathway between these states, but only the whole step as one. To circumvent this problem, several approaches have been developed, from which the classical irreversible thermodynamics (CIT) (sometimes referred to as theory of irreversible processes (TIP)) and the thermodynamics of internal variable of state (IVS) shall be discussed here in particular. The following elaborations are based on textbooks of de Groot and Mazur [14], Maugin [15], and Lebon, Jou, and Casas-Vázquez [16] and a review article of Maugin and Muschik [17].

2.3.1. Classical Theory of Irreversible Thermodynamics

A non-equilibrium state is characterized by inhomogeneity and thus gradients of its major state variables. The central assumption of the classic treatment of non-equilibrium irreversible processes is that of a local equilibrium. It is postulated that in a certain small volume of the system, the equilibrium state is present despite the gradient to the next neighboring volume. This means that the characteristic time of local equilibration is much smaller than the characteristic time of the macroscopic process, expressed by the Deborah number $\mathrm{De} = \tau_{\text{local}}/\tau_{\text{global}} << 1$. The volume cell must be small enough to fulfill this timescale condition, but large enough to be described sufficiently well as a continuum. The time needed for restoration of a local equilibrium state can be neglected in comparison to the macroscopical evolution of the system. This allows to grant the usual meaning to temperature and entropy. On the other hand, processes with a Deborah number $\mathrm{De} >> 1$ can be regarded as quasi-static. Those assumptions hold true for most thermodynamical processes in question, especially for the here regarded sintering. So, the irreversible process reduces to a sequence in time of local equilibria.

The system's state at a certain point in time is defined by the vector of state variables X. As sintering is a process at constant temperature and pressure, the thermodynamic condition of equilibrium is to be formulated in terms of Gibbs' energy $G(X)$, which is a function of the state. Temperature can be assumed constant, as the energy dissipated is small compared to the thermal energy stored in the system (sintering body and surrounding furnace), so the system does not heat up significantly. Temperature changes during the process are external changes to the system and not due to its internal evolution. Conventional sintering processes are conducted under constant atmospheric pressure, whereas in pressure-assisted sintering the applied pressure is usually kept constant by control.

The thermodynamic forces driving the evolution of the system are given as the partial derivatives of Gibbs' energy with respect to the state. The gradient of the Gibbs' energy equals zero, when the system has reached equilibrium.

$$\frac{\partial G}{\partial \boldsymbol{X}} = 0 \tag{2.8}$$

Changes in the state variables are obtained by the occurrence of respective fluxes $\boldsymbol{j}$. The dissipation $\mathscr{D}$ during evolution of the system is proportional to the product of the fluxes and their respective thermodynamic forces.

$$\mathscr{D} \sim \frac{\partial G}{\partial \boldsymbol{X}} \cdot \boldsymbol{j} \geq 0 \tag{2.9}$$

A usual (and for many processes valid) assumption is linearity between the fluxes $\boldsymbol{j}$ and the thermodynamic driving forces $\frac{\partial G}{\partial X}$ with a coefficient $\boldsymbol{C}$. This corresponds to the empirical description of diffusion by Fick's law (equation (2.2)).

$$\boldsymbol{j} = \boldsymbol{C} \cdot \frac{\partial G}{\partial \boldsymbol{X}} \tag{2.10}$$

2.3.2. Internal Variables of State

As the microscopic nature of thermodynamical systems is often too complex to be described alone by the macroscopical observable and controllable external variables of state (EVSs) $\boldsymbol{X}$ (such as temperature, pressure, internal energy and entropy), approaches only considering those neglect large amounts of internal processes and specifics of the state. An approach applied enormously successfully is that of the introduction of IVSs $\boldsymbol{x}$ which represent dissipative quantities in the system and can be freely chosen dependent on the desired precision and particularity of the model. In contrast to external variables of state, which define the macroscopic response of the system and are thus relatively easy to measure and control, internal variables of state may be measurable, but are usually out of control when setting up an experiment. The approach was mainly developed by authors like Bataille and Kestin [18], Kestin [19] and Kestin [20], Muschik [21] and Maugin and Muschik [17, 22]. Maugin [23] gives an extensive review and history.

The formalism axiomatically associates a local accompanying equilibrium state dependent on the local non-equilibrium EVSs and the IVSs to a volume cell at a respective point in time. This is an extension of the former assumption of local equilibrium in CIT. The local change in Gibbs' energy is then given by:

$$\mathrm{d}G = \frac{\partial G}{\partial \boldsymbol{X}} \cdot \mathrm{d}\boldsymbol{X} + \frac{\partial G}{\partial \boldsymbol{x}} \cdot \mathrm{d}\boldsymbol{x} \tag{2.11}$$

The physical meaning and applicability of this system to describe the real process highly depends on the choice of the IVS. Principally, one is free to choose the internal variables depending on requirements on precision, model depth and computational effort. However, they must fulfill some conditions to be appropriate. Similar to the requirements of local equilibrium in CIT, an IVS can be considered to always have their local equilibrium value if $\mathrm{De} << 1$ holds. On the other hand, if $\mathrm{De} >> 1$, the variable can be assumed constant in regard to the equilibrium condition. If neither is valid, the variable is considered to be a bad choice and another one should be selected to describe the internal state.

Another important type of variables in this notion is that of internal degrees of freedom (IDFs), which are not of pure dissipative nature. They show relevant inertia within the respective timescale, which is equivalent to $\mathrm{De} \approx 1$. IDFs must be accompanied by their own kinetic law in terms of a differential equation to be included in the simulation. They may, but do not have to, include a dissipative contribution as IVS do.

A numerical approach to obtain a system of ordinary differential equations from this formalism will be discussed in section 3.3.

2.4. Simplified Quantitative Descriptions of Sintering Processes

The mathematical description of diffusion phenomena based on potentials and empirical kinetic diffusion coefficients as discussed in section 2.2 does not encounter major difficulties. The real complexity of modelling sintering processes arises in the geometrical state and evolution of the system. Early models on particle level therefore imposed heavy assumptions on the geometry of the initial particles and of the sinter neck, to be able to obtain a concise and efficient formulations. This was especially required as powerful computer systems were not available. Such models shall be regarded in this section, more advanced approaches, however, are regarded in chapter 3.

The most common simplification applied is the assumption of a circular (2D) or spherical (3D) geometry of the particles before sintering. This has the following main benefits:

- Simple characterization of particles using a single parameter (radius or diameter).
- Absence of diffusional flows on large parts of the particle surface due to constant curvature, the only gradients are present near the neck.
- Symmetry of the particle contact with respect to the approaching axis and thus also invariance of particle rotation.

Such a system's state is sufficiently described by the radii of the involved particles, their distances (or positions) and the current size of the necks between the particles. Often, only a single pair of particles of the same size and material properties is regarded, which introduces another degree of symmetry.

These simplifications allow the analytical solution of the governing differential equations and lead to equations of the type in equation (2.12) for neck radius and equation (2.13) for shrinkage. Therein, t denotes the sinter duration, r_N the neck radius, r_P the particle radius and B, m and n are parameters depending on the acting transport mechanisms. Notable examples of such models are given by Kuczynski [24], Johnson and Cutler [25], Kingery and Berg [26], Coblenz et al. [27], and German [28].

$$\left(\frac{r_\mathrm{N}}{r_\mathrm{P}}\right)^n = \frac{Bt}{(2r_\mathrm{P})^m} \tag{2.12}$$

$$\varepsilon^n = \frac{Bt}{(2r_\mathrm{P})^m} \tag{2.13}$$

A common result from these models, alongside with dimensionless parameter studies, is the definition of the dimensionless timescale for two-particle-sintering processes as given in equation (2.14). Often the surface energy γ_S and surface diffusion coefficient D_S are referenced, but the equal is possible using the properties of the grain boundary.

$$t^\star = \frac{V_\mathrm{m} c^\circ_\mathrm{Va} D_\mathrm{S} \gamma_\mathrm{S}}{RT r_\mathrm{P}^4} \cdot t \tag{2.14}$$

Similar treatments have been given for the intermediate and final stages of sintering. In these stages the particles are usually assumed as polyhedra with cylindrical or spherical pores on the boundaries. Since these stages are mainly characterized by the decrease in pore volume, they are formulated in terms of densification. Examples for such models are Coble [29], Swinkels and Ashby [30], Kang and Jung [31] and Gouvêa [32].

The predictive value of such simple solutions is rather low because of their heavy simplifications, but they can be used to create empirical fits on experimental data.

3. Numerical Methods for Sintering Simulation

Sintering models can be distinguished in two main branches: macroscale and microscale models. The former regard the porous workpiece material as a continuum with variable density. Stresses and strains are usually modelled using viscoelastic material laws or similar while the density evolves during thermal treatment or plastic deformation (mechanical compaction). Numerical solutions are commonly obtained by use of the finite elements method (FEM). The focus of these models usually lies in the analysis of density, strain and stress distributions, as well as damage estimation, in a component with often complex geometry. Examples for these models are found in Olevsky [33], Safonov et al. [34], Kraft and Riedel [35], Al-Qudsi et al. [36], and Shi et al. [37]. These models will not be considered hereinafter, as this work concentrates on the latter kind.

Microscale models regard the interaction between distinct grains or powder particles and the evolution of their interfaces to each other and the environment. The solution space typically includes a few distinct particles or a defined representative volume element (RVE).

This chapter regards the current state of numerical methods applied to model sintering processes and similar problems. Interface description is the key problem in microscopic sintering simulation, section 3.1 will concentrate on this topic. Probabilistic approaches can be used in addition to deterministic methods or replace them entirely, as described in section 3.2. A concise approach to non-equilibrium thermodynamics is provided by the thermodynamic extremal principle (TEP), discussed in section 3.3.

3.1. Interface Descriptions

Since the tracking of the interfaces is a key problem in modeling of sintering, the numerical methods shall be classified by this feature. See figure 3.1 for a visualization of this classification. The approach taken there is also crucial for the used numerical integration procedure, the required material data and the reliability of the obtained simulation results.

The first distinction can be made into approaches with direct and indirect interface tracking. Direct tracking means that the location of the interface is directly defined by discrete points in space, which are moved during the integration procedure to describe the shape evolution of the interface. So in this family of approaches there is a sharp interface definition. A sharp interface means that the interface is considered as a defined line, where the two phases exist on the one resp. the other side of the line. The curvature of such an interface can directly be calculated by interpolation of its coordinates, for example to be used in equation (2.6). Direct interface approaches usually use the finite differences method (FDM) or the finite elements method (FEM) to represent the particle shape.

Indirect tracking, however, means that the interface location is defined by the value of an auxiliary function. The discretization is there connected to the solution space, the points are fixed in space and the motion of the interface is just realized by changes in the auxiliary function values. Indirect interface methods can further be classified in methods with sharp and diffuse interfaces.

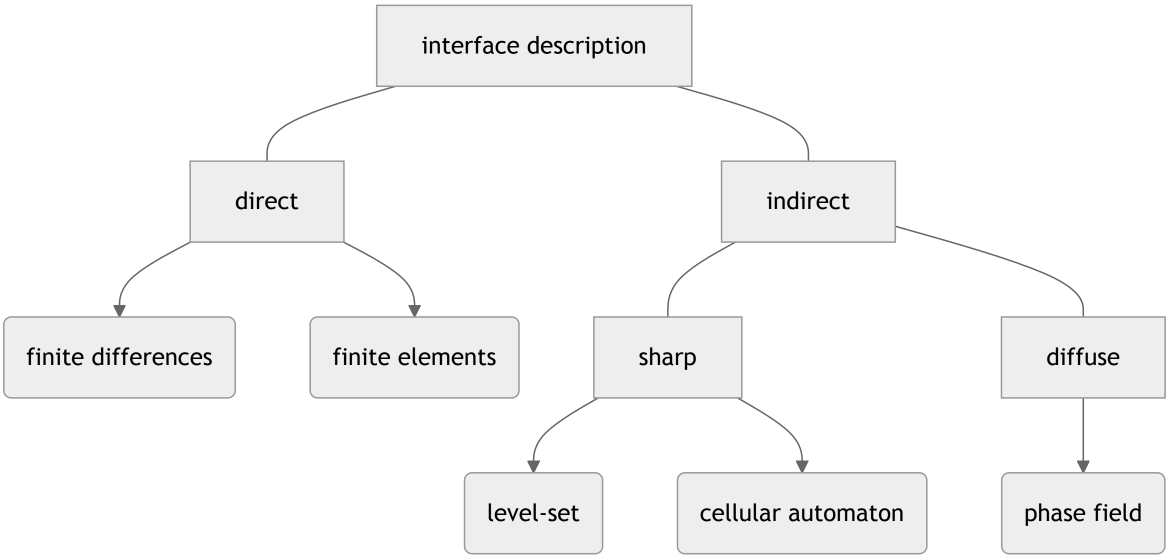


Figure 3.1: Classification of Interface Description Methods

Approaches with a sharp interface definition include the level-set method (LSM) (see section 3.1.2) and cellular automata (CAs) (see section 3.1.3) approaches.

Diffuse interfaces mean a more or less smooth transition between two phases, so that the actual position can not directly be located. This diffusivity must not be confused with the physical width of the interface. The physical width of the interface is the region where the atomic structure significantly deviates from the bulk structure. This is usually in the order of a few atomic diameters (around or below 1 nm). The width of a diffuse interface is a numerical artifact and depends on the size of the spatial discretization. Typical widths are here rather in the range around 1 µm. The diffuse interface width is not physical, but a numeric requirement to be able to calculate gradients through the interface. Diffuse interfaces are used in phase-field method (PFM) approaches (see section 3.1.4).

3.1.1. Direct Interface Approaches

Several authors have published sintering models where the interfaces are described directly by discrete points either in a finite difference or finite element scheme.

This approach has the great advantage of direct and easy computation of geometrical features such as curvatures, as the defining points are directly available. Curvature calculation is either done by application of finite differences [38, 39] or by interpolation functions [38, 40, 41, 42].

The major problem of direct interface approaches is maintaining the contact conditions resp. the geometric validity of the grain boundary. Most models, therefore, assume a heavily idealised (planar) grain boundary and rely on symmetry to simplify determination of the particle motions [38, 43, 39, 41, 44]. The problem therefore reduced to determining the evolution of the surface geometry.

Svoboda and Riedel [38] and Bouvard and McMeeking [39] are classic finite differences method (FDM)-based solutions of two equal circular particles with heavy symmetry assumptions. An advanced FDM-based model was published by Klinger and Rabkin [45, 46, 47] which supports curved grain boundaries.

Jagota and Dawson [41] published a direct FE model for two-particle sintering, however, based on viscous constitutive equations. So this is merely feasible for amorphous materials rather than crystalline ones. Léchelle et al. [42] built a direct FE model in the tradition of classic mechanical FE approaches. It regards surface and grain boundary diffusion considering the interfacial stresses as outer load and applying contact conditions in the grain boundary. This model appears to be basically able to work with particles of arbitrary shape, but was only evaluated for two spherical particles.

There is a free and open-source program by Brakke [48] aimed at modeling the evolution of different surfaces in time by minimizing the energy stepwise. However, it lacks direct support for time-coupled simulation, as the kinetics are solely determined by use of a global constant. Wakai and Okuma [49] recently built a 3D multiple-particle sintering simulation based on this program, an older version of the approach is published in [50, 44]. Their implementation includes rigid body motion only for symmetric cases, as all motion is directed towards the center point.

3.1.2. Indirect Sharp Interface Approaches Using the Level-Set Method

Tracking of interfaces using a sharp interface is possible by application of the level-set method (LSM). An auxiliary function (or level-set function) ϕ is introduced which is positive inside the regarded phase and negative outside, so $\phi = 0$ defines the location of the interface. Usually the value of the function corresponds to the signed Euclidean distance of the current point from the boundary. This enables the use of the gradient $\nabla\phi$ for example for calculation of the interface curvature. The time evolution of the auxiliary function is described with the interface velocity vector $\boldsymbol{v}$ by

$$\frac{\partial \phi}{\partial t} = \boldsymbol{v}\nabla\phi = 0 \tag{3.1}$$

Overview articles of the method in general were published for example by Osher and Fedkiw [51], Gibou, Fedkiw, and Osher [52], and van Dijk et al. [53]. Applications to sintering problems have been conducted by Bruchon et al. [54, 55] and Pino Muñoz et al. [56]. The basic approach is very similar to direct interface approaches, as diffusional fluxes are directly computed by application of Fick's law with potentials obtained from surface curvature. The velocity $\boldsymbol{v}$ is taken normal from the current surface and calculated from mass conservation incorporating the diffusional fluxes. The curvature is obtained from the gradient of the level-set function by

$$\kappa = \nabla \cdot \frac{\nabla\phi}{|\nabla\phi|} \tag{3.2}$$

The solution of equation (3.1), however, is usually only conducted in a certain region near the interface. Outside this region the solution is roughly approximated, but this does not preserve the signed distance character of the level-set function. For this reason, a renormalization procedure is applied to restore this character.

Level-set approaches give a clean indirect description of a sharp interface. This approach circumvents the logical problems inherent in diffuse interfaces, gives a smooth interface line in contrast to cellular approaches (section 3.1.3) and nevertheless has the advantage of easy indirect interface tracking. The main disadvantage of this method is the restriction to only two phases, at least no extension to multiple phases has been published yet. This is likely the main reason, why the adoption of this method for sintering problems is currently rather low. Especially, there is no numeric distinction between the particles, just between solid and void. Therefore, also the boundary between the particles is inexistent, so grain boundary diffusion cannot be described.

3.1.3. Indirect Sharp Interface Approaches Using Cellular Automata

A cellular automaton (CA) is a discretization scheme, where the solution space is discretized in finite cells, of possible arbitrary shape. The state of these cells (or sometimes voxels), for example the belonging to a phase or grain, is defined by discrete state variables. In each time step the state variables' values are changed in dependence on the governing model equations. For sintering, these governing equations are usually randomized, which leads to the kinetic Monte-Carlo-Method (kMCM), which will be discussed in section 3.2.2. In this type of model there is usually no continuous or almost continuous interface, as the interface is just the outer boundary of coherent cells belonging to the same phase or grain. This can be imagined as a low resolution pixel graphic with discrete colors.

3.1.4. Indirect Diffuse Interface Approaches Using the Phase Field Method

The phase field method is based on the idea to describe the phase distribution in a regarded solution space by the value of an auxiliary variable, usually referred to as phase field variable or order parameter. The task of tracking the interface is reduced to the task of describing the time evolution of the order parameter by use of differential equations. There are several excellent review articles on basics and application of the phase-field method (PFM) [57, 58, 59, 60, 61, 62, 63].

The method resides on a grand energy potential, either written in terms of Gibbs energy G or Helmholtz energy F, which describes the thermodynamic evolution of the system. For the example of a coupled thermal and diffusional system consisting of two phases, the total differential of the specific Helmholtz free energy f can be stated as equation (3.3), with the specific entropy s, the temperature T, the chemical potential μ and concentration c_i of species i, as well as the order parameter for the phase ξ and the thermodynamic driving force of phase transition β, which includes both the driving force of bulk transformation and of interface generation or destruction.

$$\mathrm{d}f = -s\mathrm{d}T + \mu_i \mathrm{d}c_i - \beta \mathrm{d}\xi \tag{3.3}$$

According to Allen and Cahn [64] the driving force β is given as in equation (3.4), which includes a dependency on the gradient of the order parameter ξ, which covers the change in energy due to change in interfacial areas. κ_ξ is a material dependent parameter (gradient coefficient), which is related to the interface energy γ.

$$\beta = -\left(\frac{\partial f}{\partial \xi} - \kappa_\xi \nabla^2 \xi\right) \tag{3.4}$$

Assuming kinetics proportional to the driving force (as usual for small driving forces), equation (3.5) as evolution prescription for the order parameter is obtained with M_ξ as kinetic constant (mobility constant).

$$\frac{\mathrm{d}\xi}{\mathrm{d}t} = M_\xi \beta = -M_\xi \left(\frac{\partial f}{\partial \xi} - \kappa_\xi \nabla^2 \xi\right) \tag{3.5}$$

To ensure numerical stability and accuracy, the discretization width of the numerical solution should be smaller than the interface width (at least four cells within the interface according to Qin and Bhadeshia [58]). The width of the interface can be estimated using

$$l = \sqrt{\frac{2\kappa_\xi \xi_{\mathrm{eq}}}{f(\xi) - f(\xi_{\mathrm{eq}})}} \tag{3.6}$$

For conserved order parameters, the Cahn-Hillard equation [65] can be applied instead as

$$\frac{\mathrm{d}\xi}{\mathrm{d}t} = \nabla \cdot M_\xi \nabla \left(\frac{\partial f}{\partial \xi} - \kappa_\xi \nabla^2 \xi \right) \tag{3.7}$$

Additional kinetic equations are required for the other state variables T and c_i. Temperature kinetics are usually described by equation (3.8) with the specific heat capacity c_V and the heat conductivity k, which is in fact just Fourier's heat conduction law. Diffusion kinetics, however, can be described by equation (3.9) with a law similar to the classic description of diffusion using Fick's law (see section 2.2). Note that the mobility coefficient M_{ij} is here not identical to the diffusion coeffcient D, since it relates the flux to the gradient of chemical potential μ, instead of the concentration gradient. These common laws are in fact just special cases of the Cahn-Hillard (equation (3.7)) equation, as both are conserved fields.

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{1}{c_V} \nabla \cdot (k \nabla T) \tag{3.8}$$

$$\frac{\mathrm{d}c_i}{\mathrm{d}t} = \nabla \cdot \left(M_{ij} \nabla \mu_j \right) \tag{3.9}$$

The phase field method offers a flexible approach to the description of heterogeneous thermodynamic systems, as additional energy contributions can be included easily as additional terms in the energy functional equation (3.3). Additional phases can be included by introduction of additional order parameters. Additional internal processes like chemical reactions can be regarded in the same way. The numerical treatment of all state variables in the system is very similar, the only important distinction is the for conserved fields (e.g. concentrations) and non-conserved fields (e.g. phase variables). Numerical difficulties occur due to the varying magnitude of the involved gradients. Influence of crystal orientation can also be introduced into the system by directionally dependent material parameters [66]. For sintering problems the influence of temperature gradients is usually neglected, except for Yang et al. [67]. Today, the PFM is the most used approach for simulation of sintering, but also applied in other fields of materials modeling and beyond. The main reason for this appears to be its flexibility and clean and easy formulation of multiple coupled mechanisms.

The main disadvantage of phase field models is the unphysical wide interface caused by the continuous description of phase composition by the order parameters. Therefore, material parameters in energy and kinetic functions are not directly identifiable with actual physical quantities. They depend not only on the actual material properties, but also on the numerical properties of the model, such as interface width and discretization width. The local characteristics of the (approximately) sharp real interface must be transformed to the diffuse interface while maintaining the globally observable behavior. Applications of this method to sintering problems often target on general observations on sintering behavior far from any real material data [66, 68, 69, 70, 71, 72, 73]. Biswas et al. [74] and Biswas, Schwen, and Tomar [75] determined the mobilities M from the corresponding diffusion coefficients by use of a method by Moelans, Blanpain, and Wollants [76]. Several authors [77, 74, 75, 78] used an equation system developed by Ahmed et al. [79] to determine the gradient coefficients κ and from actual material data, however still including the numerical properties.

A major difficulty in application to sintering problems is the description of rigid body motion. For modeling dense microstructures, the only possible changes are due to phase transformations and diffusional flows, since the space is fully occupied by matter. In sintering, however, there is usually a significant amount of void (or atmosphere), which is not able to withstand the motion of a solid body. The distinct grains move relative to each other due to changes in their contact geometry effected by diffusional flows along the grain boundaries. This is observed macroscopically as shrinkage, a main phenomenon of sintering processes. The movement is not directly driven by diffusional flows or phase transformations, so it cannot be included directly in the grand energy functional. Some phase-field-based sintering model neglect this mechanism, therefore not regarding the phenomenon of shrinkage [69, 77, 80, 70, 71, 72, 66, 81, 67, 78].

An approach by Wang [68] (and adopted by others [82, 74, 75]) regards the rigid body motion by additional terms in the Allen-Cahn and Cahn-Hillard equations including an advection velocity of the particles. This advection velocity is calculated from the force and torque balance on the particle volume. Forces and torques are obtained from the order parameter gradient. Shinagawa [73] used a coupled discrete element method (DEM) model to determine the rigid body motion independently of the phase field. Both models are updated from the other by an iterational process. Ivannikov et al. [83] introduced a grain boundary force similar to that of Wang [68], but did not include it in the kinetics equation, rather they set a constraint on it to be always zero. If the constraint is violated, they calculate and apply an advection vector to restore it.

The underlying numerical method to solve the field equations is usually a FDM [68, 80, 70, 71, 72, 81] or FEM [69, 77, 74, 75, 84, 83] scheme. Most models are formulated in 2D, only a few in 3D space [84, 82].

Using a FEM discretization offers the advantage of being able to use locally refined meshes according to the needs of the phase field gradient. Such methods are commonly referred to as adaptive meshing or remeshing. Especially in sintering models there are regions where the phase field function is almost constant (in the inner of particles, in the void) and regions with large gradients (near the interface). Using a fine discretization as required to describe the gradient correctly in the whole domain means a huge waste of computational efficiency. However on sintering models, the applied remeshing procedures are poorly discussed. Some authors [69, 84, 83] just claim that they have used it, without any further specification. Others [77, 74, 75] adopted a method of Tonks et al. [85] on quadrilateral elements to refine near the interface based on the current gradient of the phase field variables. Moelans, Blanpain, and Wollants [76] recommend a spectral moving mesh method of Feng et al. [86], but without any own results supporting this recommendation. Larger efforts on this topic have been conducted in the field of fracture respectively crack propagation modeling, which generally has the same caveats as the damage variable is widely constant except in the crack region. Several approaches [87, 88, 89, 90, 91, 92] use the estimated discretization error effected by the current mesh to determine the required refinement. Others [93, 94, 95] use a predictor-corrector scheme which ensures a finer discretization in the crack region, which is determined by the predicted value of the phase field variable in the next time step. Klinsmann et al. [96] used an procedure that keeps the product of the phase field gradient and the discretization width approximately constant. In this field some authors [87, 88, 89, 90] favor triangular elements over quadrilateral, since the remeshing procedure does not create hanging nodes there. All authors agree on the general computational benefit when applying adaptive meshes, although the performance of the remeshing procedure can have a significant impact on computational effort, especially when applied in each time step or even in each step of the weak form minimization procedure.

3.2. Monte-Carlo-Methods (MCM)

The term Monte-Carlo-Method (MCM) refers to a large variety of different methods with the common property to use random numbers as key feature of the algorithm. Lemieux [97] gives a detailed overview about MCMs and sampling methods used for them. Commonly, a distinction is made between Monte-Carlo sampling and Monte-Carlo simulation. The former uses random numbers following a certain distribution as input to a function to explore the values of the function in the regarded argument space. The latter uses sampling from distributions to model actual random processes. According to Lemieux [97] these are in fact just two different ways of viewing the problem, the formulation can be converted into each other. Both will be discussed in section 3.2.1. A method of this type will be applied for powder modeling in section 5.4. Another class of methods is often referred to as kinetic MCM. This term refers to modeling the evolution of a system in time by altering the system state in each time step based on random numbers and the probabilities of state change. This method was used for modelling of sintering and microstructure evolution before (e.g. [98, 99, 100, 101]), were the probability of state change usually correlates to the local chemical potential. These methods will be briefly discussed in section 3.2.2.

3.2.1. Classic Monte Carlo Methods

The basic idea behind classic MCMs is to use randomly sampled values as input to a function and analyzing the distribution of result values afterward using the methods of descriptive statistics. A classic application is the integration of multidimensional functions. Especially for higher dimensional functions, MCMs are much more effective than deterministic, grid-based integration methods like the Newton-Cotes or Gaussian formulae [97]. MCMs need fewer function evaluations to obtain a certain precision than the deterministic grid methods. The principle is to calculate the average of the function in the regarded interval by estimating the expectation value of the result observations.

Let us consider a function f which is deterministic, but is fed with a stochastic variable X and maps this to another stochastic variable Y:

$$Y = f(X) \tag{3.10}$$

The variable X is distributed according to a distribution known a-priori with the probability density function (PDF) d_X and the cumulative density function (CDF) D_X. To apply a MCM one needs to generate a sample x_i of size N which follows this distribution. The most straightforward way to accomplish this is inversion. Given a uniformly distributed variable U with samples u_i one may obtain samples of X by applying the inverse of the CDF D_X^{-1} as in equation (3.11). This method is the simplest and computationally cheapest way of generating such a sample if the inverse CDF is available and cheap to compute. For some distributions, there is no explicit formulation of the CDF, most notable for the widely used normal distribution. If inversion is not applicable, there are other methods such as acceptance-rejection and composition. These are out of the scope of this brief introduction, see Lemieux [97] for further information.

$$x_i = D_X^{-1}(u_i) \tag{3.11}$$

To obtain a sample u_i of the uniform variable U one needs a sufficiently good source of random numbers. One may use lists of real random numbers or hardware random number generators, but usually pseudo random number generator (PRNG) algorithms are applied. These are in fact not random, rather deterministic, but the number sequences generated resemble real random sequences well enough. The main advantage of these algorithms over true random number generators is their repeatability, which means, that the same sequence can be generated multiple times if the same start conditions (seeds) are used. This enables reproducible calculation results, which is especially useful for debugging und assessment. Good PRNG are characterized by good uniformity of the distribution and high cycle periods. The actual quality assessment of such generators is a complicated task and out of the scope here. Lemieux [97] gives a brief introduction to this topic and references a large number of more detailed publications. For this work the Mersenne Twister [102] was chosen as recommended there.

Evaluation of the function f multiple times with the previously obtained samples x_i as input produces the observations y_i of the variable Y. These observations are then investigated using classical descriptive statistics. Special values of interest are often the expectation $\mu(Y)$ and the standard deviation $\sigma(Y)$ or variance $\sigma(Y)^2$. Those can be estimated from the sample by equations (3.12) and (3.13). Note the $N-1$ in equation (3.13) for the sample standard deviation, because the naive approach with just N would introduce a bias into the estimation caused by the linking to the estimated expectation $\hat{\mu}(Y)$. There are several other possibilities to construct estimators for those properties, which may be more efficient (converge to required precision with less computational effort) depending on the characteristics of the problem. Such estimators are known under the term variance reduction techniques, see again Lemieux [97] for an overview on these.

$$\hat{\mu}(Y) = \frac{1}{N} \sum_{i}^{N} y_i \tag{3.12}$$

$$\hat{\sigma}^2(Y) = \frac{1}{N-1} \sum_{i}^{N} \left(y_i - \hat{\mu}(Y)\right)^2 \tag{3.13}$$

3.2.2. Sintering Simulation by Kinetic Monte Carlo Methods

The basic idea behind most kinetic MCMs (e.g. [98, 99, 100, 101, 103]) is to discretize the model space into finite volumes (voxels) with a defined discrete state, which may change with a defined probability. This type of system discretization is also referred to as cellular automaton. In each step, a voxel is chosen randomly and the probability of changing its state is calculated. Then, the state is changed conditionally by comparing the probability to the value of a random number. The term "kinetic" refers hereby to the correlation of the number of steps performed to the process time. This correlation is usually not directly obtainable, but has to be determined by comparison to experimental results, if desired.

The way to determine the probability of state changes varies in the published models. Tikare, Braginsky, and Olevsky [98, 100, 103, 104] evaluate the energy change ΔE of the system for the case that the state change has actually happened. The energy change is then correlated to the probability by equation (3.14), where a negative energy change (reduction of system energy) is always accepted, however positive ones tend to zero probability by an exponential law.

$$p = \begin{cases} \exp\left(-\frac{\Delta E}{k_{\mathrm{B}} T}\right) & \Delta E > 0 \\ 1 & \Delta E \leq 0 \end{cases} \tag{3.14}$$

Another way correlates the probability to the local surface geometry, which resembles classic sintering theory. Luque et al. [99] (although for liquid phase sintering) evaluate the number of neighboring voxels with the same or different state and correlate the probability with that count. High count of non-solid (liquid or void) voxels beneath a solid imply a high convex curvature at this point. In contrast, a high count of solid voxels beneath a non-solid one mean a high concave curvature. In Luque et al. [105], they evaluate a number of different approaches for the correlation between these counts and the probability. They remark that although lots of different approaches could be chosen, only few are feasible due to the danger of generating numerical artifacts in the system state. A similar approach was taken by Wang and Atkinson [101].

This type of models is especially useful for liquid phase sintering, since there are no voids, so state change from liquid to solid or vice versa does not violate the mass conservation law. In solid state sintering one has to make sure that for every solid voxel appearing another one is removed. Especially to model densification, void voxels have to be transported to the outer surface. Braginsky, Tikare, and Olevsky [98] remove a solid voxel at the outer surface for every vacancy annihilated at a grain boundary. This approach, although conserving the mass, appears like teleportation of mass from the surface to the inner volume and lacks therefore in physical justification.

The main difficulty of these models is the mapping of the internal time of the algorithm, often referred to as Monte Carlo steps (MCS), to the actual time of the real process. The models usually do not involve any kinetic parameters like diffusion coefficients or mobilities, as they are not required for the solution of the model. However, this complicates the transfer of the simulation results to the experiment or comparison to other simulation approaches. Usually, a linear correlation between MCS and real time is assumed, where the proportionality factor is fitted by experiments or other simpler models.

3.3. The Thermodynamic Extremal Principle

The classic formulation of the TEP was basically formulated by Svoboda and Turek [106], however dependent on the works of Ziegler [107, 108] and Onsager [109]. It is based on the assumption that the dissipation rate in the system is always maximized with respect to thermodynamic and kinetic constraints. This is equivalent to the statement, that the system always tries to approach the equilibrium as fast as possible. There are several distinct formulations of the TEP, they differ mainly in details of their formulation and the conditions of validity. Fischer, Svoboda, and Petryk [110] pointed out that Ziegler's principle is the most general one. The other can be derived from that. The following elaborations are based on Ziegler's principle in the formulation by Svoboda and Turek [106] with own modifications to meet the requirements of the current work.

The dissipation $\mathscr{D}$ can be formulated as in equation (3.15a) in dependence on the vector of external variable of state (EVS) $\boldsymbol{X}$, the vector of internal variable of state (IVS) $\boldsymbol{x}$ and the velocity vector of IVS $\dot{\boldsymbol{x}}$ (see section 2.3.2 for a brief introduction on IVS). It is a bilinear form in the velocity vector of internal state variables and the vector of thermodynamic forces. The latter can be expressed as $\partial G(\boldsymbol{X},\boldsymbol{x})/\partial\boldsymbol{x}$, where G is the Gibbs energy of the system. Note the minus sign, as the dissipation is being maximized, where the change in Gibbs energy is defined negative for spontaneous processes. The Gibbs energy is as a state function only dependent on the current state of the system, thus the thermodynamic forces do not include any kinetic constraints. Since $\mathscr{D}$ is linear in $\dot{\boldsymbol{x}}$, maximizing without further constraints is not meaningful. Therefore, an additional dissipation function $\mathscr{Q}$ is introduced, which must be always equal to $\mathscr{D}$, since both describe the dissipation of the process (compare equation (3.15b)). However, $\mathscr{Q}$ is formulated in terms of the kinetic conditions of the process and is generally a non-linear form in $\dot{\boldsymbol{x}}$, often a quadratic form. The actual form of $\mathscr{Q}$ heavily depends on the regarded process, but it commonly includes empirical kinetic material parameters such as diffusion coefficients or mobilities.

$$\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) = -\frac{\partial G(\boldsymbol{X},\boldsymbol{x})}{\partial \boldsymbol{x}} \cdot \dot{\boldsymbol{x}} \rightarrow \max_{\dot{\boldsymbol{x}}} \tag{3.15a}$$

$$\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) = 0 \tag{3.15b}$$

The constrained optimization problem in equations (3.15) can be solved using the Lagrange formalism. The problem is reformulated with the additional parameter λ as the Lagrange functional

$$\mathscr{L} = \mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) + \lambda\left(\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}})\right) \tag{3.16}$$

Setting the gradient of $\mathscr{L}$ with respect to $\dot{\boldsymbol{x}}$ and λ equal zero gives the optimum of the dissipation $\mathscr{D}$ under the given constraints:

$$\mathscr{L}_{\dot{\boldsymbol{x}}} = -(1+\lambda)\frac{\partial G(\boldsymbol{X},\boldsymbol{x})}{\partial \boldsymbol{x}} - \lambda\frac{\partial \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}})}{\partial \dot{\boldsymbol{x}}} \overset{!}{=} 0 \tag{3.17a}$$

$$\mathscr{L}_{\lambda} = \mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) \overset{!}{=} 0 \tag{3.17b}$$

Note that the derivative with respect to λ is always identical to the constraint equation.

For the simple case of $\mathscr{Q}$ being quadratic in $\dot{\boldsymbol{x}}$ the solution of the system is given in equation (3.18), which is a linear system of equations of the same size as $\dot{\boldsymbol{x}}$. Compare Svoboda and Turek [106] and Fischer, Svoboda, and Petryk [110] on this.

$$-\frac{\partial G(\boldsymbol{X},\boldsymbol{x})}{\partial \boldsymbol{x}} = \frac{1}{2}\frac{\partial \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}})}{\partial \dot{\boldsymbol{x}}} \tag{3.18}$$

Note that for the formulation above, it is required that the dissipation $\mathscr{D}$ and the dissipation function $\mathscr{Q}$ must depend on the same kinetic variables $\dot{\boldsymbol{x}}$. Often the fluxes $\boldsymbol{j}$ are used therein. Recently, Hackl, Fischer, and Svoboda [111] published a generalized formulation of the principle breaking up the need to have the same kinetic variables in the dissipation $\mathscr{D}$ and the dissipation function $\mathscr{Q}$. The following elaborations use a different notation than in the reference, which fits better to the needs of the application in chapter 6, but the meaning is generally equivalent. The dissipation $\mathscr{D}$ is defined in the same way as before, but the dissipation function $\mathscr{Q}$ does not include the velocities of internal state $\dot{\boldsymbol{x}}$, but instead the fluxes $\boldsymbol{j}$ as in equation (3.19b). Note that in this formulation the velocities $\dot{\boldsymbol{x}}$ are *not* required to be identical to the fluxes $\boldsymbol{j}$, neither must they have the same size. The relations between the velocities $\dot{\boldsymbol{x}}$ and the fluxes $\boldsymbol{j}$ are introduced by the constraints $\boldsymbol{g}$ as in equation (3.19c). The dissipation function $\mathscr{Q}$ includes non-linear forms of the fluxes $\boldsymbol{j}$ and so the fluxes are feasible targets to optimization. The state velocities $\dot{\boldsymbol{x}}$ are only present as linear forms in $\mathscr{D}$, so for each component of $\dot{\boldsymbol{x}}$ a constraint in $\boldsymbol{g}$ must be present which relates it either directly or indirectly to the fluxes $\boldsymbol{j}$. Maugin [23] clearly distinguishes between internal variables of state and internal degrees of freedom, where the first *must* be of pure dissipative nature and the latter follow defined laws but may also have a dissipative part. Internal degrees of freedom will be collected in ξ to incorporate their dissipative contributions. The associate laws must be formulated within the constraints $\boldsymbol{g}$.

$$\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) = -\frac{\partial G(\boldsymbol{X},\boldsymbol{x},\xi)}{\partial \boldsymbol{x}}\cdot\dot{\boldsymbol{x}} \to \max_{\dot{\boldsymbol{x}}} \tag{3.19a}$$

$$\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\boldsymbol{j}) = 0 \tag{3.19b}$$

$$\boldsymbol{g}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}},\boldsymbol{j},\xi) = 0 \tag{3.19c}$$

With this generalized formulation the Lagrange functional writes as in equation (3.20). Here we have more Lagrange parameters. λ_1 is equivalent to the classic formulation. The vector parameter $\boldsymbol{\lambda_2}$ for the required constraints is of the same size as $\boldsymbol{g}$.

$$\mathscr{L} = \mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) + \left(\mathscr{D}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X},\boldsymbol{x},\boldsymbol{j})\right)\lambda_1 + \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{X},\boldsymbol{x},\dot{\boldsymbol{x}},\boldsymbol{j},\xi)\cdot\boldsymbol{\lambda_2} \tag{3.20}$$

As before, the gradient of $\mathscr{L}$ is set equal to zero to obtain the constrained optimum:

$$\mathscr{L}_{\dot{\boldsymbol{x}}} = -\frac{\partial G(\boldsymbol{X}, \boldsymbol{x}, \xi)}{\partial \boldsymbol{x}} (1 + \lambda_1) + \frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{X}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \xi, \boldsymbol{j})}{\partial \dot{\boldsymbol{x}}} \cdot \boldsymbol{\lambda_2} \stackrel{!}{=} 0 \tag{3.21a}$$

$$\mathscr{L}_{\boldsymbol{j}} = -\frac{\partial \mathscr{Q}(\boldsymbol{X}, \boldsymbol{x}, \boldsymbol{j})}{\partial \boldsymbol{j}} \lambda_1 + \frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{X}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \boldsymbol{j})}{\partial \boldsymbol{j}} \cdot \boldsymbol{\lambda_2} \stackrel{!}{=} 0 \tag{3.21b}$$

$$\mathscr{L}_{\xi} = -\frac{\partial G(\boldsymbol{X}, \boldsymbol{x}, \xi)}{\partial \xi} (1 + \lambda_1) + \frac{\partial \boldsymbol{g}^{\mathrm{T}}(\boldsymbol{X}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \xi, \boldsymbol{j})}{\partial \xi} \cdot \boldsymbol{\lambda_2} \stackrel{!}{=} 0 \tag{3.21c}$$

$$\mathscr{L}_{\lambda_1} = \mathscr{D}(\boldsymbol{X}, \boldsymbol{x}, \dot{\boldsymbol{x}}) - \mathscr{Q}(\boldsymbol{X}, \boldsymbol{x}, \boldsymbol{j}) \stackrel{!}{=} 0 \tag{3.21d}$$

$$\mathscr{L}_{\boldsymbol{\lambda}_2} = \boldsymbol{g}(\boldsymbol{X}, \boldsymbol{x}, \dot{\boldsymbol{x}}, \boldsymbol{j}) \stackrel{!}{=} 0 \tag{3.21e}$$

The Jacobian matrix of this equation system is only invertible, if $\boldsymbol{g}$ provides a number of linearly independent constraints equal to the combined size of $\dot{\boldsymbol{x}}$ and ξ. Finding a general simplified equation system as done above (equation (3.18)) is here not possible due to the constraints.

4. Aim and Scope

The work aims at the following goals:

1. Develop a model for the microscopic sintering behavior of multiple particles.
2. Implement a reusable software framework for its application.
3. Investigate its numerical behavior and define feasible numerical control parameters.
4. Investigate its physical predictions, explain and compare them with literature results.
5. Develop a method for statistical representation of powders using the sintering model to allow design of powder mixtures.

4.1. Integration of the Work Within the RefraBund Project

In the research group FOR3010 RefraBund funded by the German Research Foundation (DFG), a novel refractory composite material is developed, which consists of alumina and metallic niobium respectively tantalum. The current work was conducted with a project part aimed at developing fast physically based model approaches of sintering, which are able to describe specialties of the investigated material system. These specialties are as follows:

Multi-Material-Powder The composite material consists of non-metallic (alumina) and metallic (niobium, tantalum) substances. These differ widely in their properties as interface energies and diffusivities. Especially the metallic components are prone to oxidization at high temperatures, but this atmospheric influence is out of the scope of this work.

Multi-Scale Powder Some of the developed process routes use aggregate particles which itself are alumina-metal composites [128, 129]. For the final sintering, fine-grained binder and filler powder must be added to achieve sufficient density and bonding. These large scale differences (aggregates ~1 mm, binder ~10 µm) have drastic numerical implications in terms of appropriate discretization. There is a multiscale approach in development to model these fine-grained fractions counter the aggregates as a continuum, but this work was not finished at the time writing this document. This approach will be published in later publications.

Complex Particle Shape Aggregate particles produced by crushing and sieving show complex irregular shapes [128, 129]. Another route is used to produce near-spherical beads by alginate gelation [130, 131]. Experimentally, differences in their sintering behavior have been noticed as can be explained by varying driving forces due to local geometry, but also by their surface condition.

Table 4.1: Material Data of Alumina, Niobium and Tantalum from Literature

(a) Interface Energies

Interface		Temperature T K	Energy per Unit Area γ J m^{-2}	Ref.
Al_2O_3	vac.	2123	0.905	[112]
Al_2O_3	Al_2O_3	2123	0.440	[112]
Al_2O_3	vac.	1923	1.061 ± 0.061	[113]
Al_2O_3	Al_2O_3	1923	0.747 ± 0.044	[113]
Nb	vac.	2523	2.10	[114]
Nb	Nb	2523	0.76	[114]
Nb	vac.	1773	2.55 ± 0.55	[115]
Nb	Nb	1773	0.79 ± 0.13	[115]
Ta	vac.	1773	2.68 ± 0.50	[115]
Ta	Ta	1773	0.74 ± 0.08	[115]

(b) Interface Diffusion Coefficients

Interface		Pre-Exp. Factor D_0 m^2 s^{-1}	Act. Energy Q kJ mol^{-1}	Diff. Coeff. D after Eq. 2.3 at 1873 K m^2 s^{-1}	at 2273 K m^2 s^{-1}	Ref.
Al_2O_3	air	$(7 \pm 5) \times 10^{-2}$	314 ± 21	1.22×10^{-8}	4.25×10^{-7}	[116]
Al_2O_3	air	1.5×10^{6}	544	1.01×10^{-9}	4.73×10^{-7}	[117]
Al_2O_3	vac.	50	460	7.42×10^{-12}	1.34×10^{-9}	[118]
Al_2O_3	vac.	1×10^{3}	690	5.72×10^{-17}	1.39×10^{-13}	[119]
Al_2O_3	wet H_2	2.5	565	4.38×10^{-16}	2.59×10^{-13}	[120]
Al_2O_3	dry H_2	5×10^{14}	962	7.43×10^{-13}	3.91×10^{-8}	[120]
Al_2O_3	vac.	40.5	452	1.01×10^{-11}	1.66×10^{-9}	[121]
Al_2O_3	vac.	4.2×10^{2}	494 ± 52	7.03×10^{-12}	1.87×10^{-9}	[122]
Al_2O_3	Al_2O_3	3.11[a]	419	6.42×10^{-12}	7.31×10^{-10}	[123]
Nb	vac.	0.43	193 ± 11	1.78×10^{-6}	1.58×10^{-5}	[124]
Nb	vac.	–	229	–	–	[125]
Ta	vac.	22×10^{-4}	267	7.88×10^{-11}	1.61×10^{-9}	[126]
Ta	vac.	–	279	–	–	[127]

[a] Converted under assumption of an interface thickness of $\delta = \Omega^{\frac{1}{3}} = 2.76 \times 10^{-10}\,\mathrm{m}$ with the atom volume $\Omega = 2.11 \times 10^{-29}\,\mathrm{m}^3$ [116].

Chemical Interactions Investigation in other project parts have shown that the components form ternary oxides at interfaces [132, 133, 134]. This influence will be neglected in the current model as this would require an enormous dataset about the thermodynamics of those reactions and would therefore be far beyond the scope of this work. However, the same investigation showed negligible solubility of the components in each other, so that absence of mass transfer between particle of different substance can be assumed.

Extensive literature research has been conducted for acquisition of material data (interface energies, diffusion coefficients for surface and grain boundary diffusion) but especially the data for niobium and tantalum is very sparse. Alumina is a well investigated system, with contradictory results published. Some literature results have been collected in table 4.1.

The available data of diffusion coefficients varies dramatically between the available sources. This is likely related to differing measurement procedures, purities of the substances and surface conditions. Alumina diffusion has an activation energy approximately twice as high as those of the metals, so that its diffusion coefficients vary much more pronounced with temperature. Because alumina as ionic substance consists of two elements rather than one, the question arises, whether the here used notion of diffusion as displacement of volume is limited by either the one or the other species. Prot and Monty [135] and Doremus [136] state that both species' diffusion coefficients must be equal to be able to maintain the ionic crystal structure. Paladino and Coble [137] and Cannon and Coble [123] state, however, that diffusion along grain boundaries in alumina limited by diffusion of Al atoms, as diffusion of O atoms is much faster. The latter result was explained by Doremus [136] by the occurrence of glasses formed by impurity elements at the grain boundaries, which allows circumventing the stoichiometric requirements of the ionic crystal. For this work, diffusion coefficients of Al and O in alumina will be considered equivalent. Diffusion rates for the metal-metal and the metal-alumina interfaces are missing. Generally, diffusion of niobium appears to be several magnitudes faster that of alumina. The picture on tantalum is contradictory, depending on the regarded reference, it diffuses faster, equivalently or slower than alumina, also depending on temperature.

The surface energies of the metals are approximately twice as high as of alumina at sintering temperature. Grain boundary energies for the metal-alumina interfaces are missing. Own measurements of those data were within the research group not possible for instrumentation and resource reasons. Therefore, the results presented in part III will reside on dimensionless quantities and parameters to avoid the necessity of actual material data.

4.2. Requirements to the Model

The work resides under the circumstances of the RefraBund project, where a composite refractory material consisting of alumina and metallic niobium or tantalum is developed. From these circumstances the following requirements and assumptions are derived:

Diffusion Mechanisms The model shall regard surface and grain boundary diffusion as the two main mechanisms of solid state sintering for metallic and ceramic materials. Volume diffusion is neglected, as it is usually magnitudes slower than such along surface or grain boundary. Other mechanisms like evaporation-condensation or viscose flow do not have to be expected.

Asymmetric Geometry Contacts The model shall support particles of arbitrary shape and different size. The first allows the investigation of the influence of non-ideal particle geometries. Size differences are to be limited to one or two orders of magnitude to avoid major numerical complications in terms of discretization widths. These requirements preclude the use of any symmetry assumptions which are commonly applied in previously published models.

Asymmetric Material Contacts The investigated material consists of alumina powder as well as metallic niobium or tantalum powder. The model must therefore be able to respect different material properties on particles in contact to each other with their implication of grain boundary shape and kinetic behavior. However, the materials are assumed to be insoluble in each other to avoid regard on concentration gradients within the particles. Mass transfer between particles is therefore also disregarded.

Multi-Particle-Contacts The model shall regard contacts of multiple particles, as the often applied two-particle approach neglects important effects, especially pore closing and mutual hindering of particles in their evolution. The latter is referring to the mutual influence of multiple contacts, which differ in their geometry and/or material characteristic and therefore can not be regarded as isolated.

Powder Mixture The production process involves mixing of different powder fractions to obtain optimal properties in processing and application. The development of a sintered material and its processing heavily resides on determining the ideal mixture to be used for the regarded application. The model shall be able to regard different mixtures and investigate their behavior in simulation instead of laboratory.

The model shall be restricted to 2D-space as a proof of concept, but may be extended to 3D-space as well. The latter is out of scope of this work.

4.3. Rational on Approach and Methods

The model development of the current work shall reside on a direct and sharp interface description by using the finite differences method (FDM). The main reason for this is the avoidance of a non-physically wide diffuse interface, as would come with the application of the major competitor: the phase-field method (PFM). The work shall explore the influence of concrete physical material parameters, which are included in sharp interface approaches directly, rather than requiring an additional mapping to fit into the numeric scheme. The latter would introduce additional and avoidable errors. Discretization of the bulk material is not needed, as volume diffusion is disregarded. Therefore, choosing a FEM discretization instead of FDM for the surface line only would be excessive. Application of the level-set method (LSM) can be discarded due to the lack of support for multiple phases and grains.

For getting a clean and concise mathematical formulation, as would come with the PFM, on top of the FDM representation, the thermodynamic extremal principle (TEP) shall be applied to construct the governing equations of temporal system evolution. This concept has already been successfully applied to similar problems. In the current case the complexity of the system will significantly rise in comparison to earlier applications, since the number of state variables will drastically increase. The concept has the main benefit of easy implicit incorporation of additional constraints arising from contact geometry requirements and boundary conditions.

To model the powder properties regarding size and shape of the particles, a statistical approach based on the Monte-Carlo-Method (MCM) shall be taken. Random generation of particles subject to a statistical representation of the powder characteristics shall link the microscopic model, restricted to small counts of particles, to the macroscopic world. Independent simulation of multiple draws enable furthermore parallel computing with low effort, since no attention has to be given to concurrent memory access. The problem of calculating a large representative volume element (RVE) with many particles shall be reduced to calculating multiple RVEs with fewer particles and characterizing their behavior by methods of descriptive statistics.

Part II.

Model Development

5. Model of Particles and Particle Packings

5.1. Representation of Particles by Discrete Surfaces

Continuous description of the particle surface geometry is only possible for nearly ideal geometries. For complicated geometries a discretized approach is feasible. For the current work, the concept of a node shall be introduced. A node is here considered as a discrete point of the particle surface connected with its neighbors by straight lines. The spline of those lines is defining the surface of the particle.

The location of each node in space is defined by a tuple of polar coordinates (φ, r), where φ is the angle coordinate and r the radius (distance from origin). The origin of the polar coordinates is distinct for each particle and is considered as the center of the particle, although it is generally not identical with the barycenter of the particle. The center of the particle is used as an anchor for defining a particle's position in space. With this concept the particle can be moved in space without translating the surface node coordinates and the description of node evolution is simplified, since only the local geometry must be regarded. Particle movement occurs due to diffusional fluxes in the grain boundaries, which is macroscopically observed as shrinkage. The location of a particle is defined by the Cartesian coordinates of its center x and y, as well as a rotation angle ω with the center as pole. The rotation angle always counts from the x-axis on in mathematical direction. The rotation angle also defines the origin axis of the particles local coordinate system. See figure 5.1 for illustration on this.

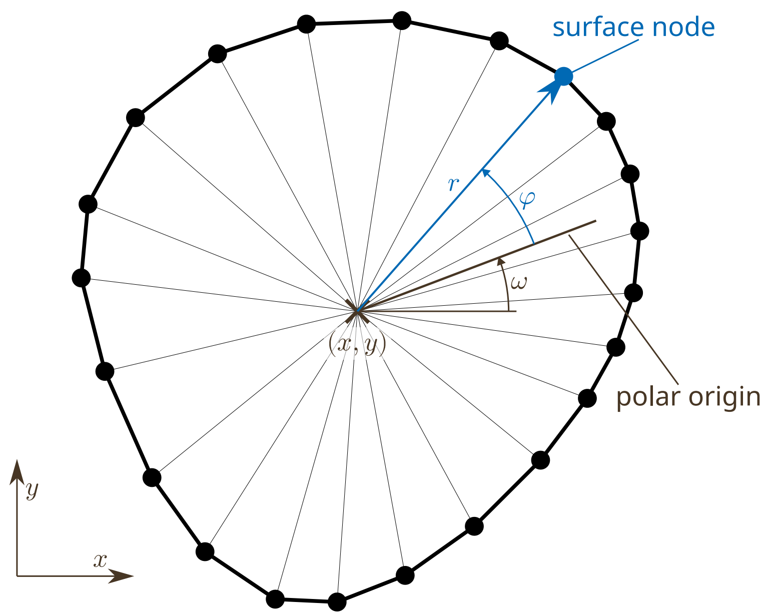


Figure 5.1: Position of a Particle and Its Local Coordinate System

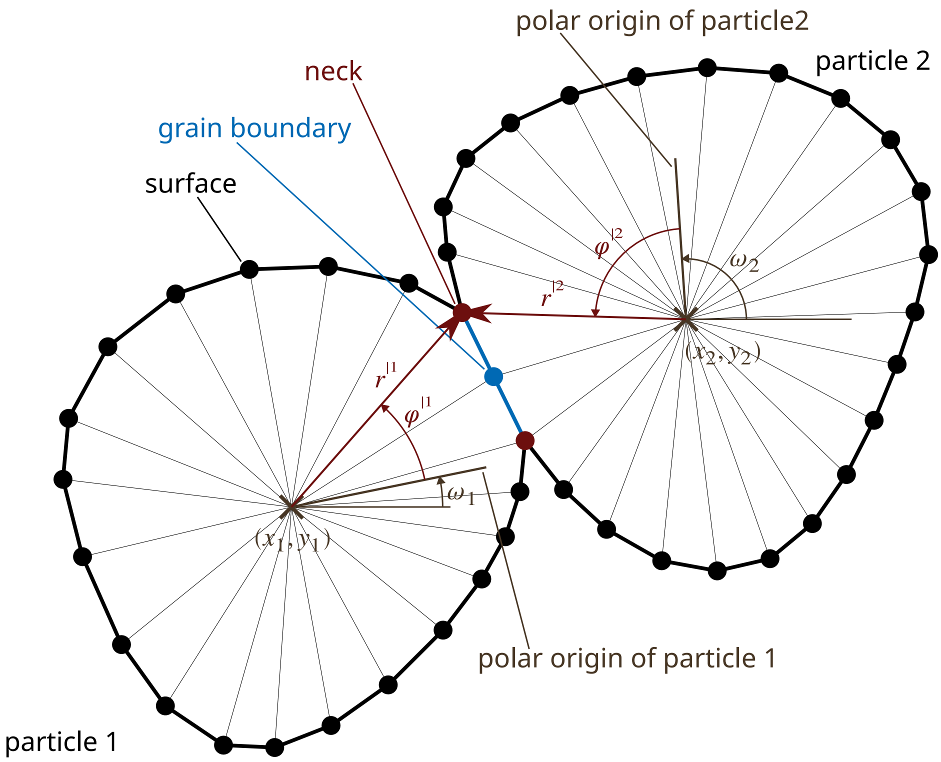


Figure 5.2: Contact of Two Particles with Respective Properties

In regard of sintering processes, the contact properties of multiple particles are investigated. The common interface of two particles in contact is commonly called a sinter neck. It consists of a grain boundary bounded by the triple points between the grain boundary and the two adjacent free surfaces. Figure 5.2 shows such a contact of two particles. Until here, three types of nodes can be identified:

Surface Nodes forming the surface of a particle in contact to atmosphere or vacuum (black).

Grain Boundary Nodes forming the grain boundary in a particle contact (dark blue).

Neck Nodes representing the triple point between grain boundary and two surfaces (dark red).

Details on the conditions at those nodes are given in the following sections. Since every particle has its own polar coordinate system, the node coordinates involved can be regarded in either particle's coordinate system. The coordinate system used shall be denoted hereinafter by $\ldots^{|\ldots}$ superscripts where necessary as demonstrated in figure 5.2, where the coordinates of one neck node are shown as $(\varphi^{|1}, r^{|1})$ and $(\varphi^{|2}, r^{|2})$. Note, that both tuples represent the same point in global space.

5.2. Geometry of a Node Element

Each node of the surface line forms an element with its neighbors, not to be confused with finite element approaches. Its geometry and important quantities are displayed in figure 5.3. Their relations used in following calculations will be derived in this section. To avoid cluttering the equations with numeric indices, the quantities belonging to the currently regarded node will be written without any index. The indices $\ldots_{\mathrm{u}}$ and $\ldots_{\mathrm{l}}$ are used to denote the upper and lower neighbor of this node.

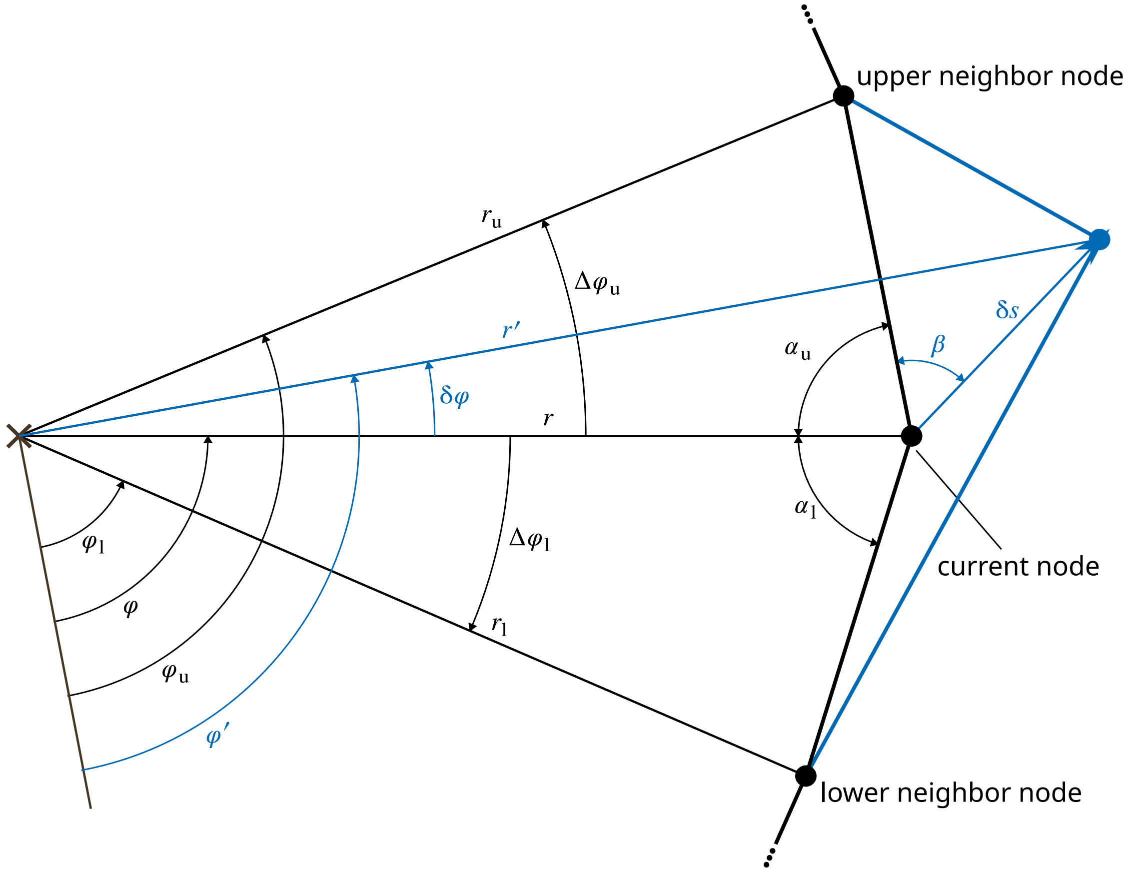


Figure 5.3: Geometry Overview of a Node Element

The surface distance between the adjacent nodes is calculated using the cosine law as in the equations (5.1), with the angle distances $\Delta\varphi_u = \varphi_u - \varphi$ and $\Delta\varphi_l = \varphi - \varphi_l$. The distances are always positive.

$$a_u = \sqrt{r_u^2 + r^2 - 2r_u r \cos\Delta\varphi_u} \tag{5.1a}$$

$$a_l = \sqrt{r_l^2 + r^2 - 2r_l r \cos\Delta\varphi_l} \tag{5.1b}$$

The angles between the surface lines and the radius vector (surface-radius-angles α_u and α_l) are calculated given the node coordinates using the cosine law as in the equations (5.2). Cosine law has been chosen over sine law due to its support for angles from 0 to π rather than $-\frac{\pi}{2}$ to $\frac{\pi}{2}$. However, when undercuts occur in the particle surface $\Delta\varphi_u$ or $\Delta\varphi_l$ may be negative, then the sign of the surface radius angle must equal that of the angle distances. This negative angle is required for the volume partial differential to change its sign accordingly as derived in section 6.3.

$$\alpha_u = \operatorname{sign}(\Delta\varphi_u) \cdot \arccos\left[\frac{r_u^2 - r^2 - a_u^2}{-2ra_u}\right] \tag{5.2a}$$

$$\alpha_l = \operatorname{sign}(\Delta\varphi_l) \cdot \arccos\left[\frac{r_l^2 - r^2 - a_l^2}{-2ra_l}\right] \tag{5.2b}$$

During simulation of sintering, the particle surface evolves by shifting the nodes to a new location in each time step. The new coordinates after shifting along a vector δs which is directed under an angle of β with the upper surface line are calculated using cosine and sine law as in the equations (5.3). The latter allows here negative changes in the angle coordinate $\delta\varphi$, which are to be expected much smaller than $\frac{\pi}{2}$. The shift vector is for now arbitrary, but will be concretized in section 6.3.

$$r' = r^2 + \delta s^2 - 2r\delta s \cos\left(\alpha_\mathrm{u} + \beta_\mathrm{u}\right) \tag{5.3a}$$

$$\varphi' = \varphi + \delta\varphi = \varphi + \arcsin\left[\frac{\sin\left(\alpha_\mathrm{u} + \beta_\mathrm{u}\right)}{r'}\delta s\right] \tag{5.3b}$$

For sintering behavior analysis and evaluation of numerical error the volume of a particle has to be determined. In the present 2D case the volume reduces to the area enclosed by the surface line, which is calculated by summing over all element triangles as in equation (5.4), where $\mathbb{N}$ is the set of all nodes the particle consists of. Note that if undercuts occur this equation is still correct since the respective elements will be counted negative, due to $\sin\Delta\varphi_\mathrm{u} < 0$.

$$V_\mathrm{P} = \frac{1}{2}\sum^{\mathbb{N}} r r_\mathrm{u} \sin\Delta\varphi_\mathrm{u} \tag{5.4}$$

5.3. Key Features of Particle Packings

Beside deeper understanding of the acting mechanisms, a main aim of sintering theory is to predict key properties of powder compacts during and after sintering, such as shrinkage and neck sizes. Correct prediction of shrinkage plays an important role in meeting required shape and size tolerances of sintered products, whereas the neck size largely determines the mechanical properties. Below, the equations to calculate these in the current model will be derived.

For two-particle models, the classic way to determine a measure for shrinkage is to calculate the decrease in distance between the particles. Most simple approaches directly include the particle distance as a state variable. For the current case it is calculated as the Euclidean distance between the centers of two particles:

$$d = \sqrt{\left(x_2 - x_1\right)^2 + \left(y_2 - y_1\right)^2} \tag{5.5}$$

With d_0 as the distance in the initial state, the shrinkage is then directly obtained via

$$\varepsilon = \frac{d - d_0}{d_0} \tag{5.6}$$

For packings consisting of multiple particles this approach is not directly applicable. One may calculate pairwise distances and the respective shrinkages. However, since multiple-particle-packings have the important feature that they are able to represent pores or voids, another approach has to be found which incorporates those.

Here, the area covered by the polygon built by the centers of the particles shall be used as a measure of the systems current size. This area is calculated by use of equation (5.7), which is commonly known as shoelace formula. For more than 3 particles in the system, only the convex hull of the particle center points must be used in this calculation, or there will be negative contributions by inner particles. Note that the points have to be ordered in mathematical rotation direction or the result will be negative.

$$A = \frac{1}{2}\sum_{i}^{\mathbb{P}} \left(y_i + y_{i+1}\right)\left(x_i - x_{i+1}\right) \tag{5.7}$$

To be able to compare this approach to shrinkages obtained by distance reduction, the square root of the polygon area is used as a measure of characteristic length in the system. Then, with A_0 as the polygon area in the initial state, the shrinkage is calculated by

$$\varepsilon = \frac{\sqrt{A} - \sqrt{A_0}}{\sqrt{A_0}} \tag{5.8}$$

Comparability of these two approaches will be discussed in section 9.2.

Neck size as the second key property is commonly described by the radius of the neck for circular or spherical models. In the current case, non-circular particle shapes will be investigated, and asymmetrical contacts will form curved neck shapes. The neck size is here defined as the half of the grain boundary length between two particles to be comparable to classic two-sphere results. Equation (5.9) formulates this approach as the sum over the set of all grain boundary nodes in the neck $\mathbb{N}_{\mathrm{GB}}$ (excluding the neck nodes). Note that each surface distance is counted twice in this formula (therefore the factor $\frac{1}{4}$), but this avoids special casing of either the lower or upper neck node limiting the regarded grain boundary.

$$r_{\mathrm{N}} = \frac{1}{4} \sum^{\mathbb{N}_{\mathrm{GB}}} a_{\mathrm{u}} + a_{\mathrm{l}} \tag{5.9}$$

For comparability, the neck size is often given as fraction of the particle radius r_{P}. Since in this work often multiple particles of arbitrary shape are present, this radius will always be a defined reference radius. This allows for comparability in parameter studies as in chapter 9 and randomized approaches as in chapter 10.

5.4. Statistical Powder Modeling

As every particle is unique, the only way to characterize a powder is by statistical quantities, either scalar properties like the mean or by a distribution. In this chapter, a statistical approach directly based on the distribution of size, shape and substance parameters shall be developed to generate inputs for the simulation. The approach resides on random sampling of particles obeying these distributions.

5.4.1. Modeling of Particle Shape Properties by a Shape Function

Common representation of particles is done using circles (2D) or spheres (3D). However, these neglect non-ideal shape features which can be of great importance. As each particle is unique, direct representation by an actual contour is not practical. The current approach will use a shape function which includes ovality and first-order waves. Multiple variants of this approach with varying depth of detail are imaginable and can be used instead of the one derived here.

The shape function will follow the general form

$$r(\varphi) = r_0 \cdot \left[\prod_i f_i(\varphi)\right] \tag{5.10}$$

with r as the local particle radius, φ as the angle coordinate and the f_i as additional terms adding a specific shape feature. If there aren't any f_i, the model will fall back to the usual circular shape

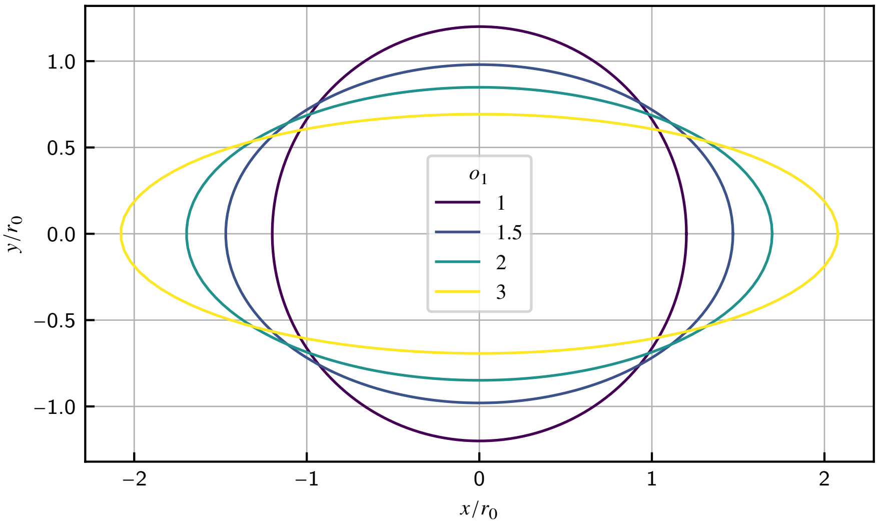


Figure 5.4: Influence of the Ovality o on the Resulting Shape

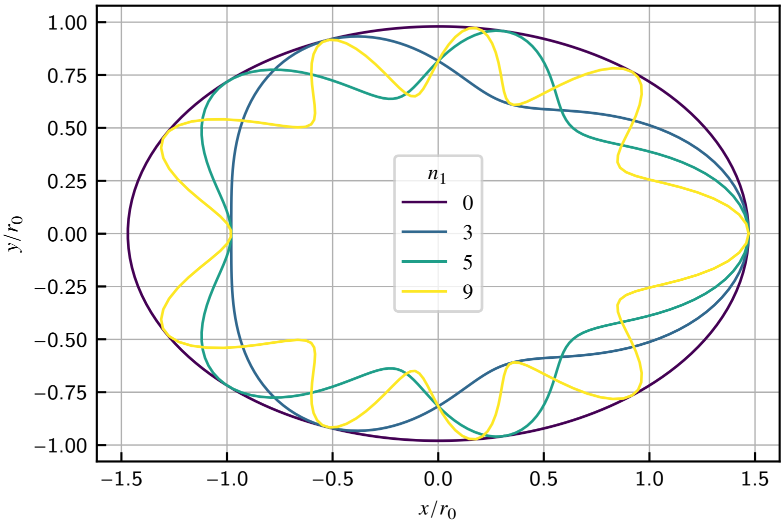


Figure 5.5: Influence of the Peak Count n on the Resulting Shape

with constant radius r_0 as the empty product has the value 1. The f_i will generally include some unknown parameters which can be used to characterize the particle shape later on. If the f_i have a notion of neutrality or deactivation, the value of f_i must be 1 in these cases.

For the regard of ovality, we define a parameter $o \in [1, \infty)$, which shall follow the common sense that

$$o = \frac{a}{b} \tag{5.11}$$

when a and b are the largest respectively the smallest measure of the particle. We can also understand a and b as the largest and smallest radii of an ellipse. In relation to the base radius r_0 we can calculate them using

$$a = r_0 \cdot \sqrt{o} \tag{5.12}$$
$$b = r_0 / \sqrt{o} \tag{5.13}$$

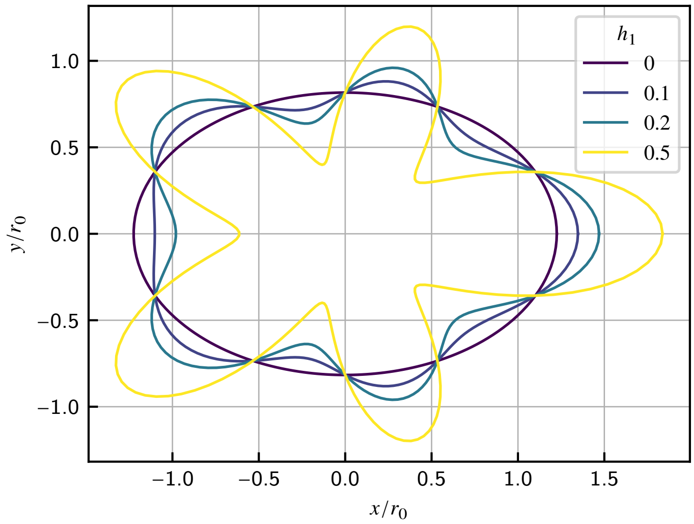


Figure 5.6: Influence of the Peak Height h on the Resulting Shape

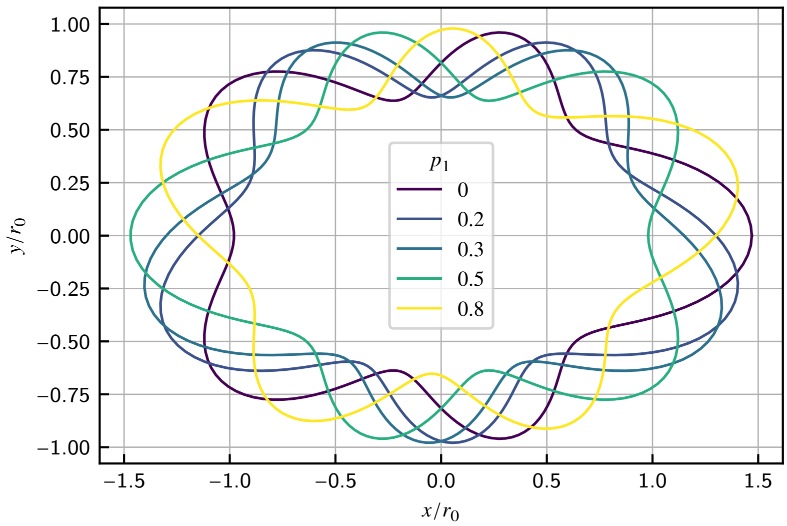


Figure 5.7: Influence of the Phase Shift p on the Resulting Shape

for which equation (5.11) holds. From the polar form of the ellipse equation

$$r(\varphi) = \frac{ab}{\sqrt{(a\sin\varphi)^2 + (b\cos\varphi)^2}} \tag{5.14}$$

we obtain, when inserting equation (5.13), the equation for the shape function of ovality f_o as

$$f_o = \sqrt{\frac{o}{o^2\sin^2\varphi + \cos^2\varphi}} \tag{5.15}$$

Figure 5.4 displays the influence of the ovality parameter on the resulting shape.

First order waves shall be modeled using a cosine function. With a relative height of the wave peaks $h_1 \in [0, 1)$, a count of waves $n_1 \in [1..)$ and a relative phase shift $p_1 \in [0, 0.5)$ we define

$$f_1 = h_1 \cdot \cos(n_1\varphi + 2\pi p_1) + 1 \tag{5.16}$$

As can be seen in figures 5.5 and 5.6, the only way to deactivate this feature is to set $h_1 = 0$, since $n_1 = 0$ is not allowed as it would introduce an ambiguity by adding a constant contribution to the radius depending on h_1 and p_1. Note especially the influence of the phase shift in figure 5.7. The phase shift enables moving the waves relative to the ovality but has a major numerical flaw. Due to its nature as an argument to the cosine function, its effect is periodic with a period of 1. If also taking particle rotation in consideration, values $p_1 \geq 0.5$ are ambiguous, too, as the same effect can be achieved by using $p_1 - 0.5$ and rotating the whole particle by 180°. So the allowed value range given above must be ensured to avoid ambiguity.

5.4.2. Modeling of Powders and Powder Mixtures

A single powder fraction can be modeled using the shape function given above by defining the statistical distribution of the base radius r_0 and the respective shape function parameters. This implicitly assumes that the shape features follow the same characteristic in all particle sizes. An actual choice of distribution functions for the shape parameters will be discussed in chapter 10. A sample of particles following these statistics can be generated by sampling from these distributions as discussed in section 3.2.

Usually, powder mixtures applied in real sintering scenarios do not consist of only one powder fraction. To obtain optimal properties, multiple fractions of certain particle size distributions are mixed together. Often they originate from different processes and therefore have different shape characteristics. They may also consist of different substances. This mixing process is here modeled by categorical distributions in a second layer to the procedure described above for simple powders. Usually, practical powder mixtures are given by volume fractions. However, the procedure here needs particle count fractions, as in each sampling process a fixed number of particles is generated.

Conversion between volume fraction x_V and count fraction x_N of a powder fraction i is done using the equivalent volume of a particle, which is only dependent on r_0, since the shape functions above preserve the volume.

$$x_{Ni} = \frac{x_{Vi}}{\pi r_{0i}^2} \tag{5.17}$$

This assumes that all powder fractions have the same packing density. The influence of varying particle shape on the packing density is neglected here, as its estimation from theory is complicated. This currently unsolved problem is out of the scope of this work.

6. Application of the Thermodynamic Extremal Principle

In the following the generalized extremal principle is applied on the current model of particles and nodes, see section 3.3 for details on the approach. As before, an index of the currently regarded particle or node is neglected where appropriate for brevity and the indices $\dots_\mathrm{u}$ and $\dots_\mathrm{l}$ are used to denote the upper and lower neighbor of a node, respectively. $\mathbb{N}$ is the set of all nodes present in the system and $\mathbb{P}$ the set of all particles, respectively.

6.1. Choice of Variables

First, the internal state variables $\boldsymbol{x}$, the internal state velocities $\dot{\boldsymbol{x}}$ and the fluxes $\boldsymbol{j}$ must be chosen. The feasibility of the approach heavily depends on this choice.

The choice of the fluxes is straightforward with the diffusional fluxes. One has two diffusional fluxes from each node j_u and j_l to the upper respectively the lower neighbor. However, the flux of the upper node to the lower and the flux from the lower to the upper are always equal due to the continuity condition. So in the following only j_u shall be regarded and j_l is substituted by $-j_\mathrm{u}$ where necessary.

The internal variables of state (IVSs) are the polar coordinates of the nodes $[r, \varphi]$ with the pole in their particle's center, since they determine the thermodynamic forces and the main aim of the simulation is to follow the time evolution of the particles and nodes. The coordinates of the particles $[x_\mathrm{P}, y_\mathrm{P}]$ are internal degrees of freedom (IDFs), since they are unambiguously defined, if the coordinates of all nodes relative to their particle are known, and also which nodes are in contact to each other. They have no inherent driving force in case of pressureless sintering. For the case of pressure assisted sintering, the pressure potential can be applied as their driving force, which is not covered in this work.

For numerical and equation formulation reasons, the internal state velocities $\dot{r}$ and $\dot{\varphi}$ are replaced by the velocities of node shift along normal and tangential surface vectors $\dot{s}_\perp$ and $\dot{s}_\parallel$. Their translation into new coordinates is directly possible using the relations obtained in the following sections.

The external state variables (such as temperature T and pressure p) are not considered in the following, since they are assumed constant.

6.2. Establishing the Equation System

In the following the necessary equations shall be composed. This includes the definition of the dissipation $\mathscr{D}$, the dissipation function $\mathscr{Q}$, required constraints and additional constraints. The resulting components of the Lagrangian gradient and the respective Jacobian matrix are omitted here for brevity, but their general structure is discussed in section 7.2.

6.2.1. Dissipation $\mathscr{D}$

The dissipation at one node is given by the product of node shifts and the respective Gibbs energy derivatives as in equation (6.1). The Gibbs energy derivatives are derived in section 6.3 for the different node types. The dissipation of the whole system is the sum of all node dissipations, since all thermodynamic forces occur at nodes. Shifting of particles alone does not follow a thermodynamic force, but is determined by the ensemble of the involved nodes.

$$\mathscr{D} = -\sum^{\mathbb{N}} \left[\frac{\partial G}{\partial s_\perp} \dot{s}_\perp + \frac{\partial G}{\partial s_\parallel} \dot{s}_\parallel \right] \tag{6.1}$$

6.2.2. Dissipation Function $\mathscr{Q}$

The dissipation function as formulation of the dissipation in terms of fluxes is defined in equation (6.2) as the sum of the product of all fluxes with their corresponding potential difference, with μ as the (unknown) chemical potential at a respective node, j_u as the diffusive flux to the upper node and the molar volume V_m of the particle substance. Regarding only the fluxes to the upper node is sufficient, since the fluxes to the lower are regarded from the lower node on.

$$\mathscr{Q} = \sum^{\mathbb{N}} \frac{\mu_\mathrm{u} - \mu}{V_\mathrm{m}} j_\mathrm{u} \tag{6.2}$$

The potential difference is obtained from the flux kinetics according to Fick's law (see section 2.2) as in equation (6.3) with the surface distance a_u, the diffusion coefficient along the interface D_u, the interface thickness δ_u, the thermal vacancy concentration c°_Va at the respective temperature, the universal gas constant R and the absolute temperature T.

$$\mu_\mathrm{u} - \mu = \frac{RTj_\mathrm{u}a_\mathrm{u}}{c_\mathrm{Va}{}^\circ \delta_\mathrm{u} D_\mathrm{u}} \tag{6.3}$$

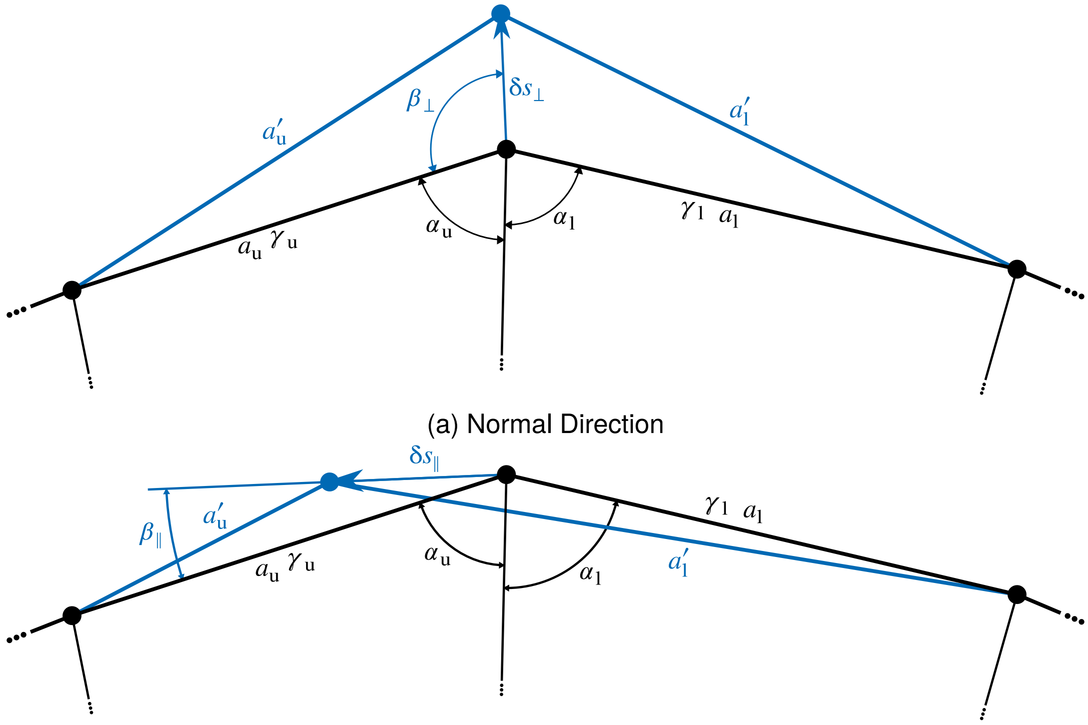
Inserting equation (6.3) in equation (6.2) leads to the final formulation of the dissipation function in equation (6.4) which is a quadratic form of j_u.

$$\mathscr{Q} = \sum^{\mathbb{N}} \frac{RT}{V_\mathrm{m} c^\circ_\mathrm{Va}} \frac{a_\mathrm{u} j_\mathrm{u}^2}{\delta_\mathrm{u} D_\mathrm{u}} \tag{6.4}$$

6.2.3. Constraints g

The required constraints needed to link node displacements and fluxes are given by the constancy of volume respectively mass at each node. The volume change at the node given by its displacement must be equal to the divergence of fluxes from and to this node. The partial derivatives of volume will be derived in section 6.3.

$$\frac{\partial V}{\partial s_\perp} \dot{s}_\perp + \frac{\partial V}{\partial s_\parallel} \dot{s}_\parallel = \dot{V} = (j_\mathrm{u} + j_\mathrm{l}) \tag{6.5}$$



(a) Normal Direction

(b) Tangential Direction

Figure 6.1: Shifting of Surface Nodes

To pin a group of particles' position in space, one of them has to be constrained at a fixed position, so that the other ones are able to move relatively to this one. This is done by constraints setting its displacement to zero as in the equations (6.6).

$$\delta x_\mathrm{P} = 0 \tag{6.6a}$$

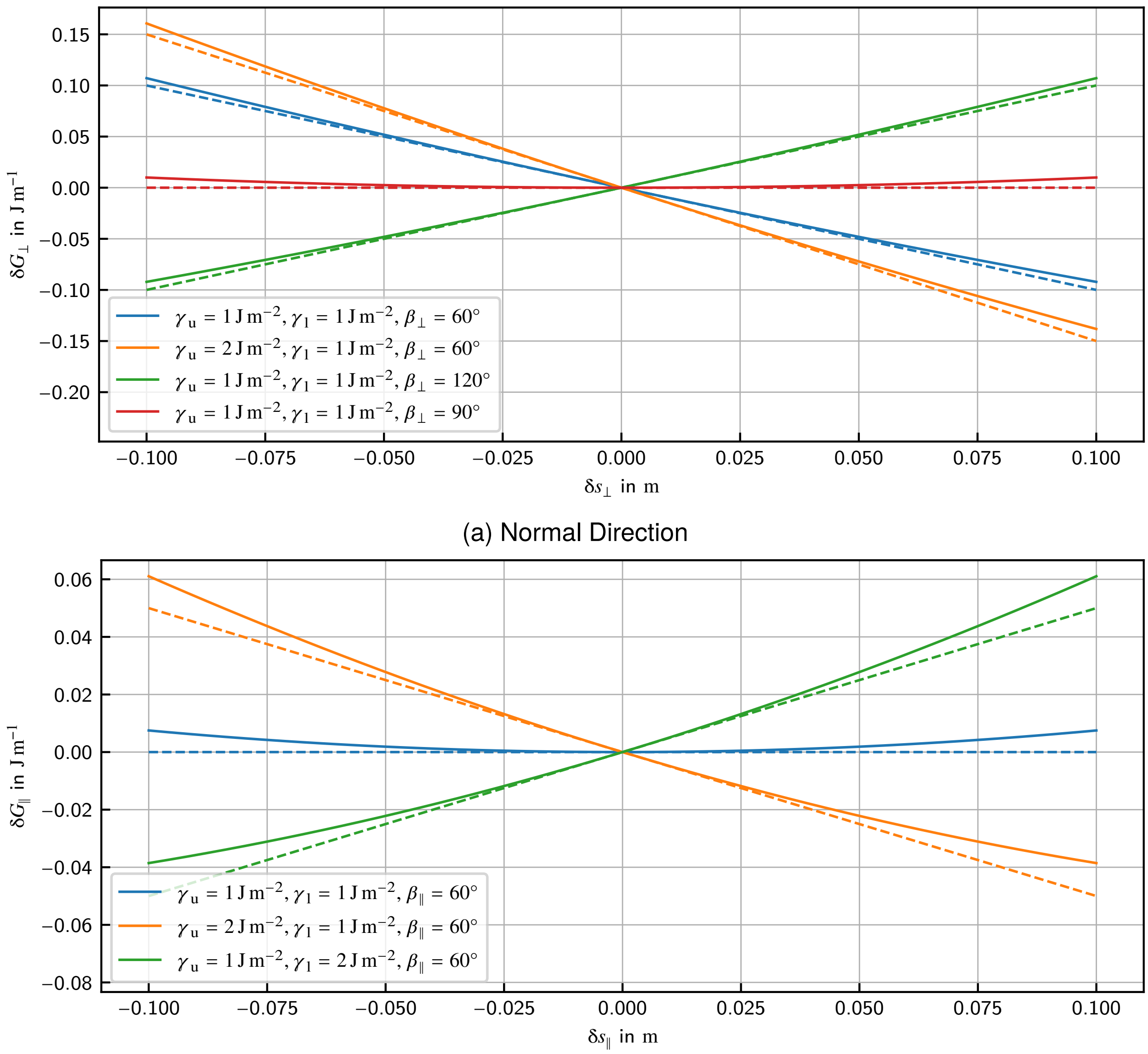
$$\delta y_\mathrm{P} = 0 \tag{6.6b}$$

To ensure the validity of the grain boundaries, so that neither voids or overlaps occur, contact constraints are to be applied to each pair of contact nodes. This is equivalent to that the node must have the identical position in global space when viewed from either of the involved particles, as already noted in section 5.1. These constraints are developed in section 6.4.

Further constraints of the like are possible, for example to enforce a certain particle movement or apply an external load on a particle to simulate assisted compaction.

6.3. Evolution of Particle Surfaces by Node Shifting

In the following, the Gibbs energy and volume gradients when shifting nodes will be derived as needed in the formulation of dissipation $\mathscr{D}$ and dissipation function $\mathscr{Q}$ given above.



(a) Normal Direction

(b) Tangential Direction

Figure 6.2: Change in Gibbs Energy Due to Node Shifting (tangents dashed)

6.3.1. Normal Shifting

The main mode of node shifting on free surfaces and grain boundaries is along the interface normal vector. Figure 6.1a shows the principal geometry of a surface node and its normal mode of shifting.

If a node is displaced in space, a change in Gibbs energy occurs due to the change in surface respectively interface area. The amount of energy bound in a surface or interface is described by the interface energy γ. Since a 2D problem is regarded, the length of the surface line a is a measure of present surface area. The change of Gibbs energy due to normal node shifting is described by equation (6.7) with the prime values as measures after shifting.

$$\delta G_{\perp} = \left(a'_{\mathrm{u}} - a_{\mathrm{u}}\right) \gamma_{\mathrm{u}} + \left(a'_{\mathrm{l}} - a_{\mathrm{l}}\right) \gamma_{\mathrm{l}} \tag{6.7}$$

The normal vector points under an angle of $\beta_\perp$ acc. to equation (6.8) to both surface lines.

$$\beta_\perp = \pi - \frac{1}{2}\left(\alpha_\mathrm{u} + \alpha_\mathrm{l}\right) \tag{6.8}$$

With a certain normal shift $\delta s_\perp$, the surface lengths after shifting are calculated by:

$$a'_\mathrm{u} = \sqrt{a_\mathrm{u}^2 + \delta s_\perp^2 - 2a_\mathrm{u}\delta s_\perp \cos\beta_\perp} \tag{6.9a}$$

$$a'_\mathrm{l} = \sqrt{a_\mathrm{l}^2 + \delta s_\perp^2 - 2a_\mathrm{l}\delta s_\perp \cos\beta_\perp} \tag{6.9b}$$

Figure 6.2a shows the change in Gibbs energy due to normal shifting with use of equations (6.7) to (6.9) with different values of $\beta_\perp$ and γ, alongside the linear approximation around $\delta s_\perp = 0$. Note that the slope is dependent on the sum of γ_u and γ_l and the sign is only dependent on $\beta_\perp$. $\beta_\perp > 90°$ means a convex surface, thus energy gain when inward shifting (negative $\delta s_\perp$). $\beta_\perp < 90°$ means a concave surface, thus energy gain when outward shifting (positive $\delta s_\perp$). $\beta_\perp = 90°$ means an even surface, thus energy loss in both shifting directions.

The partial derivative of Gibbs energy is obtained as the limit of $\delta G_\perp$ when $\delta s_\perp \to 0$:

$$\frac{\partial G}{\partial s_\perp} = \lim_{\delta s_\perp \to 0} \frac{\delta G_\perp}{\delta s_\perp} = -\left(\gamma_\mathrm{u} + \gamma_\mathrm{l}\right)\cos\beta_\perp \tag{6.10}$$

The volume derivative is obtained by calculating the area of the triangles formed by a, a' and $\delta s_\perp$ on the upper and lower side:

$$\delta V_{\perp\mathrm{u}} = \frac{1}{2} a_\mathrm{u}\delta s_\perp \sin\beta_\perp \tag{6.11a}$$

$$\delta V_{\perp\mathrm{l}} = \frac{1}{2} a_\mathrm{l}\delta s_\perp \sin\beta_\perp \tag{6.11b}$$

The partial derivative of volume is obtained as the limit of $\delta V_{\perp\mathrm{u}} + \delta V_{\perp\mathrm{l}}$ when $\delta s_\perp \to 0$:

$$\frac{\partial V}{\partial s_\perp} = \lim_{\delta s_\perp \to 0} \frac{\delta V_{\perp\mathrm{u}} + \delta V_{\perp\mathrm{l}}}{\delta s_\perp} = \frac{1}{2}\left(a_\mathrm{u} + a_\mathrm{l}\right)\sin\beta_\perp \tag{6.12}$$

6.3.2. Tangential Shifting

Regarding the tangential shifting, the direction vector is under an angle of $\beta_\parallel$ to the upper surface line, given by

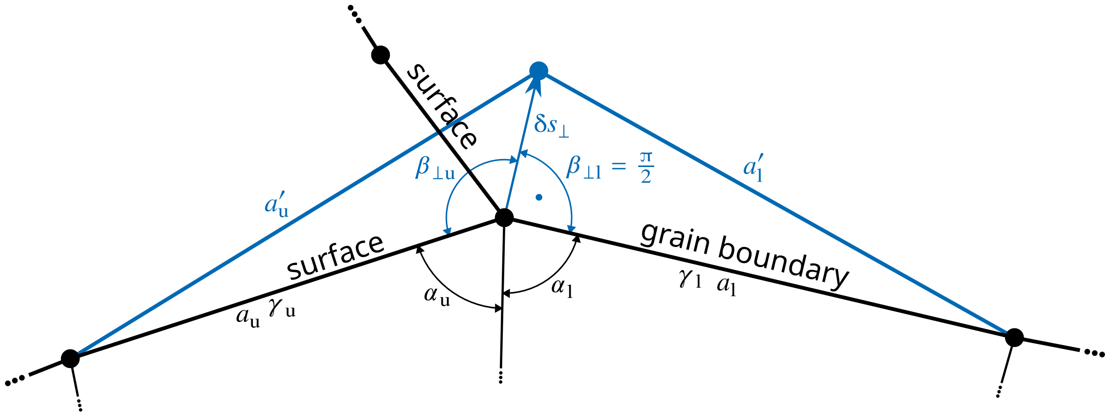
$$\beta_\parallel = \beta_\perp - \frac{\pi}{2} \tag{6.13}$$

The change in Gibbs energy is calculated similarly as in the normal case according to equation (6.14), but the shifted surface lengths are calculated as in the equations (6.15) in dependence on the tangential shift $\delta s_\parallel$. Note especially the signs in front of the cosine terms.

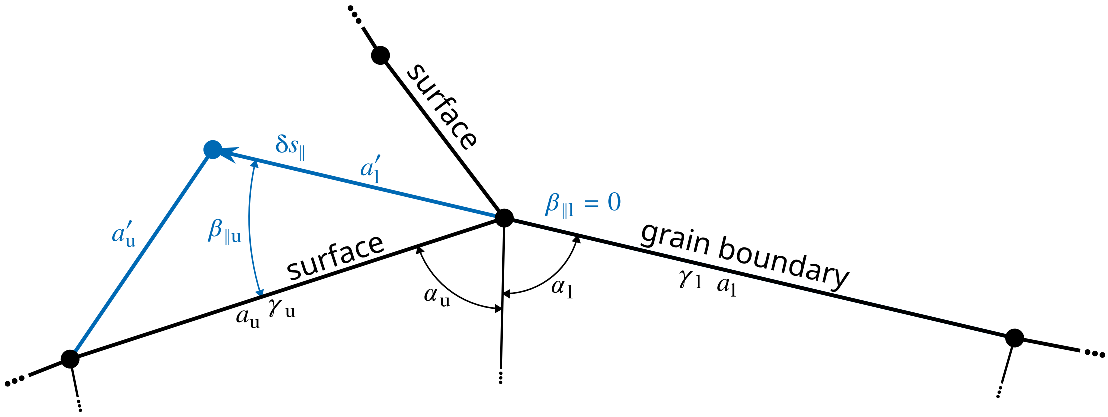
$$\delta G_\parallel = \left(a'_\mathrm{u} - a_\mathrm{u}\right)\gamma_\mathrm{u} + \left(a'_\mathrm{l} - a_\mathrm{l}\right)\gamma_\mathrm{l} \tag{6.14}$$

$$a'_\mathrm{u} = \sqrt{a_\mathrm{u}^2 + \delta s_\parallel^2 - 2a_\mathrm{u}\delta s_\parallel \cos\beta_\parallel} \tag{6.15a}$$

$$a'_\mathrm{l} = \sqrt{a_\mathrm{l}^2 + \delta s_\parallel^2 + 2a_\mathrm{l}\delta s_\parallel \cos\beta_\parallel} \tag{6.15b}$$



(a) Normal Direction



(b) Tangential Direction

Figure 6.3: Shifting of Neck Nodes

Figure 6.2b shows the change in Gibbs energy due to tangential shifting with different values of γ. The slope and monotonicity of these curves is here dependent on the *difference* of γ_{u} and γ_{l}. The convexity or concavity of the surface is here only of minor importance. Whether the interface with higher γ is located on the upper or lower side determines the monotonicity of the curves. In the special case when $\gamma_{\mathrm{u}} = \gamma_{\mathrm{l}}$, the curve has a minimum in $\delta s_{\parallel} = 0$, meaning that any shift will produce an energy loss. This is the case on all nodes except neck nodes.

Again, the partial derivative of Gibbs energy is found by the limit when $\delta s_{\parallel} \to 0$:

$$\frac{\partial G}{\partial s_{\parallel}} = \lim_{\delta s_{\parallel} \to 0} \frac{\delta G_{\parallel}}{\delta s_{\parallel}} = -\left(\gamma_{\mathrm{u}} - \gamma_{\mathrm{l}}\right) \cos \beta_{\parallel} \tag{6.16}$$

The partial derivative of volume is found similarly as in the normal case, but the contribution of the lower side must be here counted negatively, as can be seen from

$$\frac{\partial V}{\partial s_{\parallel}} = \lim_{\delta s_{\parallel} \to 0} \frac{\delta V_{\parallel\mathrm{u}} - \delta V_{\parallel\mathrm{l}}}{\delta s_{\parallel}} = \frac{1}{2} \left(a_{\mathrm{u}} - a_{\mathrm{l}}\right) \sin \beta_{\parallel} \tag{6.17}$$

6.3.3. Special Treatment of Neck Nodes

The neck nodes (triple points between two surfaces and one grain boundary) are a special case. For numerical reasons, their shifting is not taken normal and tangential to the surface line, but only

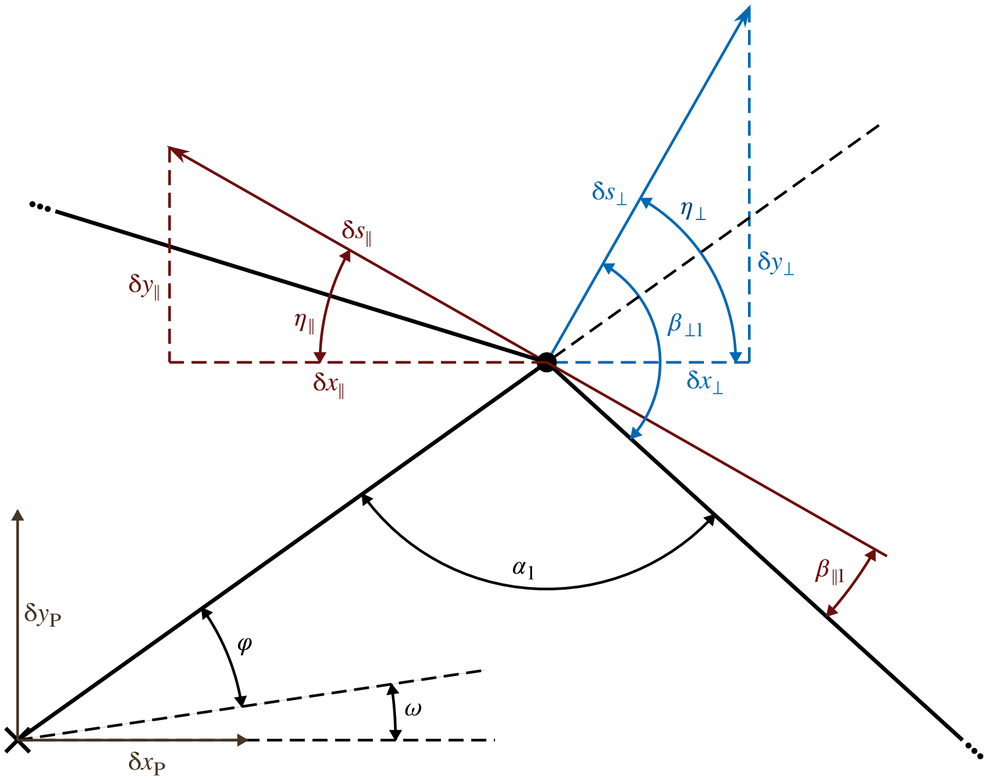


Figure 6.4: Geometric Conditions for Global Node Displacement

to the grain boundary, as this is in consistence with their primary evolution during sintering. The case, where the grain boundary is located below the currently regarded node, is shown in figure 6.3. Here, one must distinguish between the shift vector angles β to the upper and the lower surface line. For the shown case the angles are as in equation (6.19), if the grain boundary is, however, located above the node, than the values of the upper and lower angles have to be switched.

$$\beta_{\perp \mathrm{u}} = \frac{3}{2}\pi - \alpha_{\mathrm{u}} - \alpha_{\mathrm{l}} \quad \text{and} \quad \beta_{\perp \mathrm{l}} = \frac{\pi}{2} \tag{6.18}$$

$$\beta_{\parallel \mathrm{u}} = \pi - \alpha_{\mathrm{u}} - \alpha_{\mathrm{l}} \quad \text{and} \quad \beta_{\parallel \mathrm{l}} = 0 \tag{6.19}$$

The respective partial derivatives are slightly modified with the distinct angles as:

$$\frac{\partial G}{\partial s_\perp} = -\left(\gamma_{\mathrm{u}} \cos \beta_{\perp \mathrm{u}} - \gamma_{\mathrm{l}} \cos \beta_{\perp \mathrm{l}}\right) \tag{6.20}$$

$$\frac{\partial V}{\partial s_\perp} = \frac{1}{2}\left(a_{\mathrm{u}} \sin \beta_{\perp \mathrm{u}} + a_{\mathrm{l}} \sin \beta_{\perp \mathrm{l}}\right) \tag{6.21}$$

$$\frac{\partial G}{\partial s_\parallel} = -\left(\gamma_{\mathrm{u}} \cos \beta_{\parallel \mathrm{u}} - \gamma_{\mathrm{l}} \cos \beta_{\parallel \mathrm{l}}\right) \tag{6.22}$$

$$\frac{\partial V}{\partial s_\parallel} = \frac{1}{2}\left(a_{\mathrm{u}} \sin \beta_{\parallel \mathrm{u}} - a_{\mathrm{l}} \sin \beta_{\parallel \mathrm{l}}\right) \tag{6.23}$$

6.4. Contact Conditions Between Particles

In contrast to free surfaces, where the geometric evolution of a node is not constrained by the surrounding space, in grain boundaries the node shifting acts counter the solid material of the other particle. This introduces additional constraints, since the grain boundary must not form voids and must not overlap. The geometric constraints are dependent on the local evolution of the grain boundary nodes, as well as the relative position change of the particles.

Starting from a compact grain boundary, meaning that there are no holes and/or overlap present within it, the condition of maintaining the compactness of the grain boundary can be formulated as follows, if the positions of the particles are fixed:

$$\delta s_{\perp}^{|1} = \delta s_{\perp}^{|2} \tag{6.24a}$$

$$\delta s_{\parallel}^{|1} = \delta s_{\parallel}^{|2} \tag{6.24b}$$

But in reality, the positions of particles to each other change, which can be observed macroscopically as shrinkage. So we formulate instead the total displacement of a node in absolute Cartesian space, which must be equal regarded from both particles as in equation (6.25b). The total node displacement includes contributions from particle displacement (index P), normal node displacement (index $\perp$) and tangential node displacement (index $\parallel$).

$$\delta x_{\mathrm{P}1} + \delta x_{\perp}^{|1} + \delta x_{\parallel}^{|1} + \delta x_{\omega}^{|1} = \delta x_{\mathrm{P}2} + \delta x_{\perp}^{|2} + \delta x_{\parallel}^{|2} + \delta x_{\omega}^{|2} \tag{6.25a}$$

$$\delta y_{\mathrm{P}1} + \delta y_{\perp}^{|1} + \delta y_{\parallel}^{|1} + \delta y_{\omega}^{|1} = \delta y_{\mathrm{P}2} + \delta y_{\perp}^{|2} + \delta y_{\parallel}^{|2} + \delta y_{\omega}^{|2} \tag{6.25b}$$

As can be seen from figure 6.4, the angles of projection of the normal and tangential node shifting to the global cartesian axes are

$$\eta_{\perp} = \omega + \varphi + (\alpha_1 + \beta_{\perp 1} - \pi) \tag{6.26a}$$

$$\eta_{\parallel} = \left(\pi - \alpha_1 + \beta_{\parallel 1}\right) - \omega - \varphi \tag{6.26b}$$

With these and further identities on the trigonometric functions the displacements of the node are given as follows:

$$\delta x_{\perp} = -\delta s_{\perp} \cos\left(\omega + \varphi + \alpha + \beta\right) \tag{6.27a}$$

$$\delta y_{\perp} = -\delta s_{\perp} \sin\left(\omega + \varphi + \alpha + \beta\right) \tag{6.27b}$$

$$\delta x_{\parallel} = \delta s_{\parallel} \cos\left(\omega + \varphi + \alpha + \beta\right) \tag{6.27c}$$

$$\delta y_{\parallel} = \delta s_{\parallel} \sin\left(\omega + \varphi + \alpha + \beta\right) \tag{6.27d}$$

Inserted in equation (6.25b) this leads to two additional constraints for each contact node (grain boundary or neck node), which define the particle displacements $\delta x_{\mathrm{P}i}$ and $\delta y_{\mathrm{P}i}$ of each particle except one. Given one grain boundary node (only normal displacement allowed) and two neck nodes per contact, the equation system is unambiguously defined as we have 6 additional constraints (2 per node) for 2 particle displacements and 4 tangential neck node displacements. Introducing more grain boundary nodes creates more constraints but no additional unknowns (as their normal displacement and flux are covered by the volume balance respectively the optimization). This leads to an overdetermined system which forces the solution to zero shrinkage (contact nodes are not displaced at all). Also allowing the grain boundary nodes to be displaced tangentially leads to an underconstrained system as the contact conditions are not able to determine the particle position anymore. A solution to this issue has not been found at the time writing this work, so the following applications will be restricted to two neck nodes and one grain boundary node per particle per contact. This also forbids multiple contacts between the same particles, which is impossible for circular particle geometries but may be for arbitrarily shaped ones.

7. Implementation of the Model

7.1. Architecture of the Simulation Program

The model is implemented in a C# program library including

- data structures that represent sintering processes, particles, nodes and their properties
- a planar geometry library based on hierarchically defined coordinate systems
- a solver routine with adjustable subroutines for the diffusion problem as described in section 7.2
- remeshing routines as described in section 7.3
- basic algorithms for contact determination and creation (not described here)
- some more infrastructure (e.g. data storage, plotting, …).

The package is designed with extensibility in mind. Therefore, most subroutines like remeshing, integration scheme and step estimation can be selected or replaced by the user if necessary. The user can build a simulation by

- defining the initial state of the particle system
- defining one or multiple processing steps at constant temperature (stepwise constant temperature history approximation)
- selecting quantities (state variables and fluxes) to be evolved in the system as well as respective constraints or use the default set
- selecting the numerical routines or use the default set.

The code is made available to the public [138].

7.2. Numerical Solution Procedure

7.2.1. Time Integration Scheme

The time integration scheme follows a simple explicit Eulerian forward procedure. This method however, is prone to numerical instabilities, which are often observed as alternating zigzag like surface structures (see chapter 8). Therefore, the time step width must be limited. As the normal or tangential displacements of a node in one step are bad measures due to the varying distances between the nodes, it was chosen to use the surface angles (which are a measure for curvature) as indicator for time step width.

The angle difference is calculated in accordance to the equations in section 6.3 with $\delta s = \dot{s}\delta t$ via

$$\max_i^{\mathbb{N}} \max_j^{[\mathrm{u},\mathrm{l}]} \max_k^{[\perp,\|]} \arcsin\left(\frac{a_{ij}}{a'_{ij}} \sin \beta_{ijk}\right) < 0.01\,\mathrm{rad} \text{ to } 0.1\,\mathrm{rad} \tag{7.1}$$

7.2.2. Step System Solution

The application of the TEP leads to a non-linear system of equations, although most equations are linear therein with a few exceptions, most notably the main dissipation equality constraint. The system size is dependent on the count of particles present in the simulation and the count of points defining their surface. Typically, it contains from a few hundred to a few thousands of components. Solution of this system is accomplished using the standard Newton method, as this method is known for fast convergence thus sparing evaluations of the system. This method requires the evaluation of the systems gradient resp. its Jacobian matrix, which is sparse and follows the general structure shown in figure 7.1. Newton's method is, however, also known for its vulnerability to the choice of the initial solution estimation (see below). As the regarded system only includes linear, bilinear and quadratic terms, non-convergence does not have to be expected here. However, the system may have two solutions caused by the quadratic terms, so the procedure may converge to the wrong one.

Most notably, in a particle system without external loads, the trivial solution (overall dissipation is zero) is a valid solution of the equation system, but the minimum of dissipation rather its maximum. Convergence to the trivial solution usually occurs, when the estimated overall dissipation is too small by several orders of magnitude. The reasoning behind this is that the overall dissipation at necks is often underestimated by the step estimator. During testing, it has been noticed that in cases where the solution procedure converges to the trivial solution, doubling the initial solution estimate often pushes the procedure to converge to the right solution. This is a simple and fast trick to avoid this issue.

The calculation of a Newton step requires the solution of the linear system in equation (7.2), where $J_{\mathscr{L}}$ is the Jacobian of the Lagrangian and $\nabla\mathscr{L}$ its gradient. The system is solved in each step by LU-factorization using an algorithm specialized for sparse matrices of Davis [139] implemented in the CSparse.NET package [140].

$$J_{\mathscr{L}}(x) \cdot \delta x = -\nabla\mathscr{L}(x) \tag{7.2}$$

Solution Estimation

The estimation of the step solution is a crucial point to ensure fast and reliable convergence of the Newton procedure. The solution on free surfaces is easily obtained with high precision by first estimating the vacancy concentrations using equation (2.4), but expressing the chemical potential by the Gibbs energy gradient of equation (6.10) as:

$$\hat{\Delta\mu} = \frac{\partial G}{\partial \delta_\perp} \cdot \frac{2V_\mathrm{m}}{a_\mathrm{u} + a_\mathrm{l}} \tag{7.3}$$

Next, the fluxes are obtained using Fick's first law (equation (2.2)). By balancing the fluxes and dividing by the volume gradient, the normal displacement of a node is obtained as:

$$\delta\hat{s}_\perp = (j_\mathrm{u} - j_\mathrm{l})\left(\frac{\partial V}{\partial s_\perp}\right)^{-1} \tag{7.4}$$

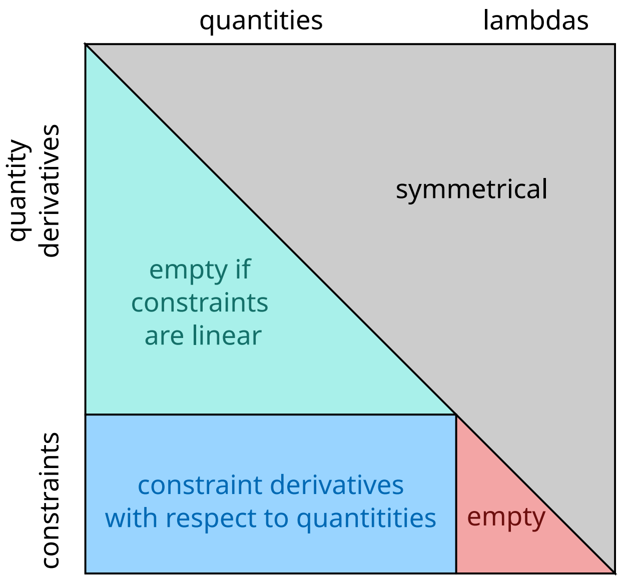


Figure 7.1: General Structure of the Systems Jacobian Matrix

When simulating the evolution of a single particle free in space, this estimate solves the step directly. A single Newton step is then only needed to obtain the Lagrange multipliers. When a sinter neck is present, the situation becomes much more complicated, as the neck points feature two modes of displacement with distinct volume and energy gradients and the requirement of maintaining contact. The issue is solved using a fixed-point iteration procedure, whose starting point is the simple estimation described above. The target variable is the flux between the neck node and the central grain boundary node, denoted here as $\hat{j}$.

In the first approximation, the normal displacement of the neck nodes must be equal to that of the grain boundary node to fulfill contact conditions. Then, using the flux balance, the tangential displacement of the neck node is calculated as in equation (7.5), with j_S as the estimated flux to the adjacent surface node calculated as above and i as the iteration index.

$$\delta\hat{s}_{\parallel i} = \left(\hat{j}_i - j_\mathrm{S} - \frac{\partial V}{\partial s_\perp}\delta\hat{s}_{\perp i}\right)\left(\frac{\partial V}{\partial s_\parallel}\right)^{-1} \tag{7.5}$$

The estimated total dissipation on the neck node is then given by:

$$\hat{\mathscr{D}}_i = \frac{\partial G}{\partial s_\perp}\delta\hat{s}_{\perp i} + \frac{\partial G}{\partial s_\parallel}\delta\hat{s}_{\parallel i} \tag{7.6}$$

Using the form of the dissipation function in equation (6.4) the new flux estimate is obtained as in equation (7.7). The sign term is required to support negative dissipations, which occur when the system reaches a quasi-stationary state of the surface near the neck in later stages characterized by small fluctuations. This flux is then inserted in the iteration again in equation (7.4) for the grain boundary nodes and equation (7.5) for the neck node till the fixed point is reached.

$$\hat{j}_{i+1} = \operatorname{sign}\hat{\mathscr{D}}_i \cdot \sqrt{\frac{V_\mathrm{m} c_{\mathrm{Va}^\circ}}{RT}\frac{D_\mathrm{u}}{a_\mathrm{u}}|\hat{\mathscr{D}}_i|} \tag{7.7}$$

7.3. Remeshing Procedures

Remeshing plays a central role in maintaining the solution procedure's stability and lowering computational effort. As discussed in section 7.2.1 the time step size achievable without producing instabilities is related to the displacement angle of a node and thus to the discretization width of the surface.

In the current implementation two distinct types of remeshing procedures are used: remeshing the free surface and remeshing near a neck node. They will be discussed below.

7.3.1. Free Surface Remeshing

The main aim of this procedure is lowering the surface node count where possible and increase it where necessary to be able to choose a time step size as large as possible and lower computational effort. It is not required to make the simulation work, but significantly accelerates the solution progress. Surfaces with even profile (low curvature) need less points to be sufficiently approximated as surfaces with high curvatures. The flatness of a surface at a node is measured by the angle between its adjacent surface lines. A surface is flat if this angle equals 180°, and is curved if this angle deviates from that. Positive and negative curvatures are treated equally. So, with a threshold chosen by the user, the criterion for considering a node for deletion is as in equation (7.8). Too low thresholds lead to fluctuating surfaces in flat regions were surface transport is numerically performed by alternatingly shifting the nodes outwards and inwards in each time step. This mechanism typically limits the time step width to avoid instable escalation of the fluctuation.

$$|\alpha_\mathrm{u} + \alpha_\mathrm{l} - \pi| < 0.01\,\mathrm{rad} \text{ to } 0.05\,\mathrm{rad} \tag{7.8}$$

A second check as in equation (7.9) is applied that the deletion would not lead to too large distance between the remaining nodes. So, too fast coarsening and lack of influence of the chosen initial discretization width is avoided. $\bar{a}$ is the current mean distance between the surface nodes of the regarded particle. The allowed relative distance can be chosen to control the speed of discretization coarsening during simulation. Adjacent nodes are never deleted in the same run. Surface nodes directly adjacent to a neck are excluded, too, since they are treated by the neck remeshing procedure.

$$\frac{1}{2}\left(a_\mathrm{u} + a_\mathrm{l}\right) < (1.5 \text{ to } 3)\bar{a} \tag{7.9}$$

If the local curvature is high as determined by equation (7.10), new nodes are inserted in the middle of the adjacent surface lines above and below. Lower values of the control parameter lead to finer discretization of curved surfaces.

$$|\alpha_\mathrm{u} + \alpha_\mathrm{l} - \pi| > 0.5\,\mathrm{rad} \text{ to } 0.8\,\mathrm{rad} \tag{7.10}$$

7.3.2. Neck Neighborhood Remeshing

This procedure is required to make the simulation work as the neck nodes approach their adjacent surface nodes when the neck is growing. If they come too close, the respective surface node must be deleted or the simulation gets stuck in a singularity. The surface node is deleted if the distance between it and the neck node falls below an adjustable fraction of the mean surface distance $\bar{a}$ as determined by equation (7.11). Higher values of the control parameter lead to large time steps but inhibit the representation of detailed near-neck surface profiles.

$$a_{\mathrm{NS}} < (0.3 \text{ to } 0.5)\bar{a} \tag{7.11}$$

Part III.

Model Application and Validation

8. Investigation of the Model's Numerical Behavior

The studies performed in this chapter aim at the investigation of the model's numerical behavior before transition to investigations on its physical implications. The test case regarded is identical in all of these. A pair of circular particles of equal size and equal material properties is used. The exact values of material data are not important, as all data is displayed and analyzed in dimensionless manner. The most important point not covered by the normalization is the ratio of grain boundary and surface energy, which is taken as $\frac{\gamma_{\mathrm{GB}}}{\gamma_{\mathrm{S}}} = 0.5$, as a difference in them is necessary to achieve a main driving force. The diffusion coefficients on both types of interfaces are taken as equal.

This choice was made as the case of two equal particles is the standard case of all microscopic sintering models. Additionally, during development and testing it has been found that circular particles are more prone to numerical instabilities due to their smooth surface, which is in or near the equilibrium in regions far from the neck. This implies that there are always regions of large and of negligible driving force present in the current state. Regions near the local equilibrium tend to show fluctuations with alternating gradient signs and perhaps instabilities, as due to the local formulation this is the only way to represent long-distance effects in the model.

8.1. Study on the Influence of the Time Step Size

In section 7.2.1 the limit of the time step width to use during time integration was given. As currently no theoretical limit for stability can be given, the free parameter has to be chosen empirically. This study aims at investigating the numerical behavior of the simulation with different maximum displacement angles per time step and fix one for the remains of this work.

8.1.1. Construction of the Study

The maximum displacement angle is varied between 0.001 rad to 0.1 rad. Surface remeshing is deactivated to keep the finely discretized surface and avoid superposition of the effects. Neck remeshing is necessarily activated with a distance limit parameter of 0.5, arbitrarily chosen.

Targets of investigation are the time consumption (duration) of the differently parametrized simulations, the step count needed and the evolution of the step width's during the simulation. As measure for precision, their agreement in shrinkage and the overall volume loss are regarded.

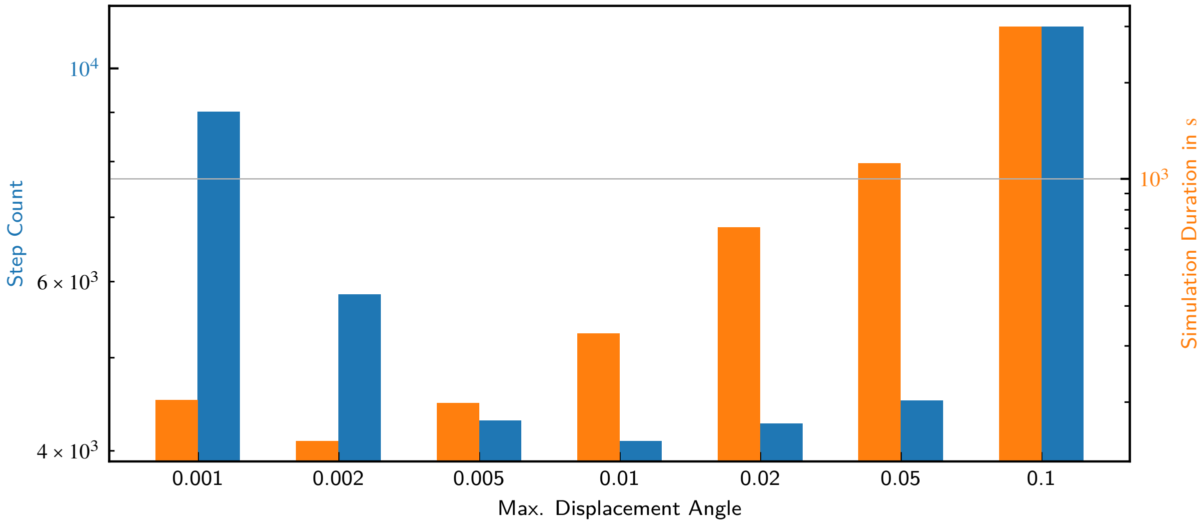


Figure 8.1: Step Counts and Simulation Time Consumption in Dependence on the Maximum Displacement Angle

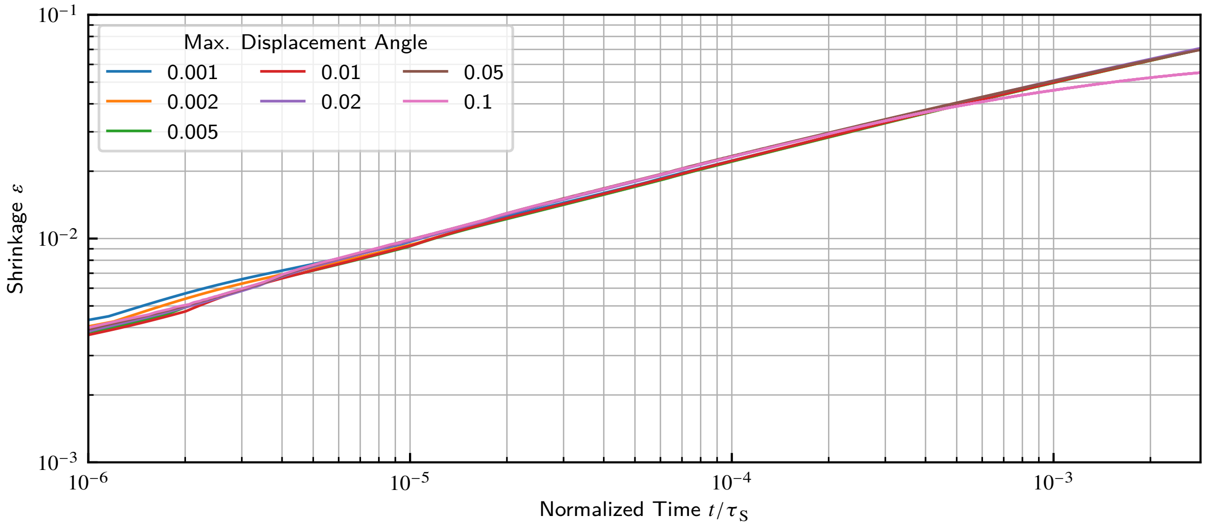


Figure 8.2: Shrinkage in Dependence on the Maximum Displacement Angle

8.1.2. Results and Discussion

Figure 8.1 shows the count of calculated time steps alongside with the total time consumption for the simulations. The most striking point is that the step count does not directly correlate with the duration. The step count shows a minimum around 0.02 rad, whereas the time consumption is constantly low till 0.02 rad and then rises rapidly. This is explained through the superimposed influence of the time needed to calculate one time step. With small time steps, the solution of the current time step is close to the one of the previous, which is used as initial guess in the Newton procedure. Thus, the routine needs fewer iterations to determine the solution. With large time steps the new solution is more distant, especially when fluctuations of even surfaces occur in later stages. This means slower convergence of equation solving. Moreover, from the calculation logs can be observed that the convergence to the trivial solution (as described in section 7.2.2) more frequently occurs when the parameter takes higher values.

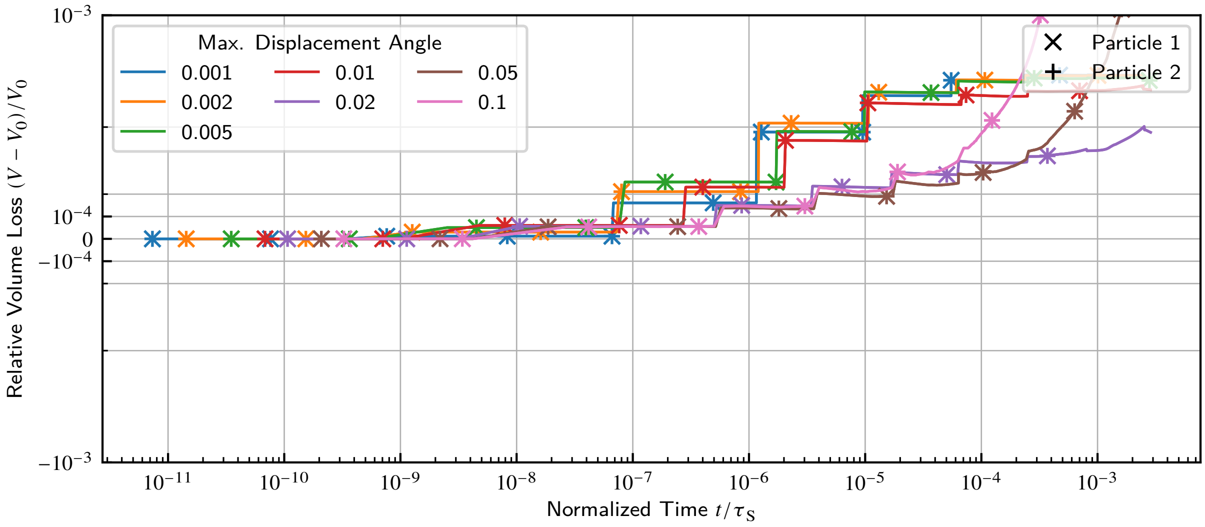


Figure 8.3: Volume Loss in Dependence on the Maximum Displacement Angle

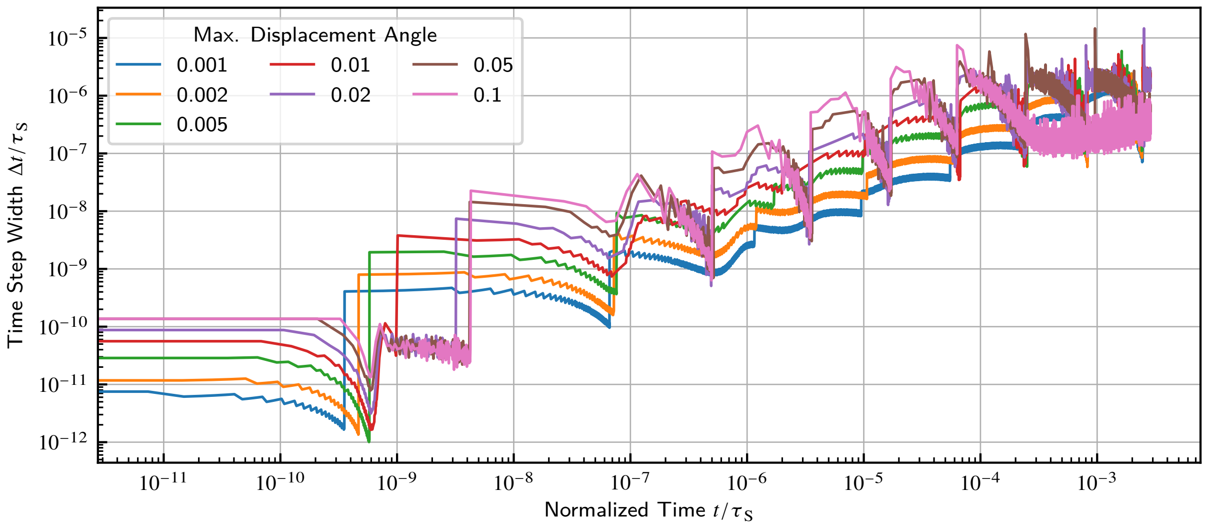


Figure 8.4: Time Step Width in Dependence on the Maximum Displacement Angle

Figure 8.2 shows the shrinkage curves obtained from the parameter variation. Except for the highest 0.1 rad they are all in well agreement. This indicates, that the latter is definitely to high to obtain accurate results. In the initial stage, the differences are larger than in the final stages, since the conditions change faster there and so a smaller step width could be desirable. However, these differences are equalized so that the results are in well agreement in the later stages nevertheless.

Figure 8.3 shows the loss of volume in the system during simulation in dependence on the time step parameter. For the values below 0.01 rad the major source of error are the remeshing steps (jumps in the shown curves), between which the error does not alter very much. For the higher values the error originating in the actual time integration is much more pronounced. Especially for 0.05 rad and above the error escalates in the later stages. This can be explained by the occurrence of significant fluctuations on the particle surface (numerical instabilities).

Figure 8.4 shows the evolution of the time step width used in the simulations for the distinct cases. It is conspicuous that simulations with the parameter values that were noticed as instable previously, also a high variance in time step width is present. For lower values the time step width follows the general trend that after neck remeshing has deleted a surface node, the step jumps up and then decreases quite continuously till the next remeshing as the neck node approaches the next surface node. This indicates, that an early neck remeshing would be desirable for efficient computation. This issue will be addressed in detail in section 8.2.

The discussed results show, that lower values of the maximum displacement angle are generally preferable. Higher values lead, as expected, to numerical errors, however do not improve the time consumption of the solution due to convergence issues of the Newton procedure. Small time steps allow faster solution due to fewer iterations per step and provide better precision in general. With regard to the high memory usage for result data storing the maximum displacement angle 0.005 rad is chosen for the followings investigation as it provides the best trade-off. Lower values did not provide notable better performance in the test case but produce significantly higher amounts of data.

8.2. Study on the Influence of Neck Remeshing

As stated previously, the procedure of remeshing near the neck is absolutely necessary to make the simulation work. However, there is a free parameter determining at which distance a surface node near the neck may be deleted. Targeting numerical efficiency it is desirable to have this parameter as large as possible as the time step decreases as already noticed in section 8.1. Too large limits, however, are expected to decrease the precision especially in regard on geometrical features near the neck. In the previous studies the limit was arbitrarily chosen as 0.5. In this study the parameter will be altered to get an idea of its effects.

8.2.1. Construction of the Study

The case matrix of the study has two dimensions. The first is the count of nodes per particle in the initial state, which is inverse proportional to the discretization width. The levels used are 50, 100 and 200 nodes per particle. The second is the parameterization of the neck remeshing procedure (section 7.3.2). The levels of the distance limit parameter used are 0.3, 0.5 and 0.7. Surface remeshing is deactivated as before.

Targets of investigation are as before the time consumption (duration) of the simulations, the step count needed and the evolution of the step width's during the simulation. As measure for precision, their agreement in shrinkage and the overall volume loss are regarded.

8.2.2. Results and Discussion

Figure 8.5 shows the count of calculated time steps alongside with the total time consumption for the simulations. In general, the step count and time consumption show an exponential dependence on the node count. Aggressive remeshing (higher limit values) is able to decrease the effort by factors from 2 to 5.

Figure 8.6 shows the shrinkage curves obtained from the parameter variation. In late stages the distinct simulations are in good agreement. Especially depending on the node count, the agreement in early stages is much weaker. This can be explained by disregarding of local geometrical

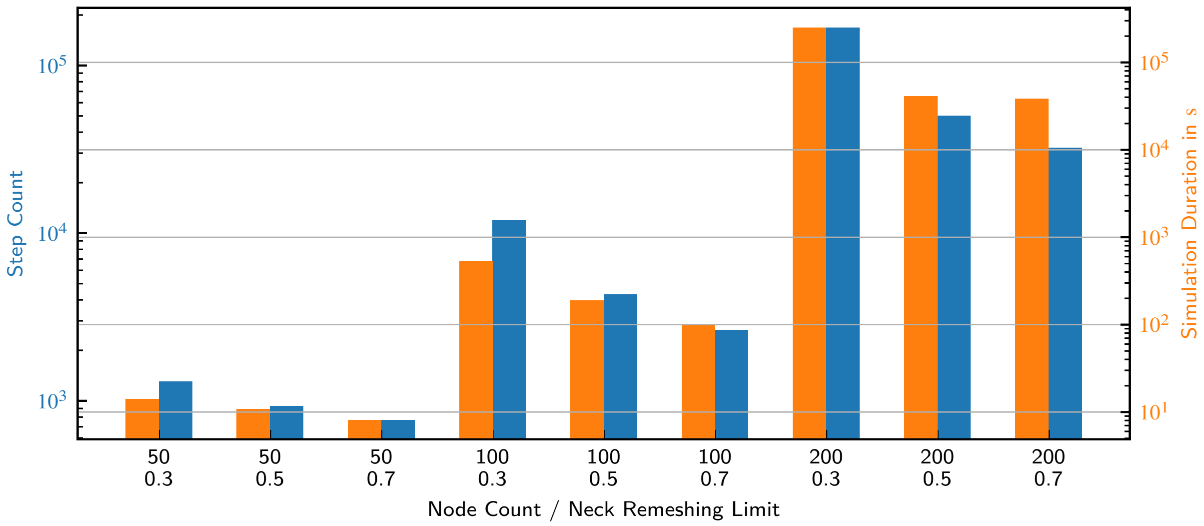


Figure 8.5: Step Counts and Simulation Time Consumption in Dependence on the Neck Remeshing Limit

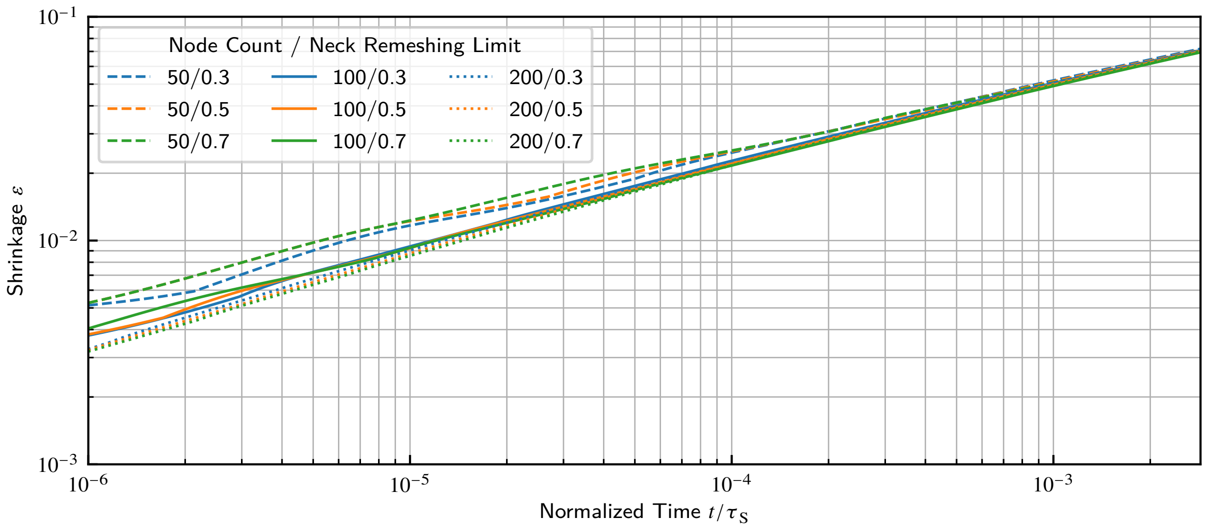


Figure 8.6: Shrinkage in Dependence on the Neck Remeshing Limit

features near the neck when using coarse discretization. When the gradients are smaller in late stages this influence has less effect. However, within the node count groups the agreement is good, with low influence of the neck limit parameter.

Figure 8.3 shows the loss of volume in the system during simulation in dependence on the parameters. As has to be expected, the numerical volume loss shows significant dependence on the node count and remeshing regime. Coarse discretization and aggressive remeshing lead to higher volume losses. All stay in the order of 1×10^{-3} and thus far below the shrinkage by diffusional flows observed and can therefore be tolerated.

Figure 8.8 shows the evolution of the time step width used in the simulations for the distinct cases. The achievable time step size is, as expected, increasing with the reduction of node count. Aggressive remeshing generally allows higher time steps in later stages and avoids the mentioned decrease in time step width when neck node and the next surface node approach.

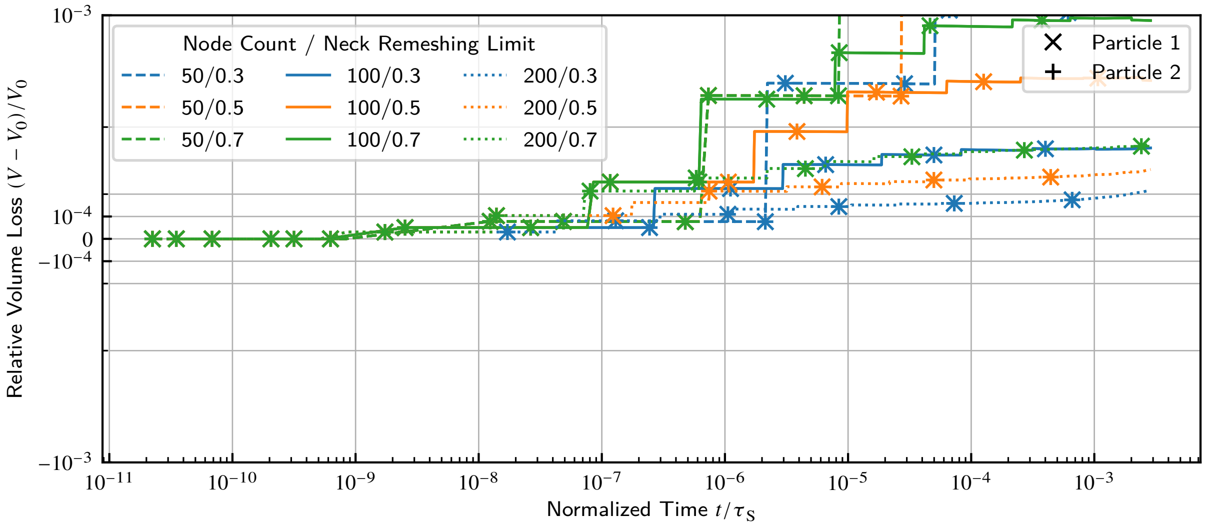


Figure 8.7: Volume Loss in Dependence on the Neck Remeshing Limit

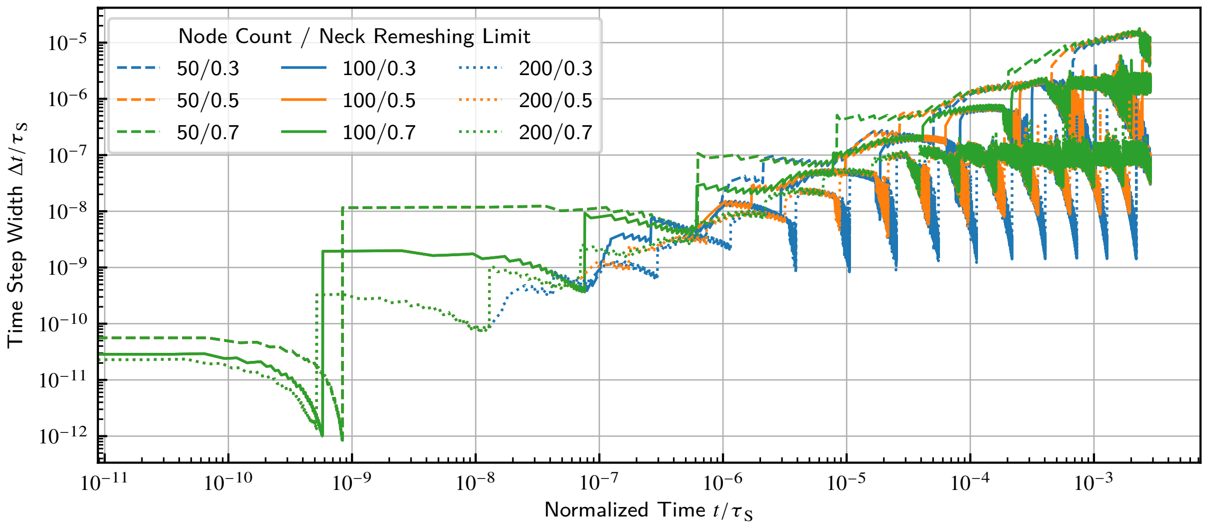


Figure 8.8: Time Step Width in Dependence on the Neck Remeshing Limit

The obtained results show, that if one is interested in the behavior in early stages, a high node count has to be used. If the regard is only paid on late stages, low node counts lead to equivalent results. The aggressiveness of neck remeshing is of minor importance regarding the simulation accuracy, but is able to significantly decrease the computational effort. For the sake of efficiency, as to be expected, node counts as low as possible are preferred, as the effort increases exponentially with the node count.

8.3. Study on the Influence of Surface Remeshing

The choice of discretization width is always a trade-off between precision of the numerical procedure, and its computational cost. Usually one aims at the finest discretization required and the coarsest possible. Often in simulation tasks, this choice has to be made one time before running the simulation. In the current case, these needs and possibilities highly vary during progress of

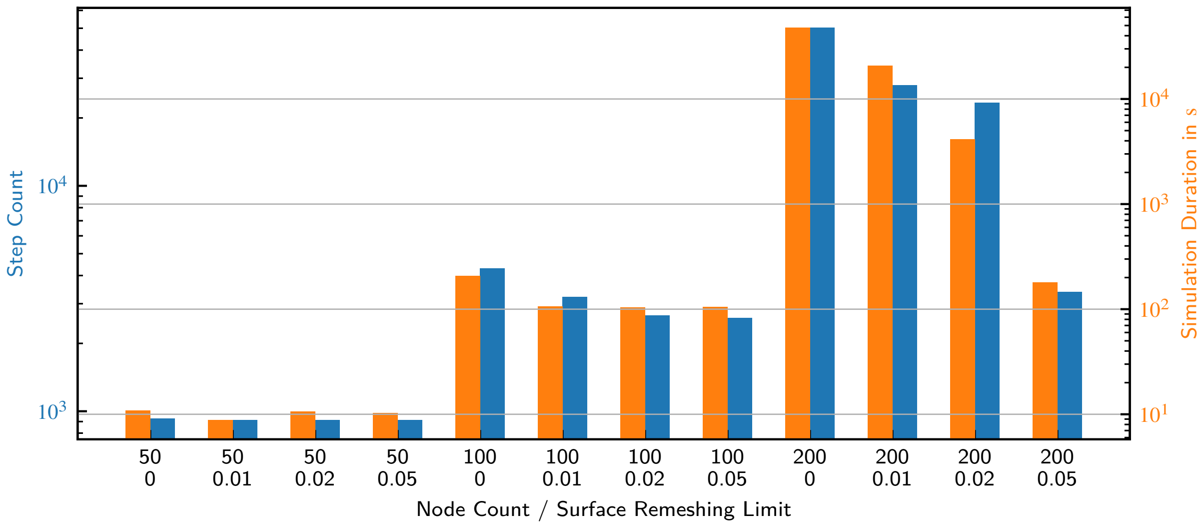


Figure 8.9: Step Counts and Simulation Time Consumption in Dependence on the Surface Remeshing Limit

simulation. So in the beginning of the sintering process, where curvatures are high a fine discretization is desirable to not miss the details. In later stages, curvatures have decreased, and fine discretization leads to unnecessarily high effort.

This study aims at evaluating the influence of the surface remeshing procedure described in section 7.3 on the precision of time integration and providing a decent choice of the parameters to use in the following investigations.

8.3.1. Construction of the Study

The case matrix of the study has two dimensions. The first is the count of nodes per particle in the initial state, which is inverse proportional to the mean discretization width. The levels used are 50, 100 and 200 nodes per particle. The second is the activation and parameterization of the surface remeshing procedure (section 7.3.1). The levels used are deactivated and the angle thresholds of 0.01 rad, 0.02 rad and 0.05 rad. As the surface follows a circular shape, high influence of node adding has not to be expected here, so it is disregarded. The surface distance limit is fixed to the default of 3.0. Neck remeshing is necessarily activated with a distance limit parameter of 0.5, as was determined as appropriate in section 8.2.

Targets of investigation are as before the time consumption (duration) of the simulations, the step count needed and the evolution of the step width's during the simulation. As measure for precision, their agreement in shrinkage and the overall volume loss are regarded.

8.3.2. Results and Discussion

Figure 8.9 shows the count of calculated time steps alongside with the total time consumption for the simulations. The general trend is similar to that of neck remeshing discussed in the previous section. Step count and time consumption increase with node count and decrease with the limit parameter. A special observation here is the 200 nodes/0.05 rad limit run which shows a drastic drop in the effort compared to the other runs. The reason for this is, that in this case the limit is

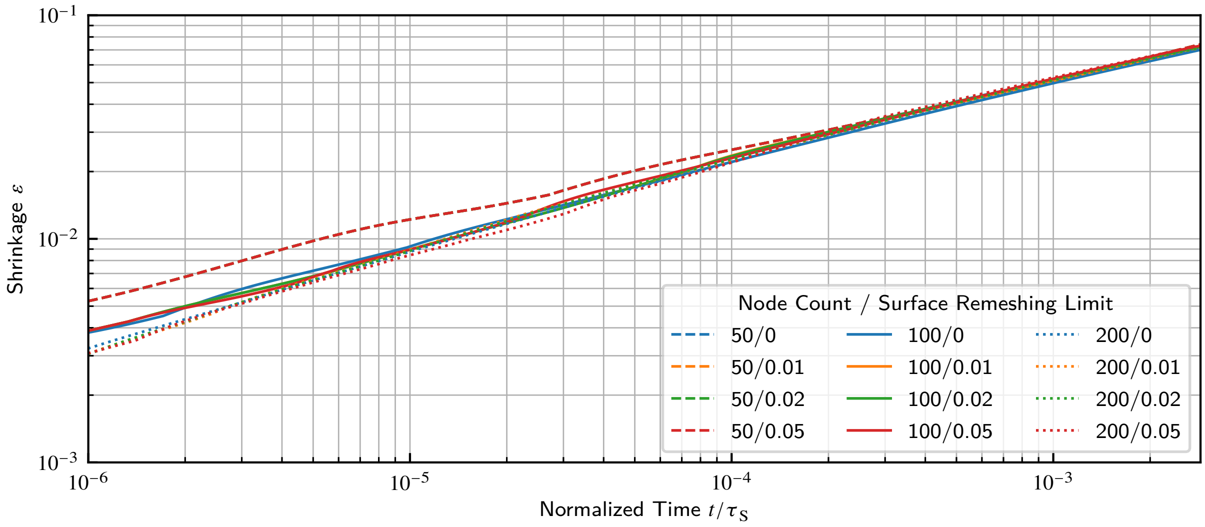


Figure 8.10: Shrinkage in Dependence on the Surface Remeshing Limit

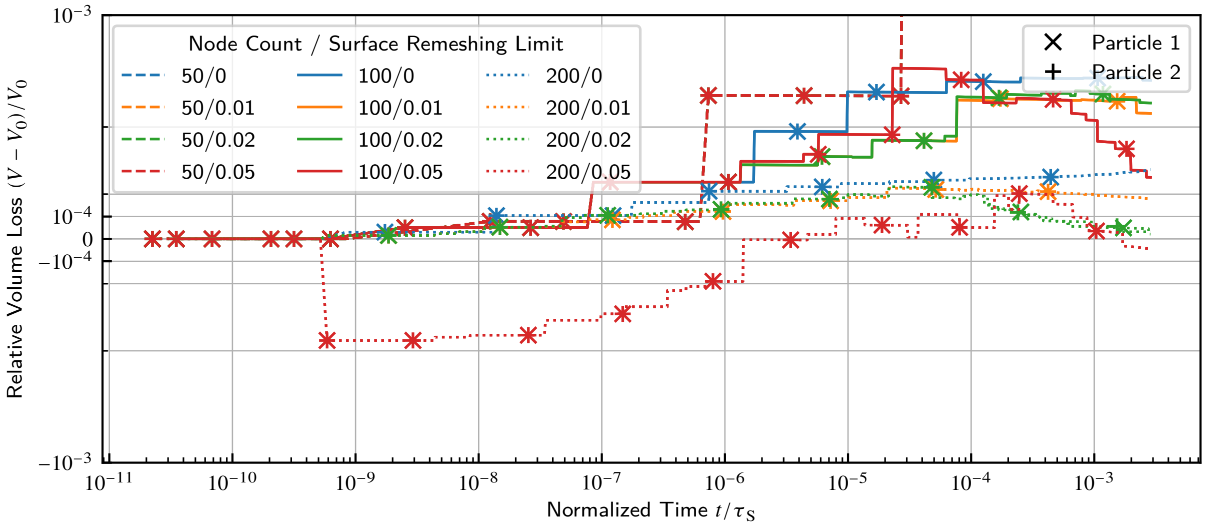


Figure 8.11: Volume Loss in Dependence on the Surface Remeshing Limit

high enough to remesh the free surface of the circular shape down to approximately 100 nodes in the first remeshing run. So this case is roughly equivalent to the 100 nodes/0.05 rad limit run.

Figure 8.10 shows the shrinkage curves obtained from the parameter variation. As above, the trend is generally equivalent to the neck remeshing. The low count runs show the same errors in the early stages, whereas the good agreement in late stages is present, too. Note that the 200 nodes/0.05 rad still shows the accurate results of the other 200 nodes runs, despite the aggressive remeshing.

Figure 8.3 shows the loss of volume in the system during simulation in dependence on the parameters. As to be expected, the general trends are similar to the neck remeshing, too. The special 200 nodes/0.05 rad case shows an extraordinarily low volume loss in late stages, as the effect of neck remeshing losses is canceled out by the counter-signed effect of the initial rigorous surface remeshing step, which was noticed above.

Figure 8.12 shows the evolution of the time step width used in the simulations for the distinct cases.

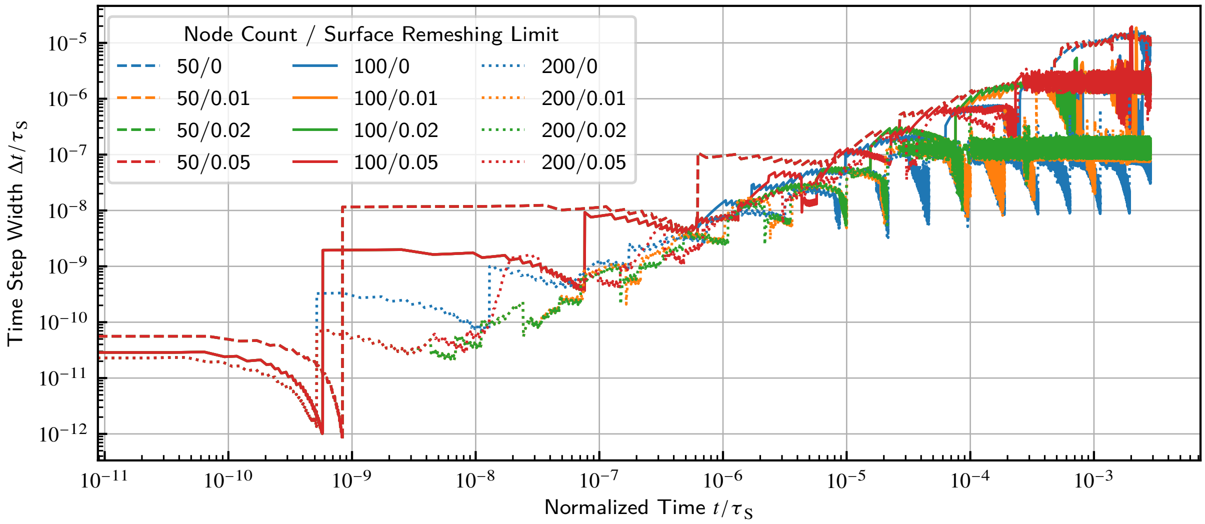


Figure 8.12: Time Step Width in Dependence on the Surface Remeshing Limit

As to be expected, the general trends are similar to the neck remeshing, too. The special 200 nodes/0.05 rad case shows the same behavior as the other 200 nodes cases in the early stages but much higher time steps than those in later stages.

The results regarding surface remeshing generally show the same trends as with the neck remeshing. The case of high initial node counts and aggressive surface remeshing shows that it could be preferable to use high initial node counts and use an aggressive remeshing step to create the initial state. With this procedure fine discretization is only maintained in regions where it is necessary (high curvatures).

8.4. Conclusions from Numerical Investigations

From the previous investigations the overall strategy applied in the following investigations is as follows:

1. fine initial discretization to capture the detailed shape features near the neck in the initial states
2. aggressive remeshing of the free surface to allow for large time steps in late stages
3. aggressive remeshing of the neck neighborhood to avoid small time steps while running into the singularity.

A fourth strategy is derived from these to combine the advantages of fine and coarse discretization where needed:

4. introduce a protected region of about 10 nodes around the neck where surface remeshing is suppressed.

This last strategy enables taking advantage of coarsening free surfaces by remeshing aggressively but preventing removing too many nodes near the neck to be able to represent the neck geometry precisely.

9. Investigation of the Model's Physical Behavior

9.1. Study on the Influence of Key Physical Parameters

The simulation of an asymmetrical two-particle-contact involves a variety of physical geometry and material parameters, which highly influence the sintering behavior and macroscopic response of the observed system. A widely used approach to characterize a system is by its dimensionless parameters. The following study investigates the influence of the dimensionless parameters present in an asymmetrical two-particle-system.

Table 9.1 lists the varied dimensionless parameters and their respective value ranges, where the indices 1 and 2 denote the first resp. the second particle. The study includes:

Symmetrical Contacts with varied ratios of interface energies and diffusion coefficients between surface and grain boundary

Asymmetrical Geometry Contacts with circular particles of different sizes and oval particles in different contact modes

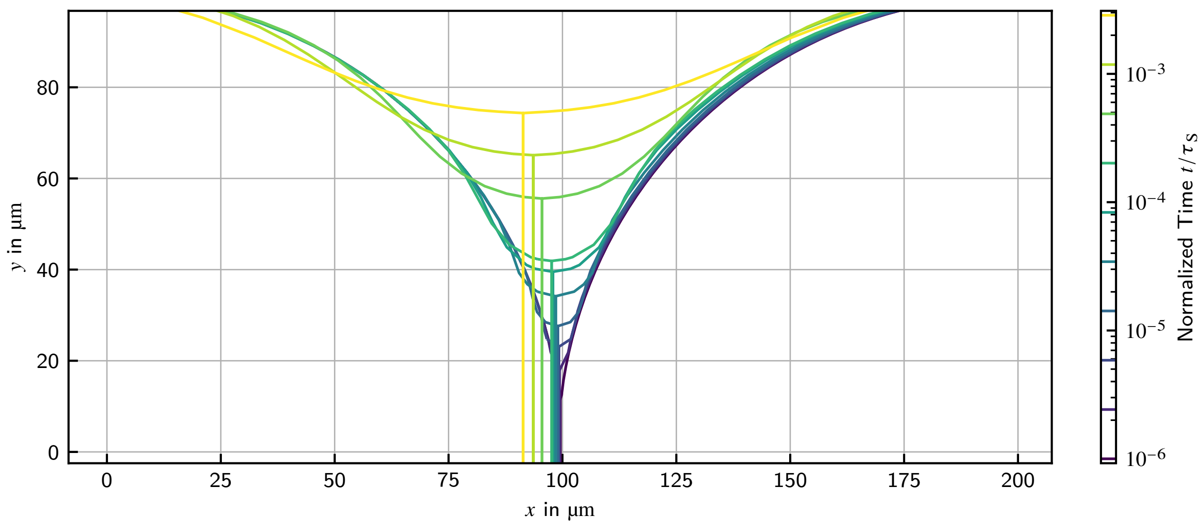
Asymmetrical Material Contacts with varied ratios of interface energies and diffusion coefficients on both sides

Further details on the study conditions are given in the following respective sections.

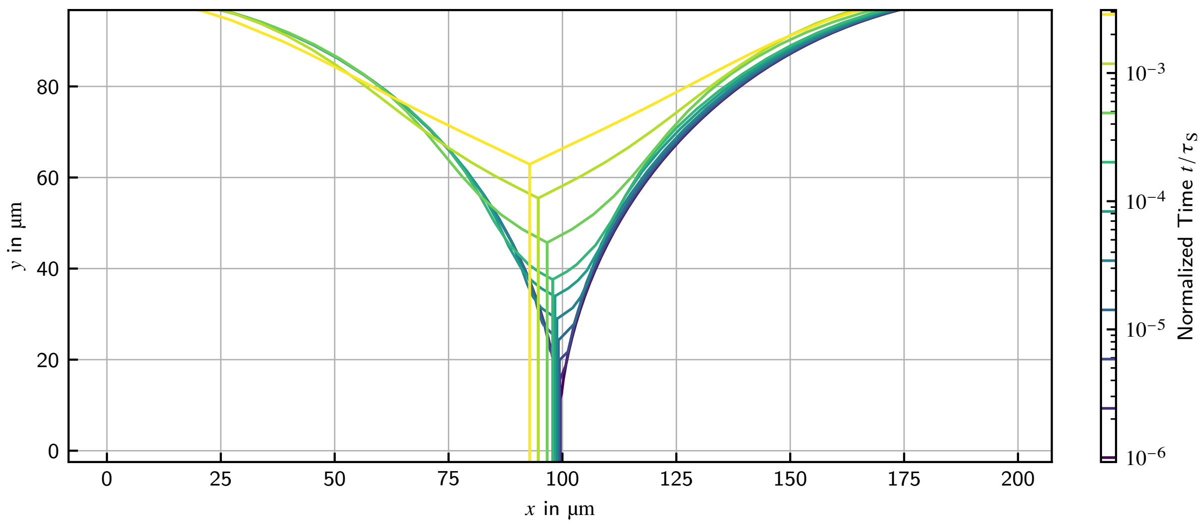
The initial contact of the particles is always created by setting the distance of the particles equal to 1 % of the reference particle's radius r_1 less, than would be needed for the particles to just touch. Then, the intersection of the particles' surface lines is calculated, and the grain boundary is placed as a straight line between the surface line intersection points.

Table 9.1: Investigated Dimensionless Parameters and their Range

Parameter	Min	Max	Scale	Remarks
$\gamma_{\mathrm{GB}}/\gamma_{\mathrm{S}}$	0.1	0.9	lin	
$D_{\mathrm{GB}}/D_{\mathrm{S}}$	0.01	10	log	
r_2/r_1	1	10	lin	
$\gamma_{\mathrm{S2}}/\gamma_{\mathrm{S1}}$	1	10	lin	
D_2/D_1	1	100	log	surface and grain boundary diffusion scaled equally
o	1	3	lin	several contacts tip vs. flank



(a) $\gamma_{GB}/\gamma_S = 0.1$



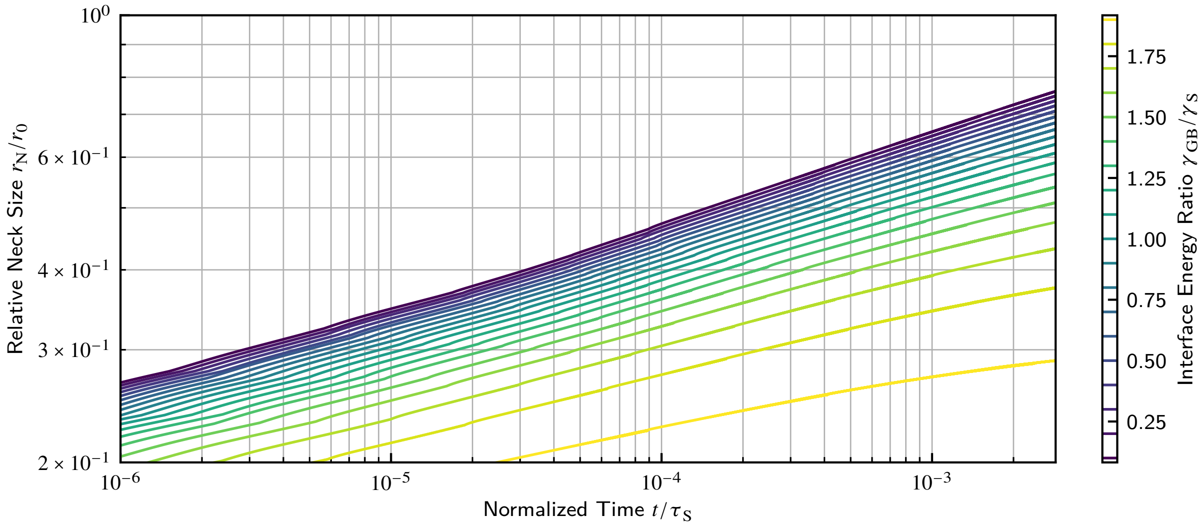
(b) $\gamma_{GB}/\gamma_S = 0.9$

Figure 9.1: Geometry Evolutions for Selected Surface-Boundary Energy Ratios

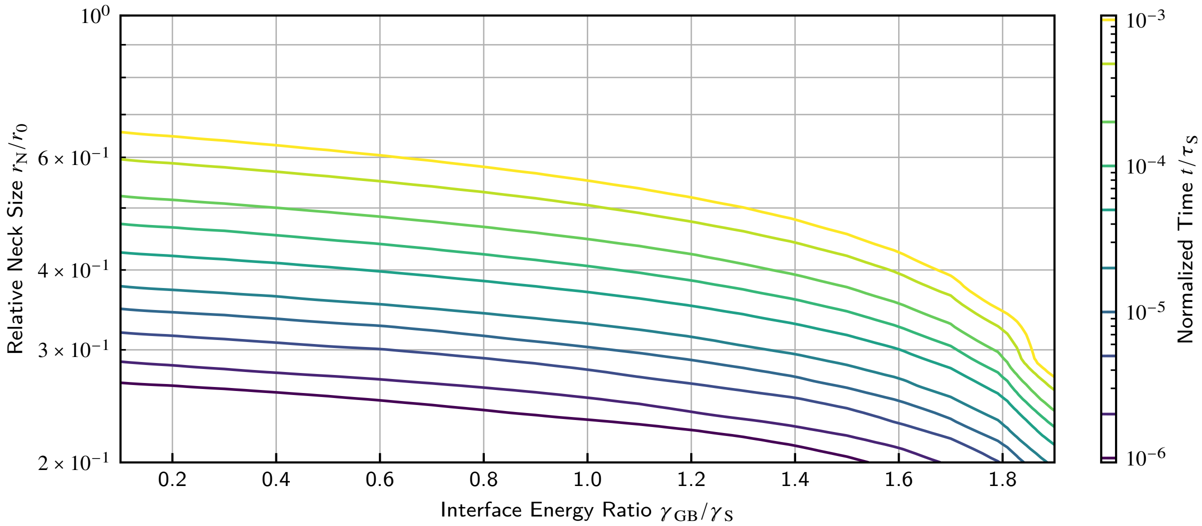
9.1.1. Surface-Boundary Energy Ratio γ_{GB}/γ_S

This study investigates the effect of the ratio between grain boundary and surface energy. Low values mean a high driving force for generating a grain boundary in favor of surface consumption. Values near 1.0 mean there is no energetic benefit of creating a grain boundary and any driving force is purely of geometric nature. In reality, usually $\gamma_{GB}/\gamma_S < 0.5$ for metallic and ceramic substances [5]. The initial particle geometry and the material parameters are taken equal on both particles.

Figure 9.1 shows two selected evolutions of the contact geometry. Note the large dihedral angle between the surfaces at low interface energy ratios and the small one at high ratios, since in the latter case the local stationary state around the neck is reached earlier. When $\gamma_{GB}/\gamma_S \to 1$ the dihedral angle approaches 120° as is proposed by equation (2.7). Accordingly, when $\gamma_{GB}/\gamma_S \to 0$ the dihedral angle approaches 180°. With high energy ratio (large dihedral angle), the particle surface around the neck has a smooth concavely curved shape. With low energy ratio (small



(a) Evolution Curves



(b) Time Isolines

Figure 9.2: Neck Size in Dependence on Surface-Boundary Energy Ratio

dihedral angle) the surface near the neck appears almost flat with transition into the convex surface on the backside of the particles. The large dihedral angle also corresponds with large achievable neck sizes.

Figure 9.2a shows the dependence of the neck size on the interface energy ratio. The neck size decreases notably with the energy ratio. A larger decrease appears as the energy ratio approaches 2. The main influences suspected here are the decrease in driving force (decrease of γ_{GB}/γ_S) in conjunction with the geometric effect of the smaller dihedral angle which have additive effect. As $\gamma_{GB}/\gamma_S \to 2$, the driving force vanishes completely, as there is no energetic benefit in growing the neck in all possible geometric configurations.

Figure 9.3a shows the dependence of the shrinkage on the interface energy ratio. The dependence of shrinkage on the energy ratio is similar to that of neck size. In addition to the previously described driving force effect, there is a counter-acting influence of a longer diffusion path and a larger volume displacement needed, when the neck is larger.

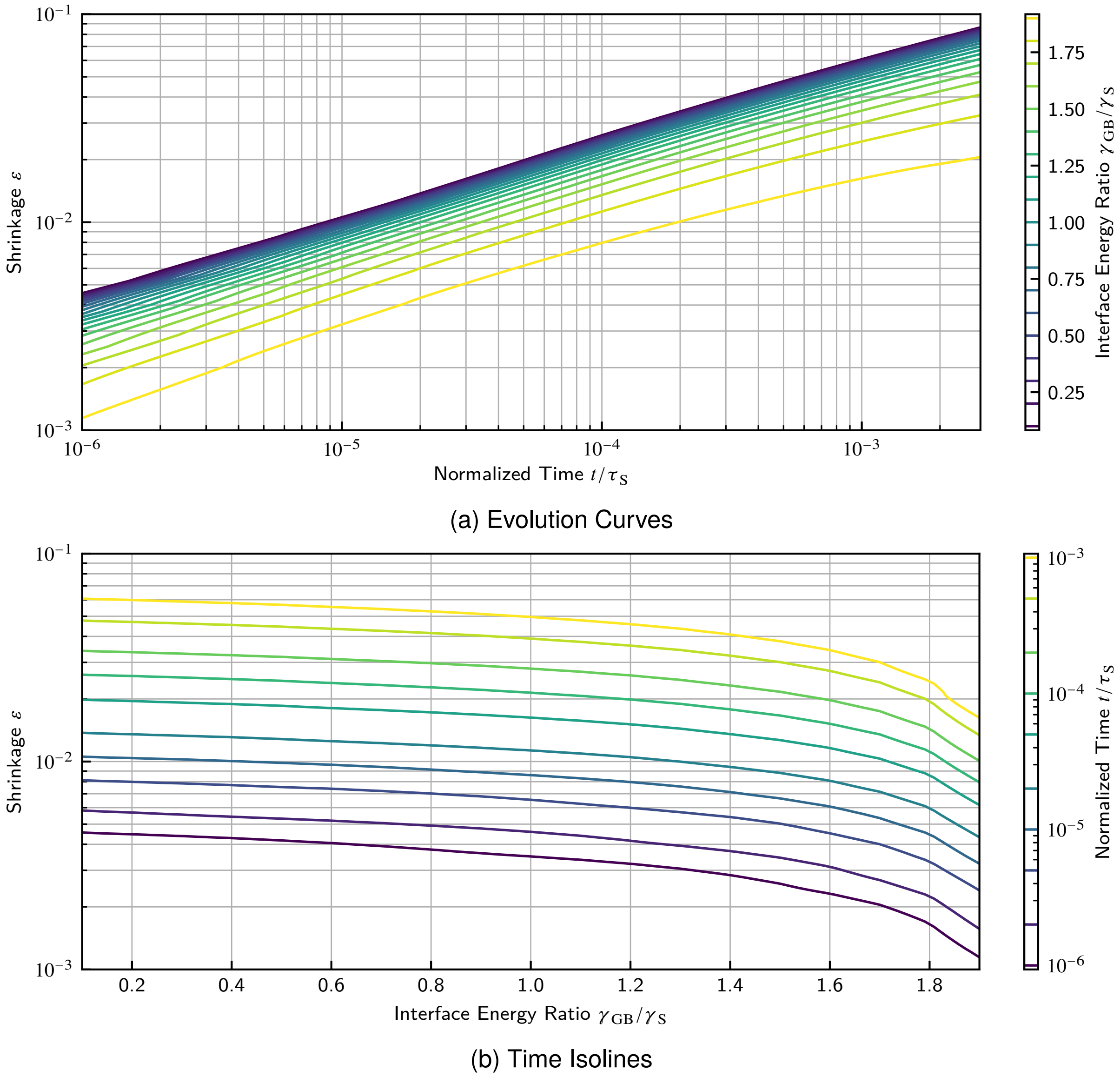


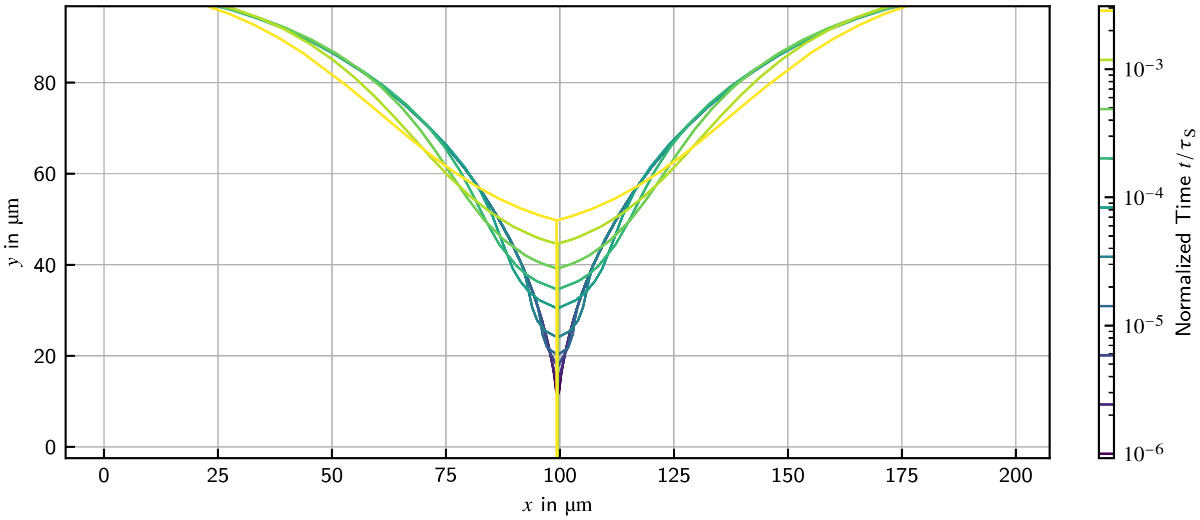
Figure 9.3: Shrinkage in Dependence on Surface-Boundary Energy Ratio

These findings are in accordance to several other studies on the influence of the grain boundary energy ratio such as [141, 142, 143, 144, 145, 32].

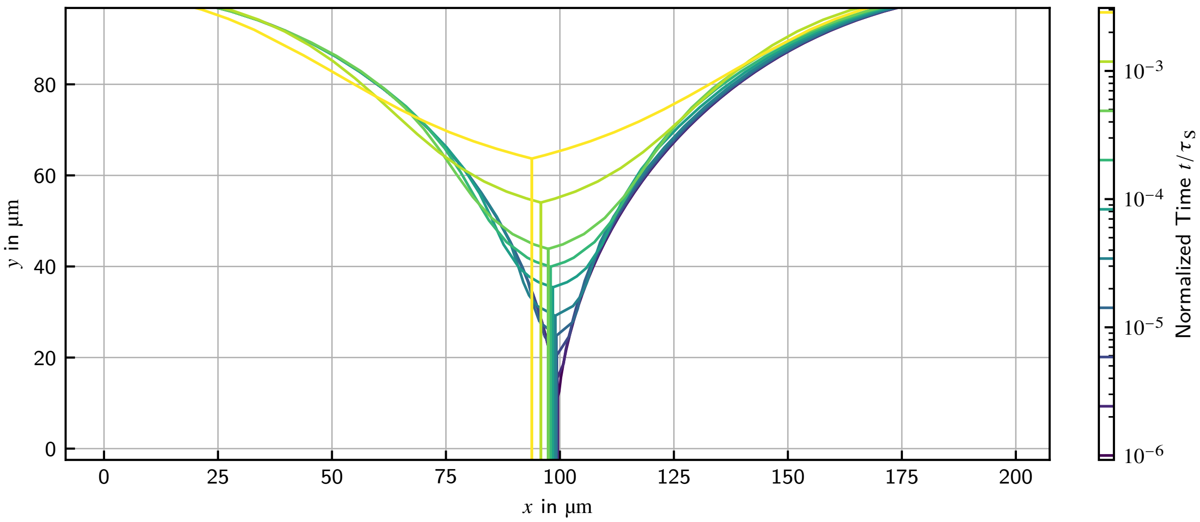
9.1.2. Surface-Boundary Diffusion Ratio D_{GB}/D_S

This study investigates the effect of the ratio between grain boundary and surface diffusion rate. This parameter is the key factor on the relation between neck size evolution and shrinkage, as grain boundary diffusion is the only mechanism present in this model able to remove volume from between the particles and thus effect the shrinkage. The initial particle geometry and the material parameters are taken equal on both particles.

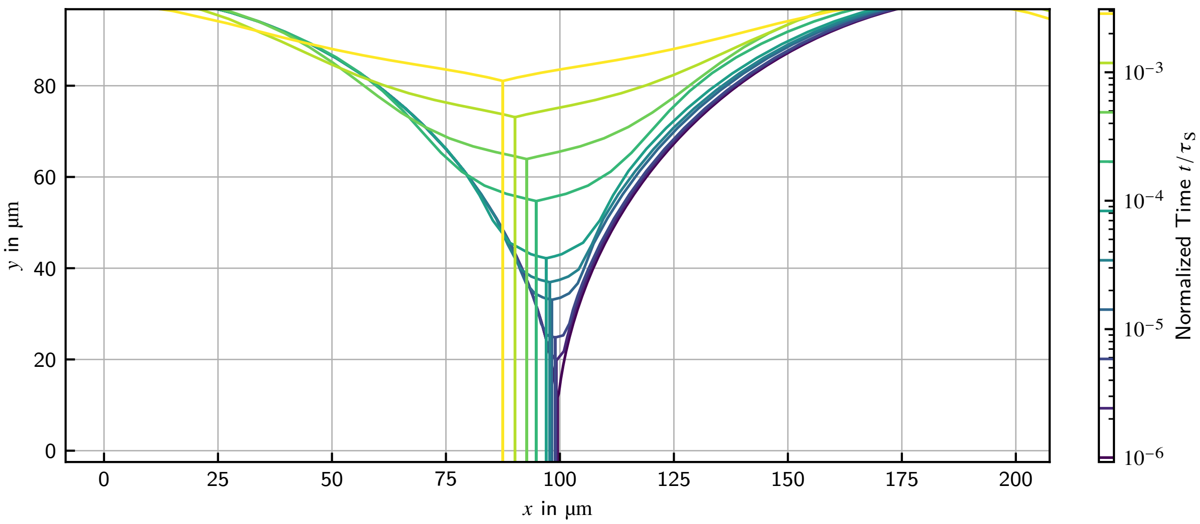
Figure 9.4 shows three selected evolutions of the contact geometry. One may directly observe the lack of noticeable particle movement at low diffusion ratios and the difference in neck size obtained depending on the contribution of boundary diffusion.



(a) $D_{GB}/D_S = 0.01$

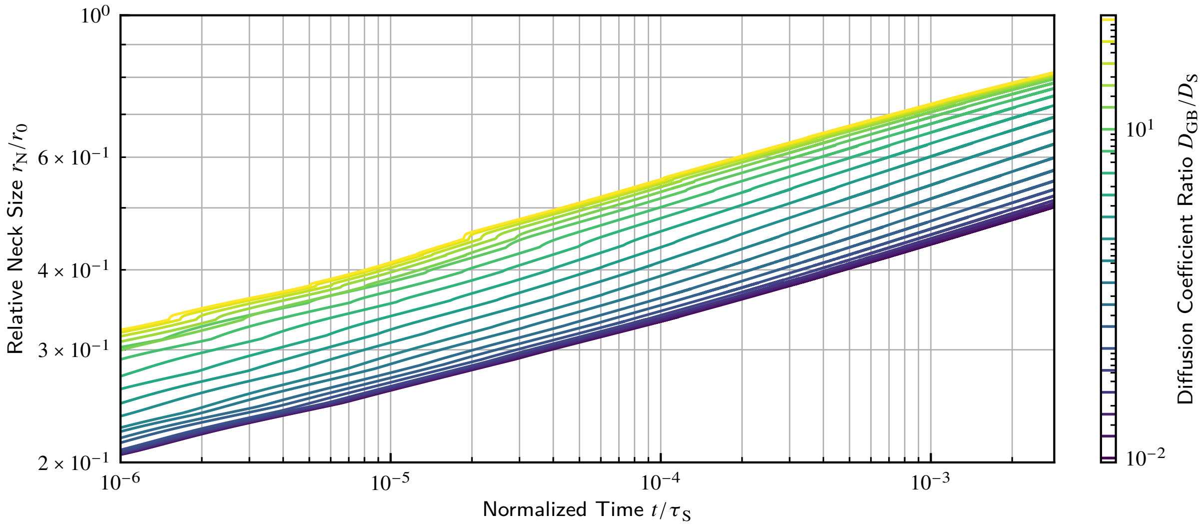


(b) $D_{GB}/D_S = 1.0$

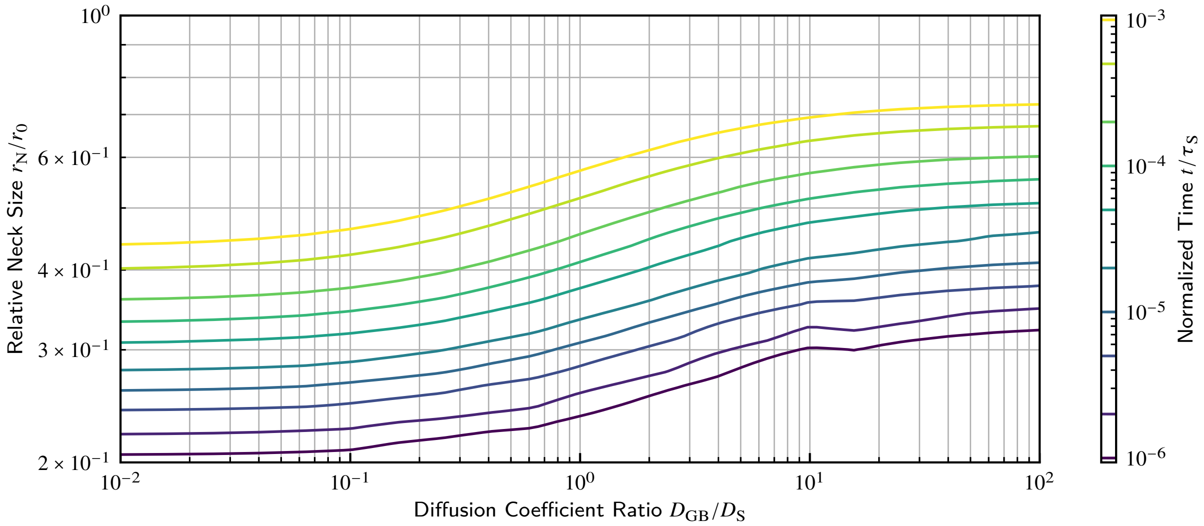


(c) $D_{GB}/D_S = 100.0$

Figure 9.4: Geometry Evolutions for Selected Surface-Boundary Diffusion Ratios



(a) Evolution Curves

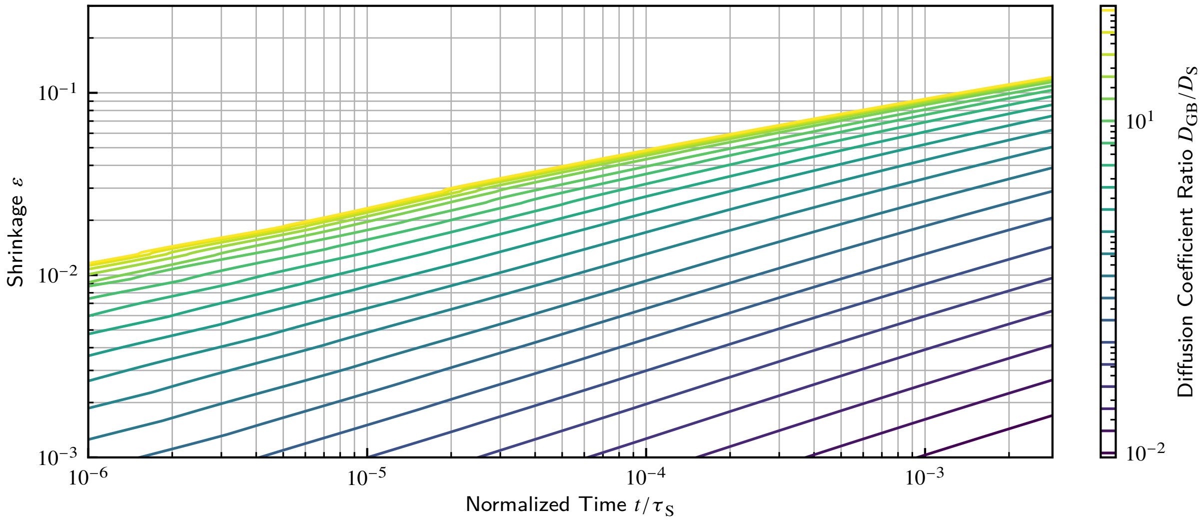


(b) Time Isolines

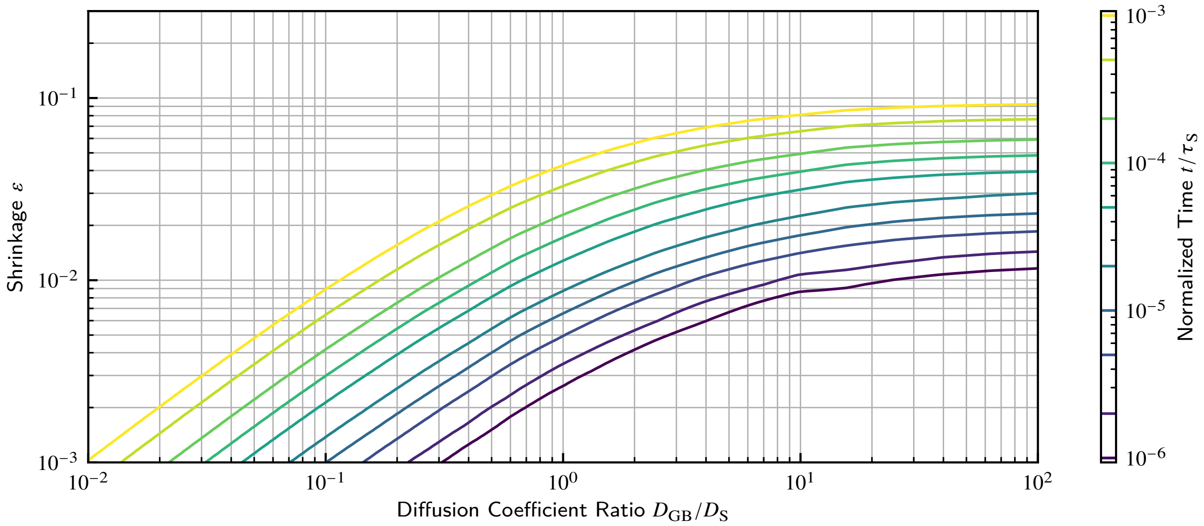
Figure 9.5: Neck Size in Dependence on Surface-Boundary Diffusion Ratio

Figure 9.5a shows the dependence of the neck size on the diffusion ratio. The diffusion ratio has principally an accelerating effect on neck growth. The neck growth is accelerated by the additional volume fluxes via grain boundary diffusion. Additionally, as shrinkage occurs due to boundary diffusion, the volume needed to fill the neck is lowered due to the particles' approaching. However, this effect is limited at both sides symmetrically at ratios of 0.01 and 10, respectively. If either diffusion rate is much larger than the other, neck growth rate appears to be limited by the other mechanisms. Simulation results of Svoboda and Riedel [38] and Wakai and Brakke [50] appear to support these findings, although this issue was not discussed in those publications.

Figure 9.6a shows the dependence of the shrinkage on the diffusion ratio. The shrinkage largely increases with the diffusion ratio as the grain boundary diffusion is the only mechanism able to effect shrinkage. Although, the benefit from accelerated grain boundary diffusion decreases and appears to reach a limit at higher ratios, when grain boundary diffusion is much faster than surface diffusion. A first influence on this is the larger diffusion path at high neck sizes, which also increase with the diffusion ratio as states above. Also, shrinkage progress at high boundary diffusion rates



(a) Evolution Curves



(b) Time Isolines

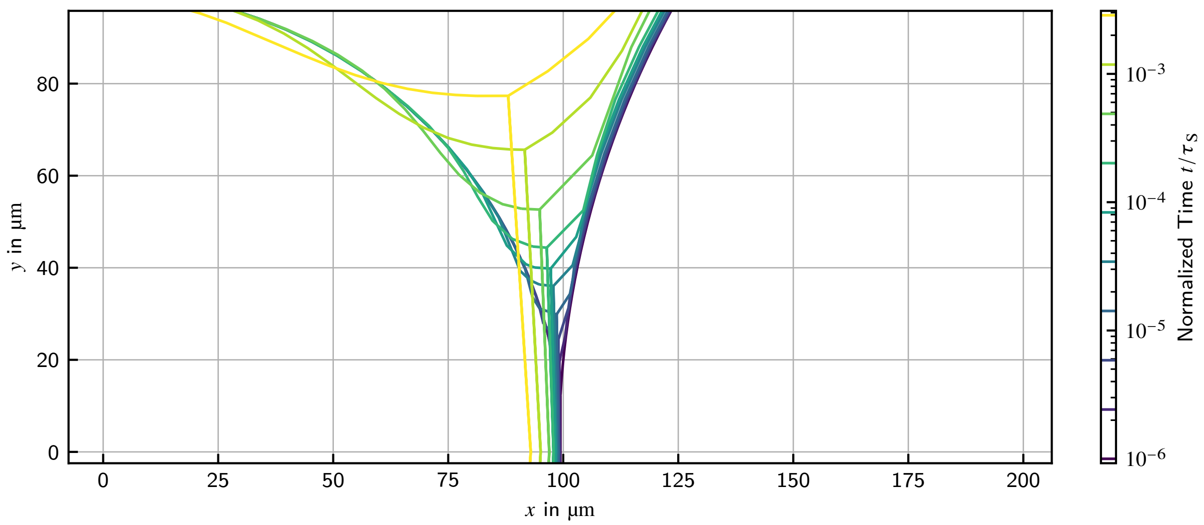
Figure 9.6: Shrinkage in Dependence on Surface-Boundary Diffusion Ratio

appears to be limited by the rate of surface diffusion, too, as then volume is distributed from the neck along the surface.

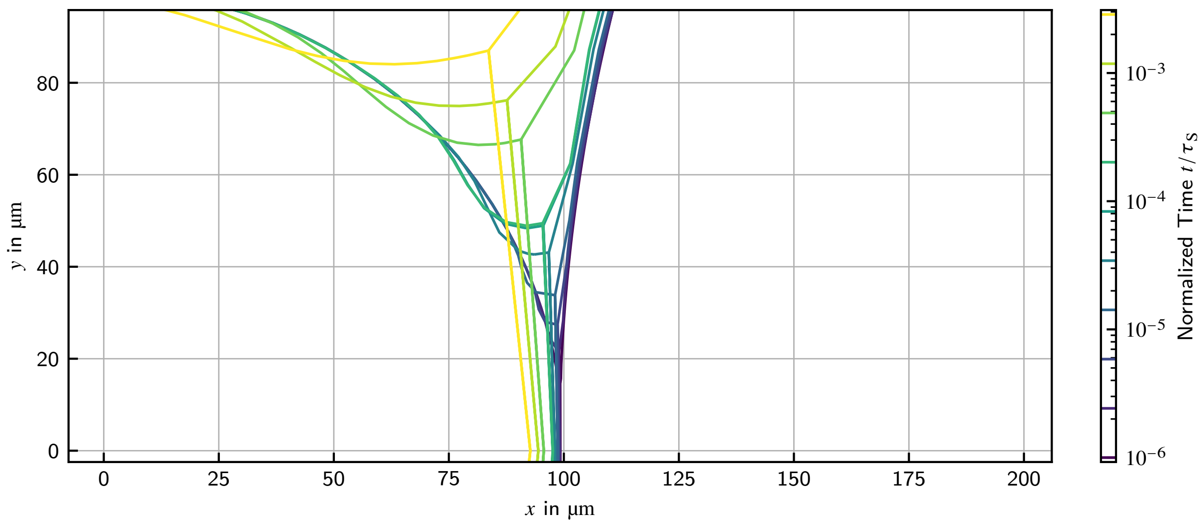
9.1.3. Particle Size Ratio r_2/r_1

This study investigates the effect of sintering unequally sized circular particles. The material parameters are taken equal on both particles with a grain boundary energy of $\gamma_{GB} = 0.5\gamma_S$.

As expected and observed in the previous studies, at particle size ratio of 1 the contact is and stays symmetrical during the whole process. Figure 9.7 shows two selected evolutions of the contact geometry of higher particle size ratios. When increasing the size ratio, the contact becomes asymmetric and the grain boundary between the particles exhibits a curvature. This curvature is what would drive the consumption of the smaller particle by the larger, what is generally referred



(a) $r_2/r_1 = 2$



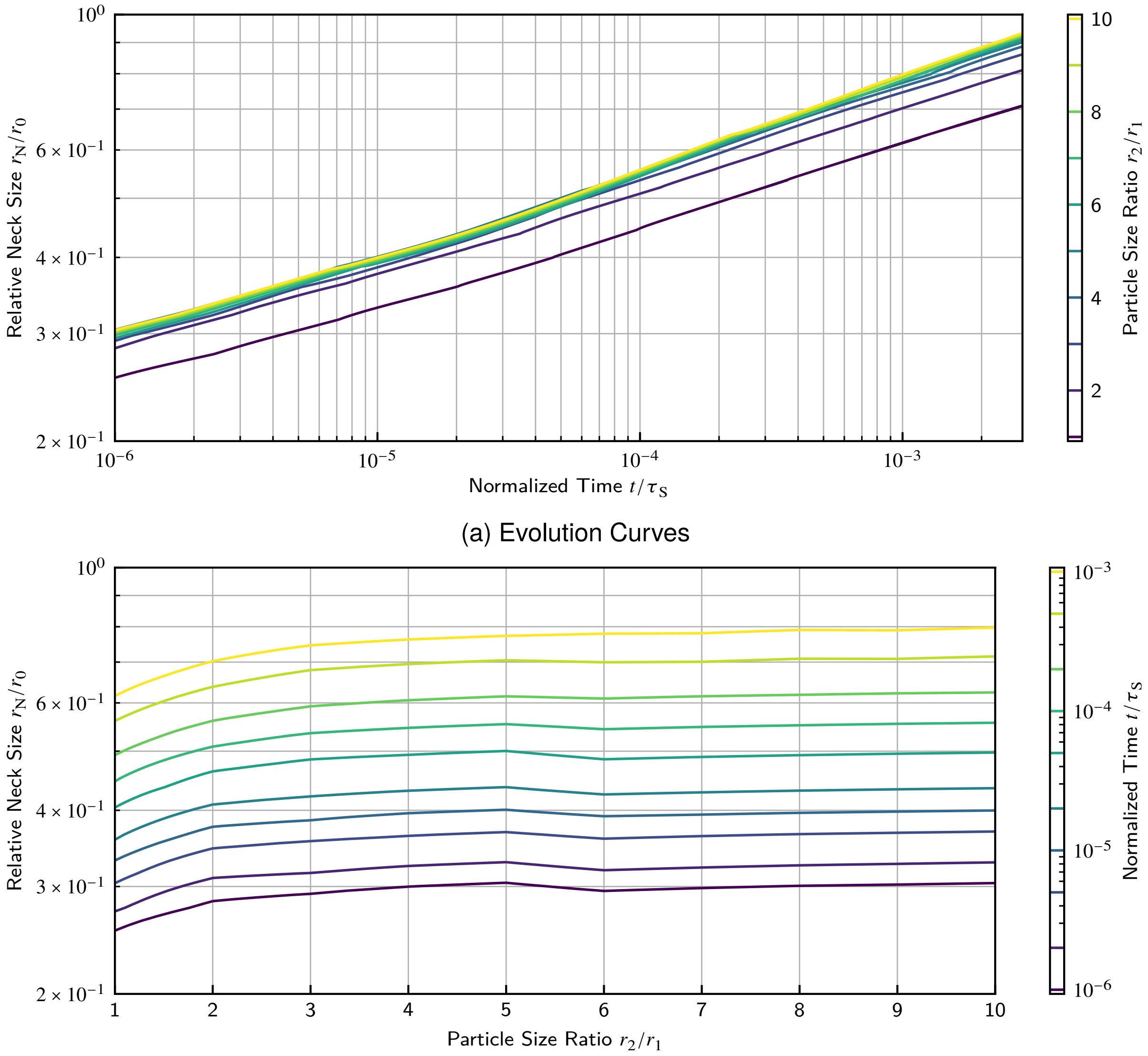
(b) $r_2/r_1 = 4$

Figure 9.7: Geometry Evolutions for Selected Particle Size Ratios

to as grain growth. However, the present model does not include diffusional fluxes cross the grain boundary.

Figure 9.8 shows the dependence of the neck size on the particle size ratio. Until ratios of 3 the neck growth is accelerated, as the gap between the particles is narrowed by the less curved surface of the large particle, thus fewer volume is necessary to grow the neck. Above, the acceleration is remarkably smaller, as the neck size is strongly limited by the size of the smaller particle. Compare therefor the geometry of the neck in Figure 9.7. The same limitation was reported by Pan et al. [146] and Kumar et al. [147], whose models however take volume fluxes cross the boundary into account.

Figure 9.9 shows the dependence of the shrinkage on the particle size ratio. Generally, the shrinkage observed decreases with the size ratio, where the influence is higher at low ratios. Several contributions are present here. The higher proportion of inner particle volume in the distance between the particle centers decreases the relative shrinkage created by equal absolute particle



(a) Evolution Curves

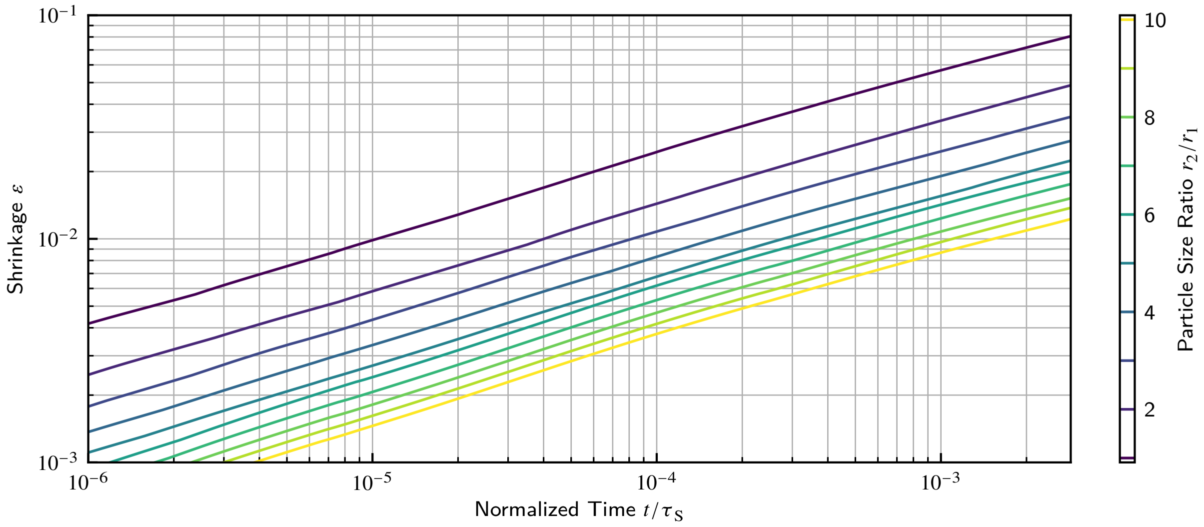
(b) Time Isolines

Figure 9.8: Neck Size in Dependence on Particle Size Ratio

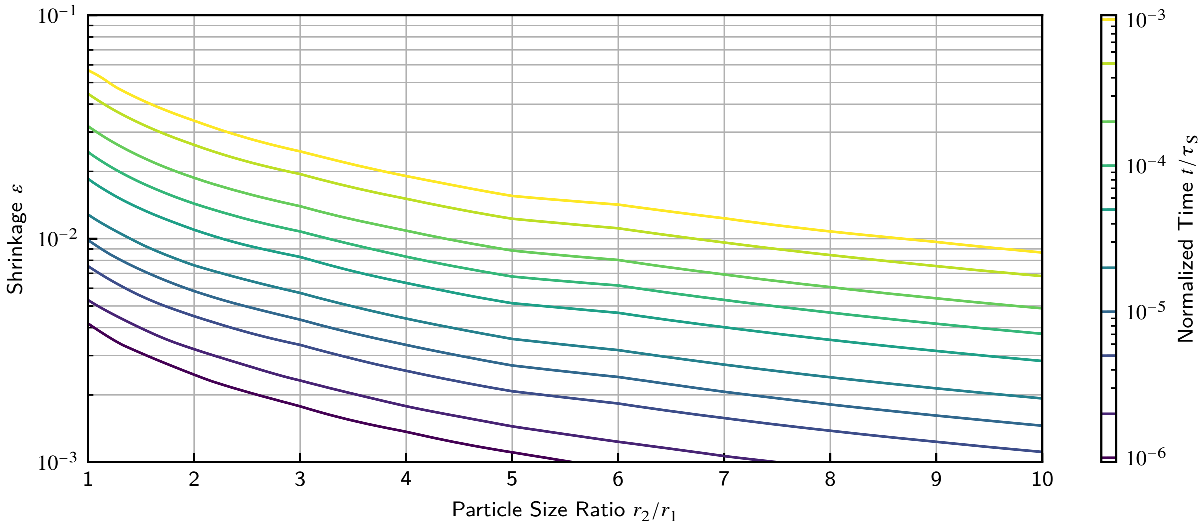
displacement. The larger neck means a longer grain boundary diffusion path and a higher volume to displace.

9.1.4. Asymmetric Surface Energy Ratio γ_{S2}/γ_{S1}

This study investigates the effect of sintering particles of different substance by means of a difference in their surface energy. The initial particle geometry and remaining material parameters are taken equal on both particles with a grain boundary energy of $\gamma_{GB} = 0.5\gamma_{S1}$. The study is limited to ratios up to $\gamma_{S2}/\gamma_{S1} = 3$, because above this point the model showed numerical instability. At high surface energy ratios an undercut forms on the lower surface energy particle's surface for thermodynamic reasons as discussed below. Due to the mathematical construction of the model, volume can be removed from a surface node in such an undercut despite there is too little volume left between the surface and the grain boundary, so after a critical time step, surface line and grain



(a) Evolution Curves

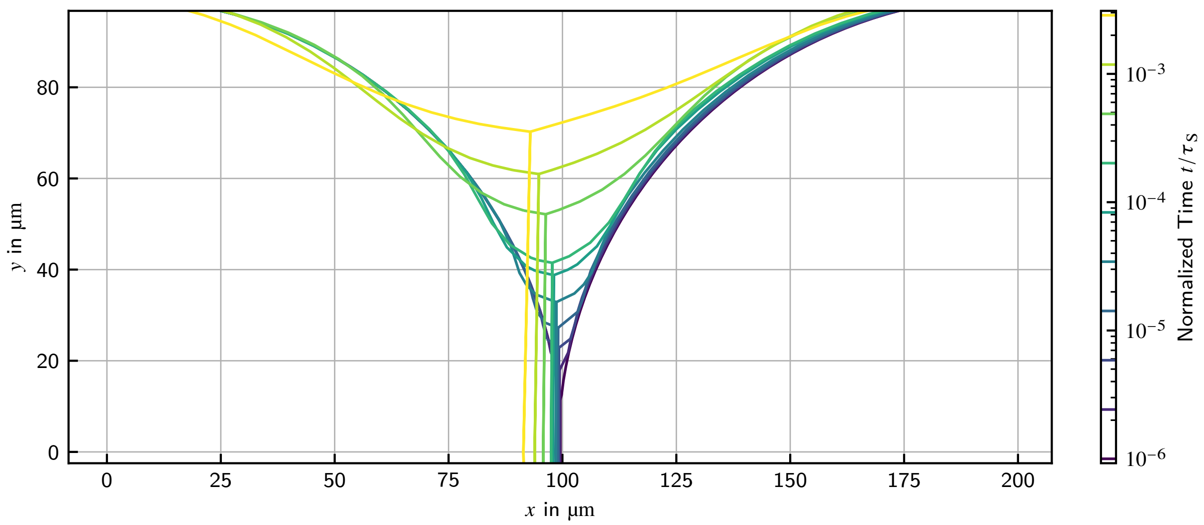


(b) Time Isolines

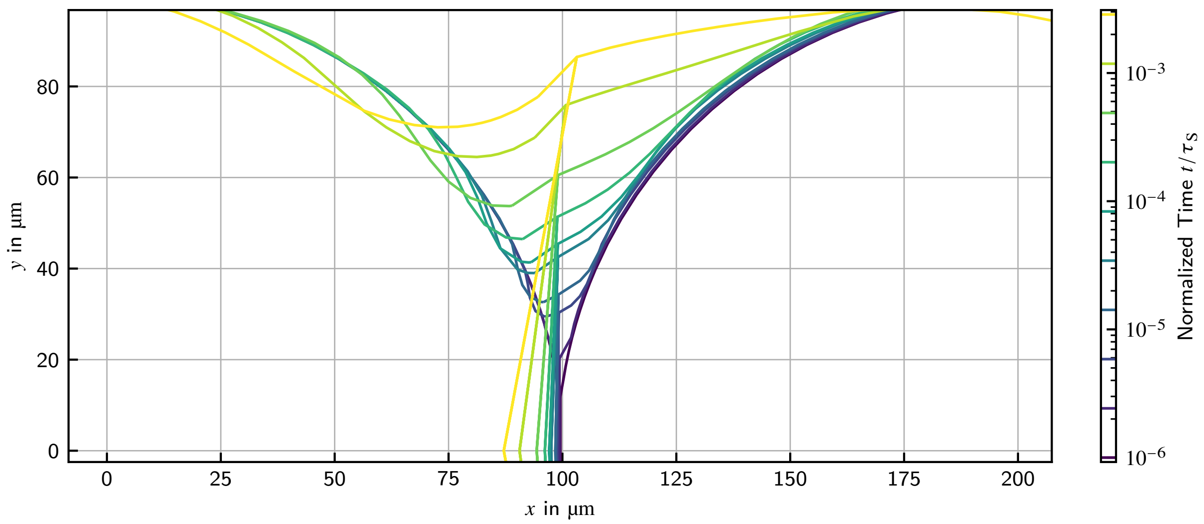
Figure 9.9: Shrinkage in Dependence on Particle Size Ratio

boundary cross. This is of course physically impossible, but at the time of writing this thesis, the issue was not solved.

Figure 9.10 shows two selected evolutions of the contact geometry. Note especially the undercut occurring near the neck on the surface of particle 1. Due to the high surface energy of the second particle it is thermodynamical beneficial to grow the neck while creating additional surface of the first particle. As the surface diffusion is too slow to deliver enough volume to the neck, an undercut evolves. This process is similar to the relation between surface energy and grain boundary in general determining the dihedral angle (see section 9.1.1), with the main difference that here three different interface energies are in action. This leads to an asymmetric neck geometry. Young's equation (equation (2.7)) is only applicable for $\gamma_{S2}/\gamma_{S1} < 1.5$ as for higher ratios equilibrium is not possible (see section 2.2). Remarkably for $1 < \gamma_{S2}/\gamma_{S1} < 1.5$ the equilibrium dihedral angle between the surfaces ψ_{GB} was present in later stages in the results, but the angles between surfaces and grain boundaries ψ_1 and ψ_2 did not fit to those proposed by equation (2.7). So for example the simulation for $\gamma_{S2}/\gamma_{S1} = 1.25$ (see figure 9.10a) showed a $\psi_{GB} \approx 158°$, which is in coincidence



(a) $\gamma_{S2}/\gamma_{S1} = 1.25$



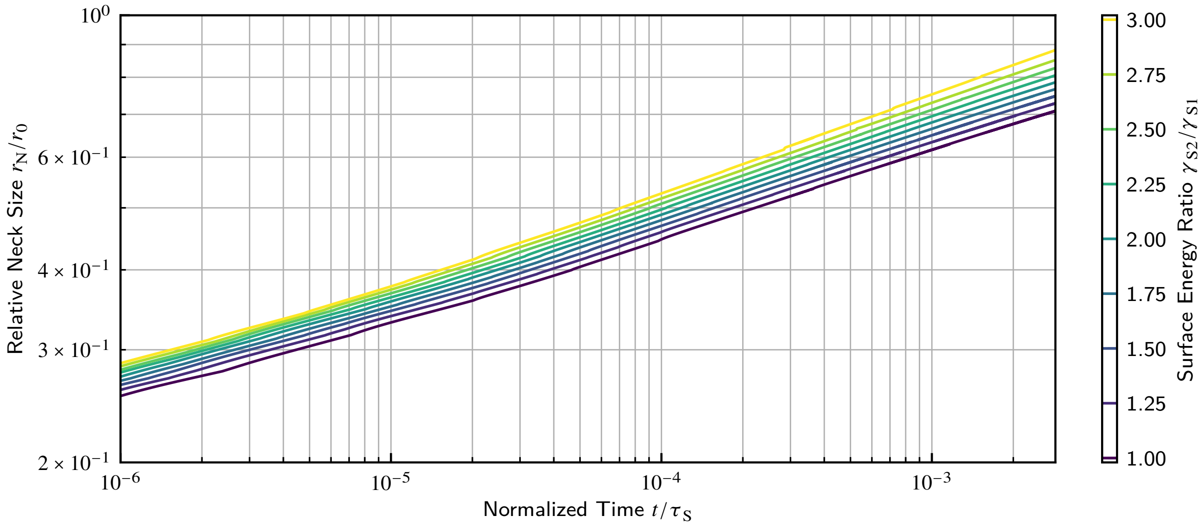
(b) $\gamma_{S2}/\gamma_{S1} = 3$

Figure 9.10: Geometry Evolutions for Selected Asymmetric Surface Energy Ratios

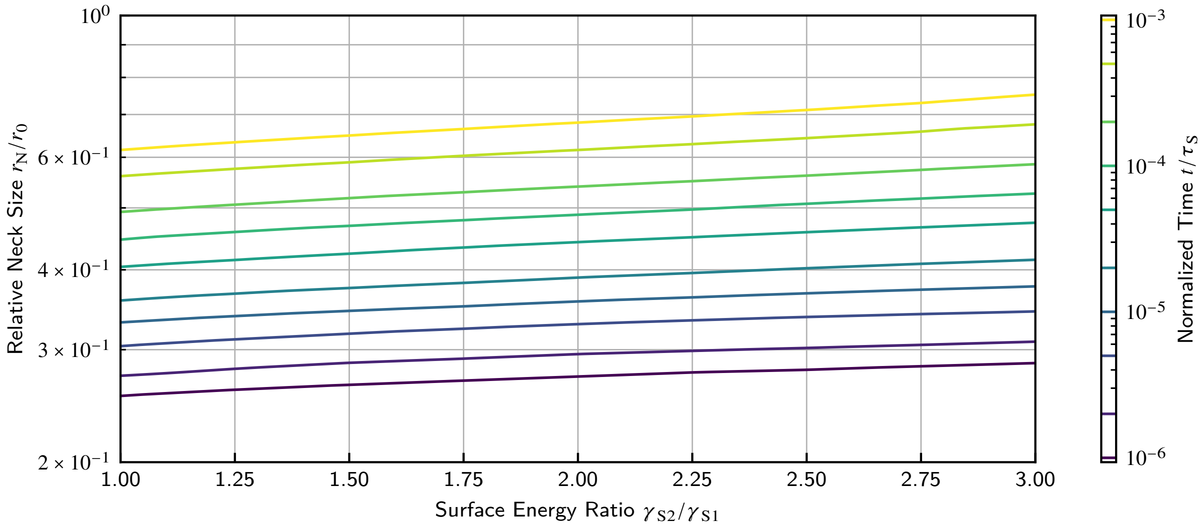
with Young's equation proposal, but the simulated angles to the grain boundary were $\psi_2 \approx 96°$ (left) and $\psi_1 \approx 106°$ (right) compared to the proposed 72° and 130°. The main reason for this is suspected in the coarse discretization of the grain boundary by a single node (see section 6.4 for the reasoning behind this). When imagining a smoothly curved grain boundary, the fit would be closer.

Figure 9.11a shows the dependence of the neck size on the asymmetric surface energy ratio. Neck size increases slightly with the surface energy ratio, as it becomes more beneficial to grow the neck in favor of lowering the second particle's surface, although more surface of the first particle is created.

Figure 9.12a shows the dependence of the shrinkage on the asymmetric surface energy ratio. Shrinkage increases slightly with the surface energy ratio, too, since the driving force for grain boundary diffusion of the second particle is increased. Also, the neck undercut creates a driving force for the first particle to deliver more volume to the undercut neck region, which is partially



(a) Evolution Curves



(b) Time Isolines

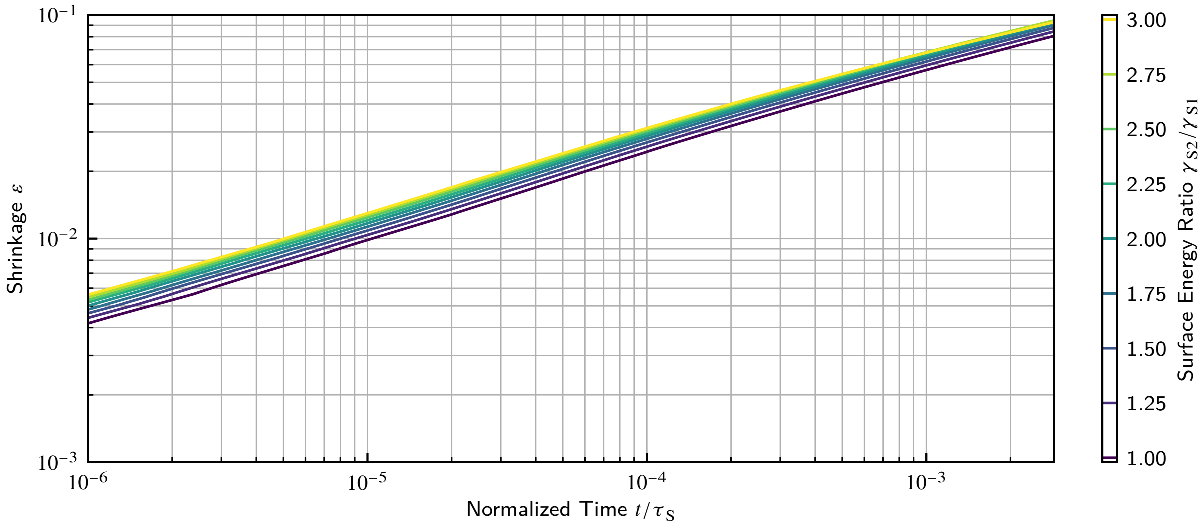
Figure 9.11: Neck Size in Dependence on Asymmetric Surface Energy Ratio

served by diffusion from the grain boundary.

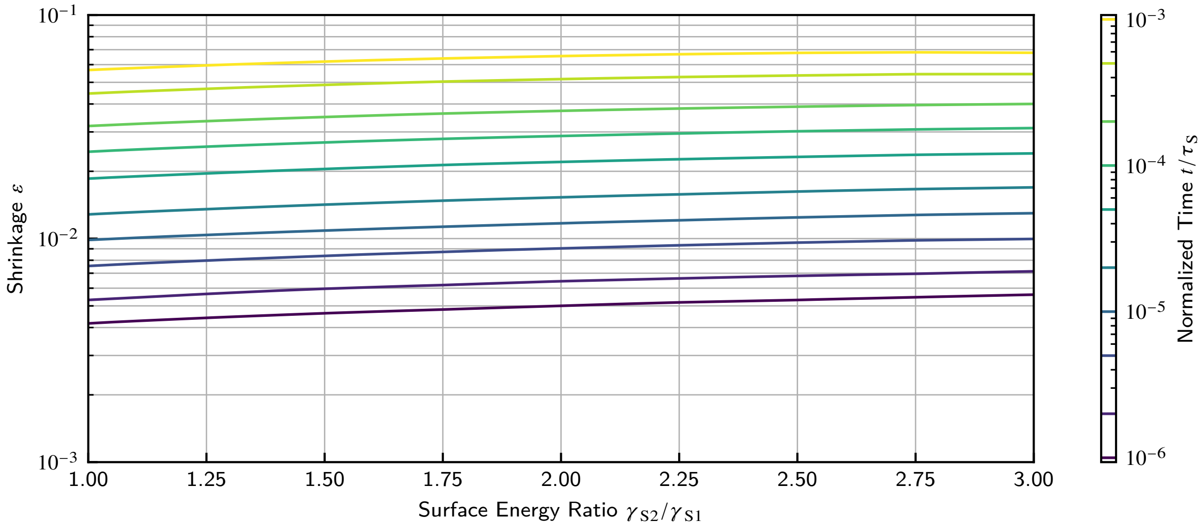
9.1.5. Asymmetric Diffusion Ratio D_2/D_1

This study investigates the effect of sintering particles of different substance by means of a difference in their diffusion rate. The diffusion rates on surface and grain boundary of one particle are taken equal, but vary compared to the other particle. The initial particle geometry and remaining material parameters are taken equal on both particles with $\gamma_{GB} = 0.5\gamma_S$.

Figure 9.13 shows two selected evolutions of the contact geometry. Note especially the increasing curvature of the grain boundary with increasing diffusion ratio, as the contribution by diffusion from the second particle is higher. This allows the second particle to creep around the first one. The surface of the second particle is flatter as diffusion is able to equalize it faster. The shape of the grain boundary clearly shows a limitation of the model, since currently only one grain boundary



(a) Evolution Curves



(b) Time Isolines

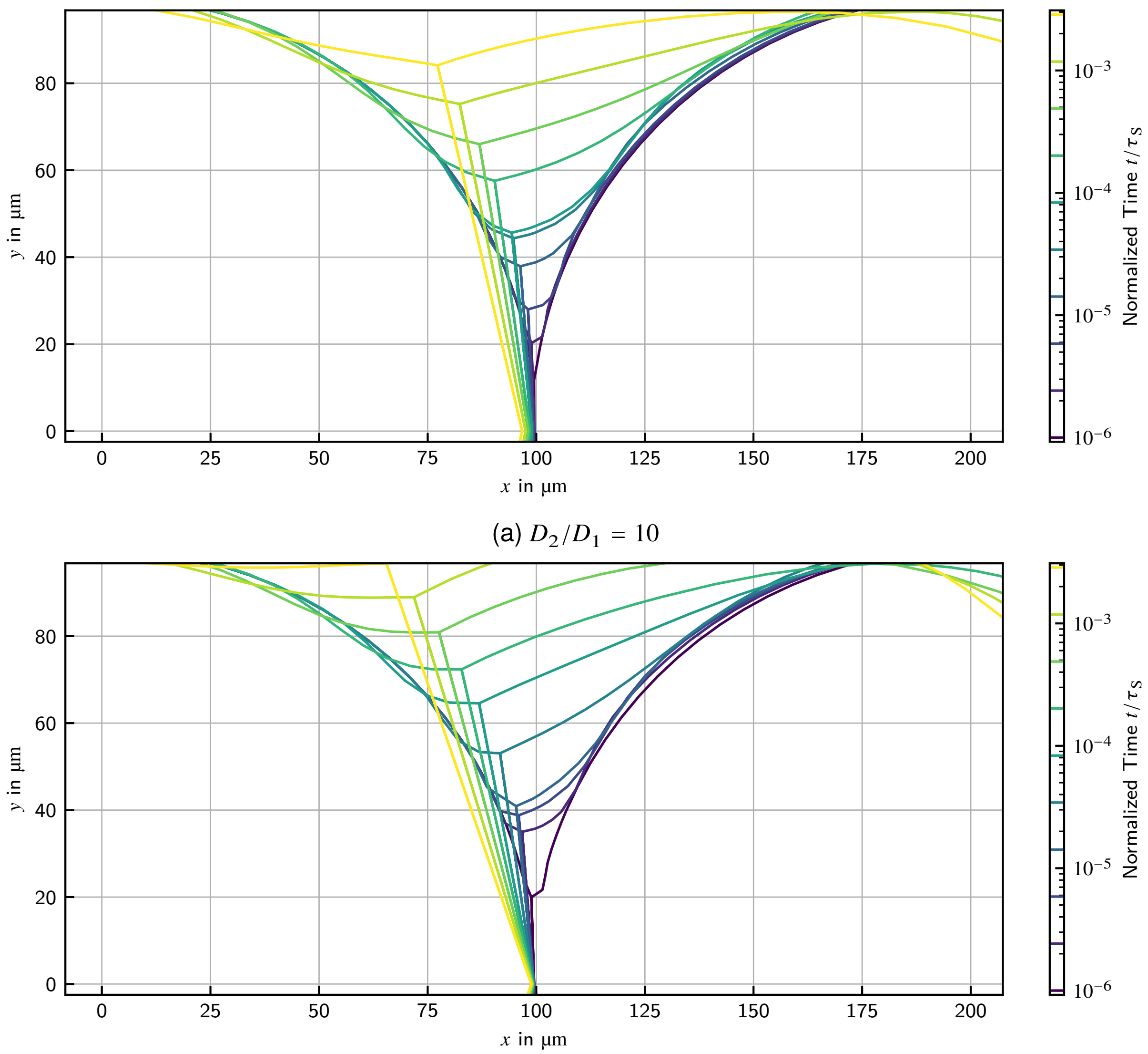
Figure 9.12: Shrinkage in Dependence on Asymmetric Surface Energy Ratio

node is supported (see section 6.4). The coarsely discretized grain boundary poorly represents the real conditions, where the original surface shape of the less active particle would be approximately maintained while the active one creeps around.

Figure 9.14a shows the dependence of the neck size and Figure 9.15a the dependence of the shrinkage on the asymmetric surface diffusion ratio. Both are increasing with the diffusion ratio, as the contribution of the active particle in terms of surface as well grain boundary fluxes is increased.

9.1.6. Ovality o

This study investigates the influence of non-ideal particle geometry by ovality on the sintering behavior. Oval particles have two additional degrees of freedom regarding the initial contact by rotation, here three distinct cases will be selected:



(a) $D_2/D_1 = 10$

(b) $D_2/D_1 = 100$

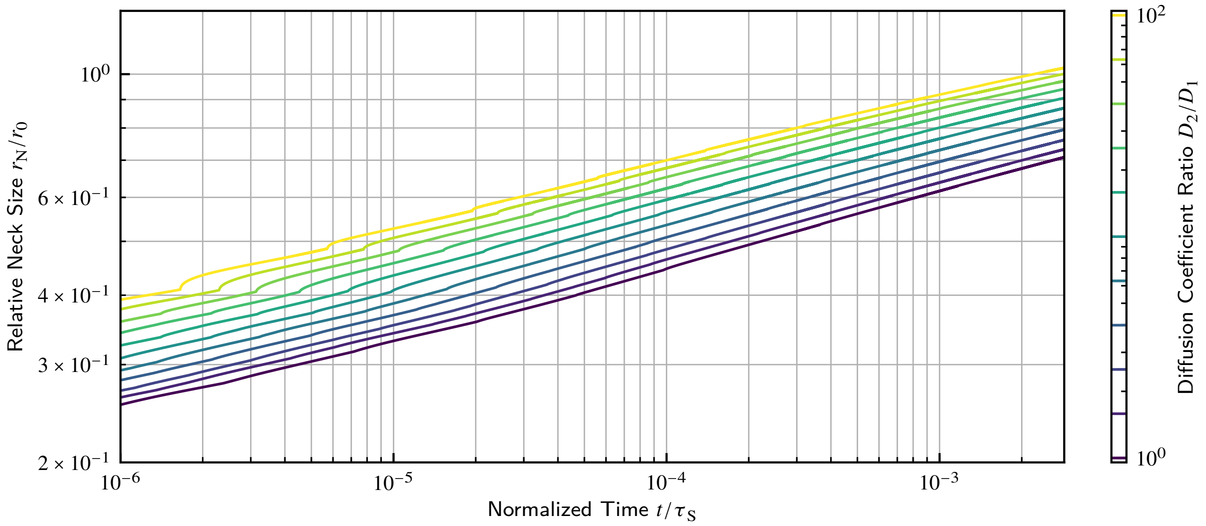
Figure 9.13: Geometry Evolutions for Selected Asymmetric Surface Diffusion Ratios

1. Both particles touch on their tip side (highest curvature side).
2. Both particles touch on their flank side (lowest curvature side).
3. The tip of one particle touches the flank of the other.

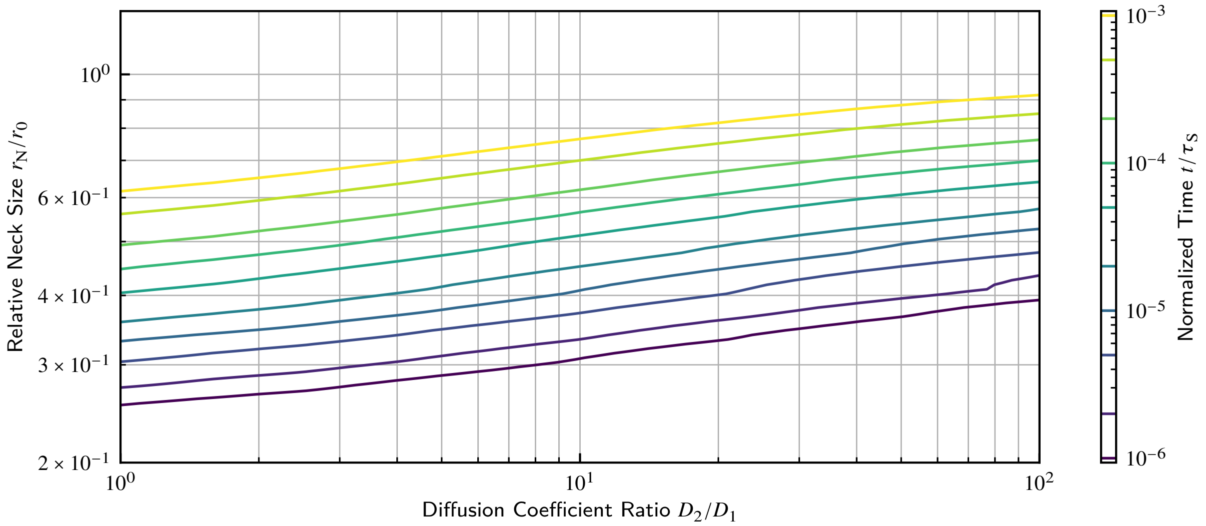
Tip-Tip-Contact

Figure 9.16 shows two selected evolutions of the contact geometry. The plots clearly show the limitation of neck growth by the small diameter of the ellipse. The fill volume needed to grow the neck is increased by ovality due to the wide surface profile around the neck.

Figure 9.17 shows the dependence of the neck size on the particle ovality. The neck size heavily decreases with the ovality as the fill volume is increased, and the neck growth is limited by the small diameter of the ellipse. The oval particles behave here similarly to a particle of smaller size.



(a) Evolution Curves



(b) Time Isolines

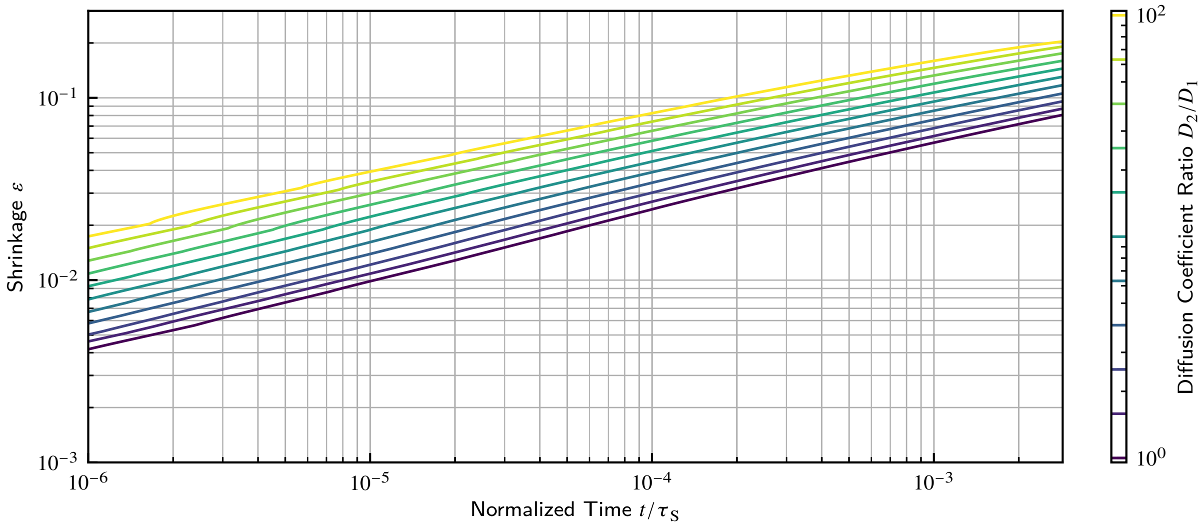
Figure 9.14: Neck Size in Dependence on Asymmetric Surface Diffusion Ratio

Figure 9.18 shows the dependence of the shrinkage on the particle ovality. The shrinkage slightly increases with the ovality. There are two counter-acting effects here. First, the diffusion paths are shorter due to the small neck. Second, the distance between the particle centers is increased due to the oval shape, which increases the denominator of shrinkage. The latter influence dominates, apparently.

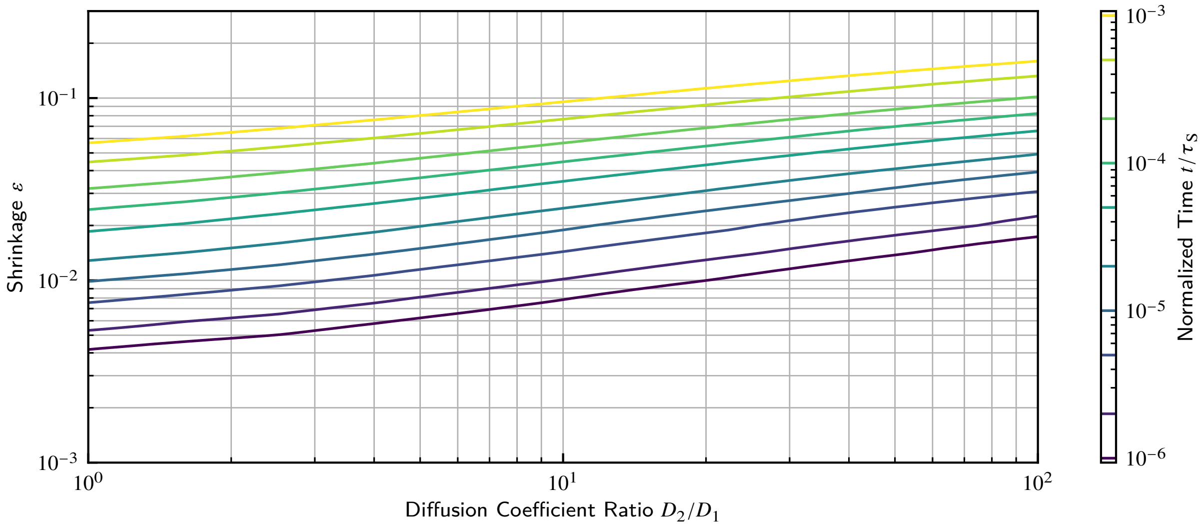
Flank-Flank-Contact

Figure 9.19 shows two selected evolutions of the contact geometry. The plots clearly show the limitation of neck growth by the large diameter of the ellipse. The fill volume is decreased by ovality due to the narrow gap between the particle surfaces.

Figure 9.20 shows the dependence of the neck size on the particle ovality. Neck size is heavily increasing with ovality, as the volume to fill is small and it is here limited by the large diameter of



(a) Evolution Curves



(b) Time Isolines

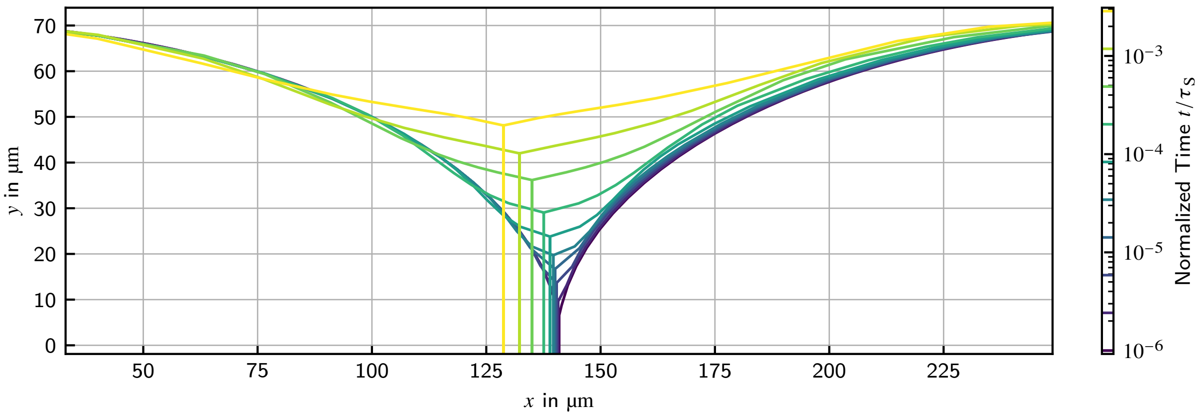
Figure 9.15: Shrinkage in Dependence on Asymmetric Surface Diffusion Ratio

the ellipse. The particles behave like larger ones.

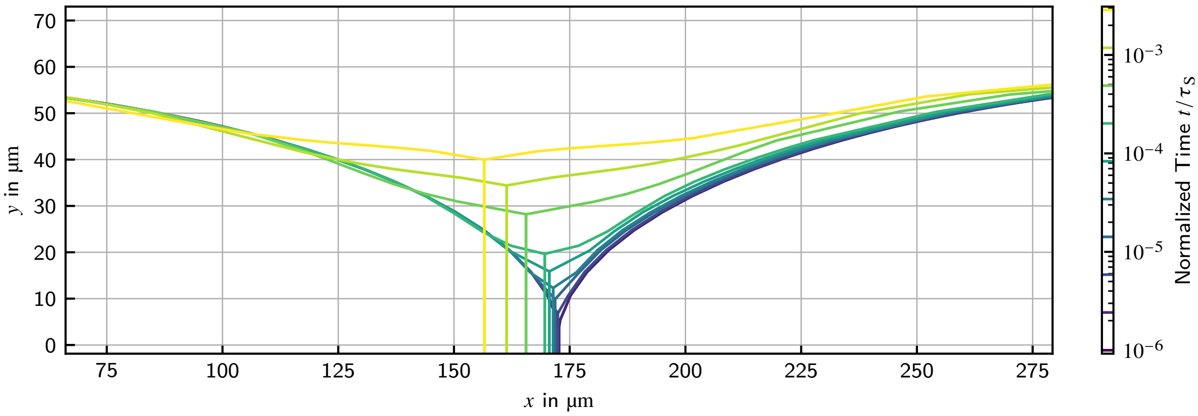
Figure 9.21 shows the dependence of the shrinkage on the particle ovality. The shrinkage shows a slight decrease with the ovality. The reasoning is contrary to the tip-tip case, as here longer diffusion paths and smaller particle distance are in effect. The length of the diffusion path appears to dominate.

Tip-Flank-Contact

Figure 9.22 shows two selected evolutions of the contact geometry. The neck size is limited by the small diameter of the ellipse. The grain boundary is heavily curved due to surface fluxes of the particle in flank position. This would lead to consumption of the tip-positioned particle by the flank-positioned one if cross-boundary fluxes were allowed by the model. So, although the particles are of equal size, the contact resembles that of unequally sized particles.



(a) $o = 2$



(b) $o = 3$

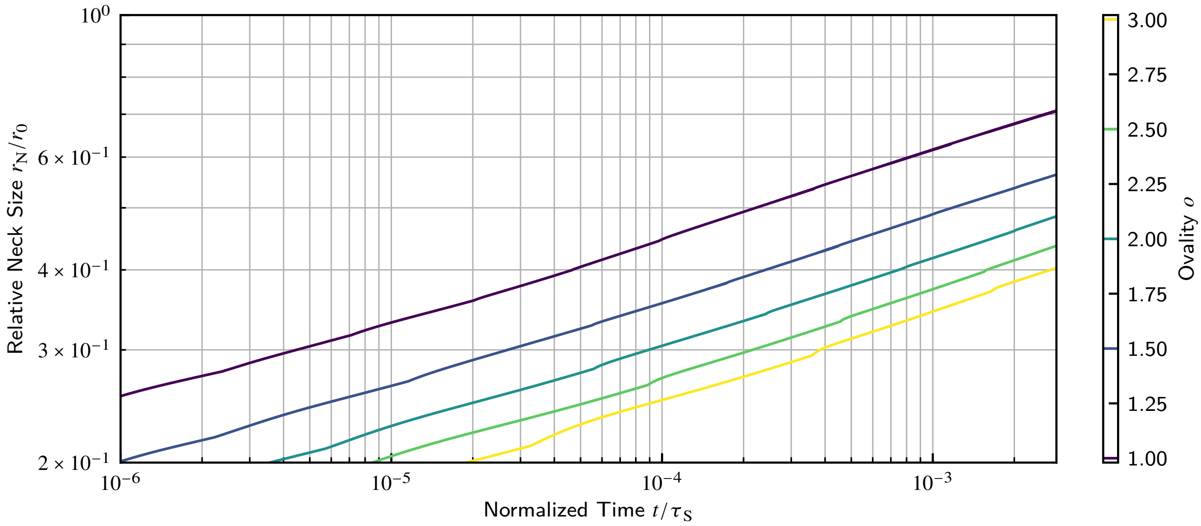
Figure 9.16: Geometry Evolutions for Selected Ovalities in Tip-Tip Contacts

Figure 9.23 shows the dependence of the neck size on the particle ovality. As in the tip-tip case, neck size decreases with ovality, but much less pronounced. Also larger neck widths compared to tip-tip are possible, the neck may even grow larger than the small ellipse diameter.

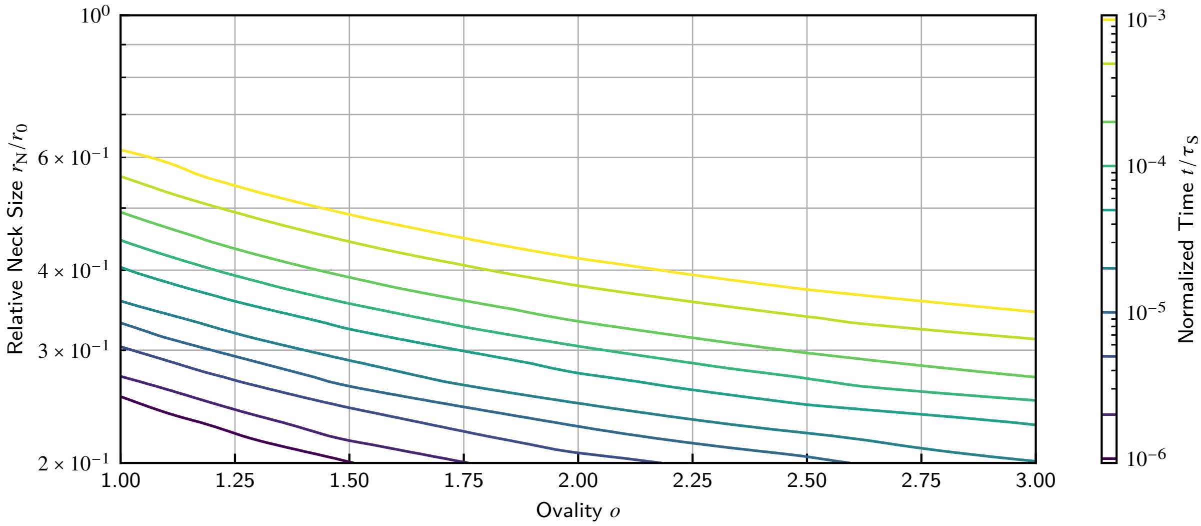
Figure 9.24 shows the dependence of the shrinkage on the particle ovality. The shrinkage increases slightly with the ovality similarly to the tip-tip case. Although the shorter diffusion paths are acting here, too, the initial distance stays constant over the whole parameter range since the increase by the tip-positioned particle is equalized by the decrease by the flank-positioned one.

Summary

The influence of ovality on shrinkage is to be rated as rather low. Depending on the contact configuration there is only a slight increase or decrease with the ovality. Contrary, the effect on neck size is large. The particles behave there like smaller or larger particles depending on the contact configuration. The notion of an apparent radius can be introduced, which is the radius of an equivalent circular particle in regard on neck size evolution. The neck size is always limited by the smaller of the apparent particle radii.

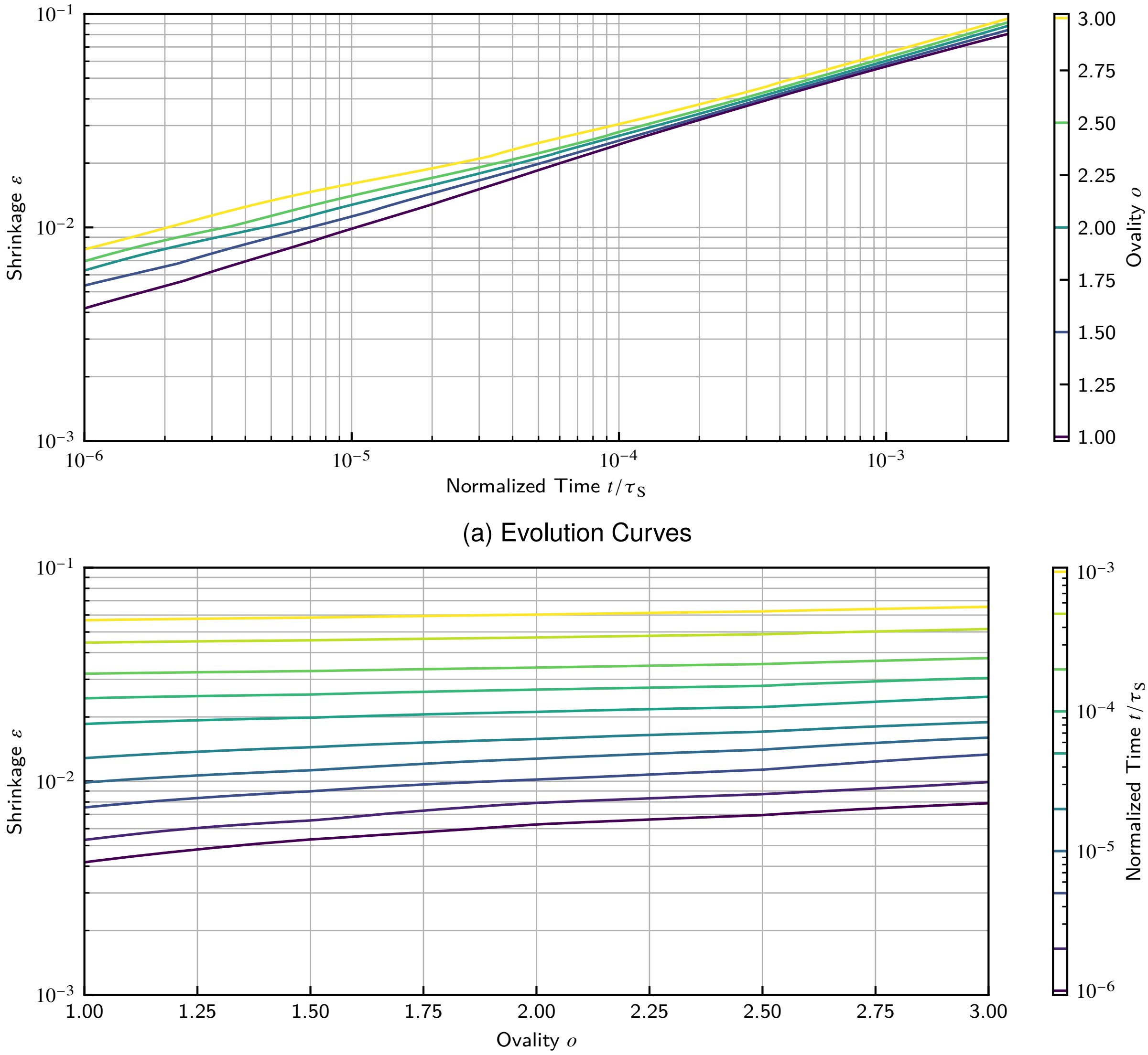


(a) Evolution Curves



(b) Time Isolines

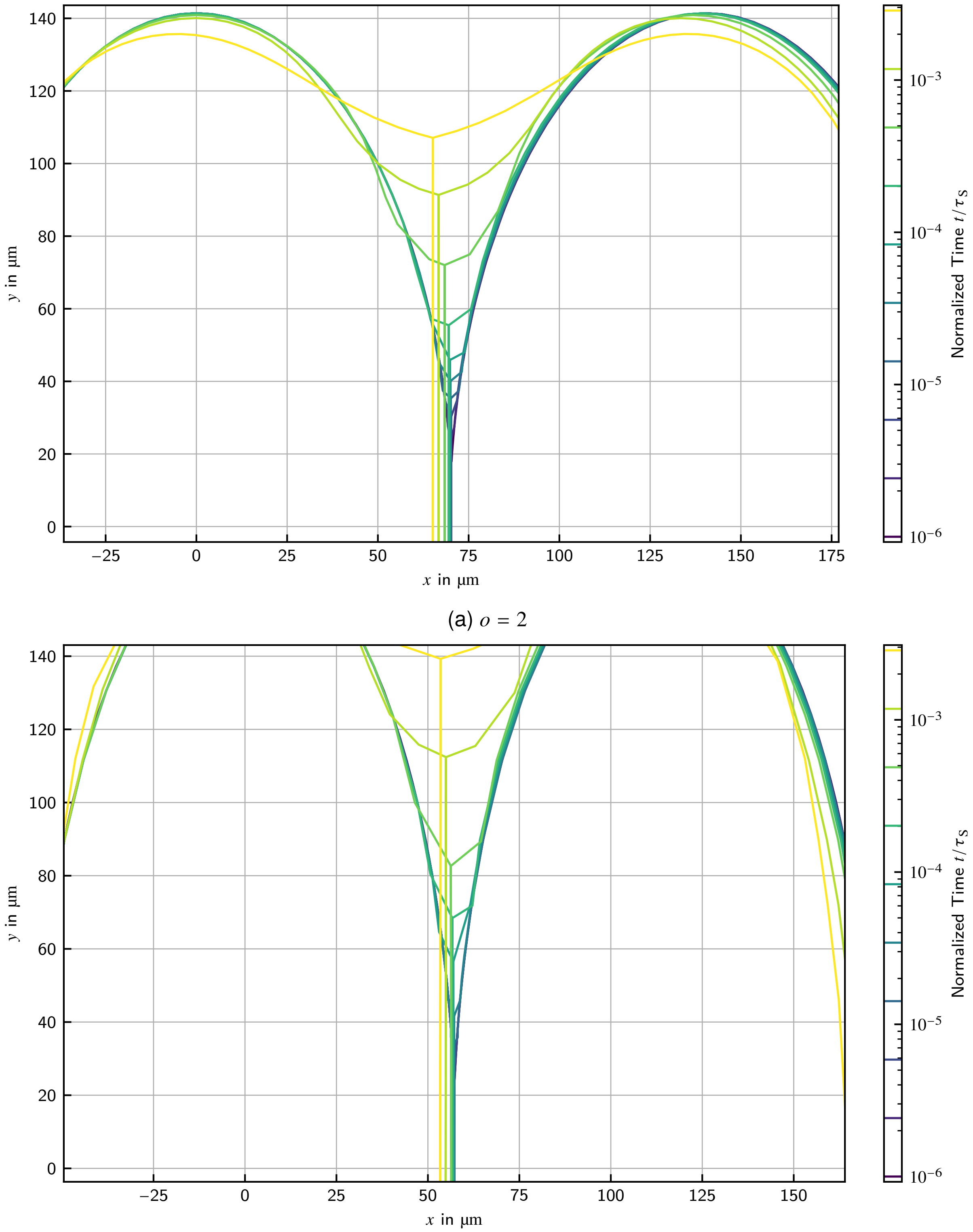
Figure 9.17: Neck Size in Dependence on Ovality in Tip-Tip Contacts



(a) Evolution Curves

(b) Time Isolines

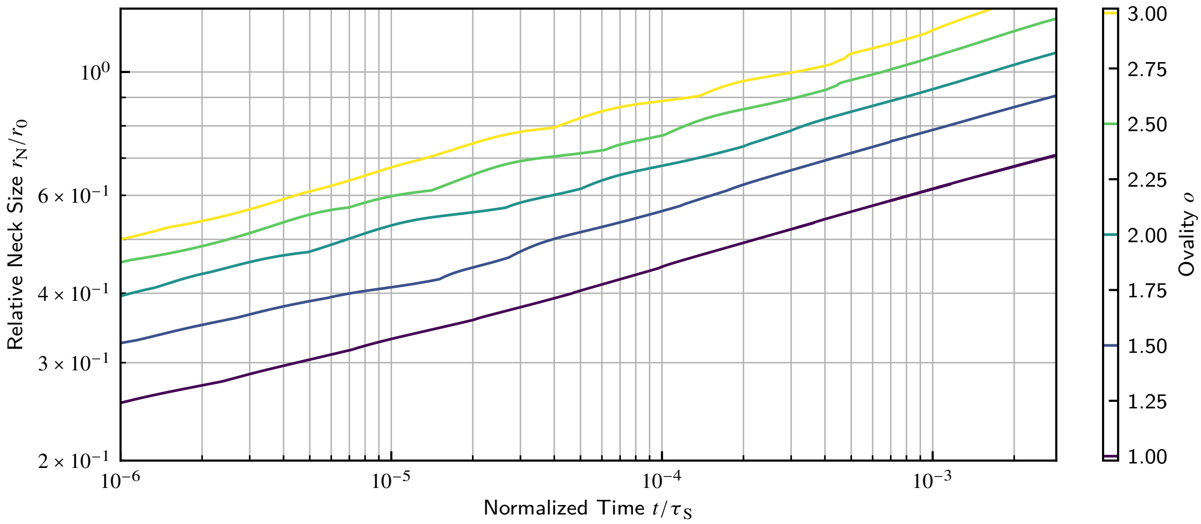
Figure 9.18: Shrinkage in Dependence on Ovality in Tip-Tip Contacts



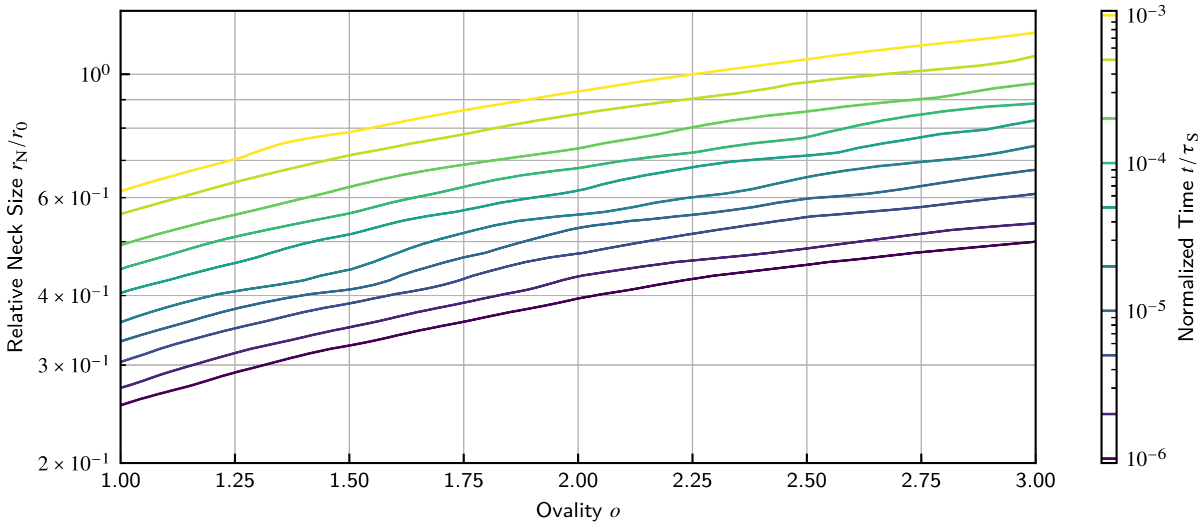
(a) $o = 2$

(b) $o = 3$

Figure 9.19: Geometry Evolutions for Selected Ovalities in Flank-Flank Contacts

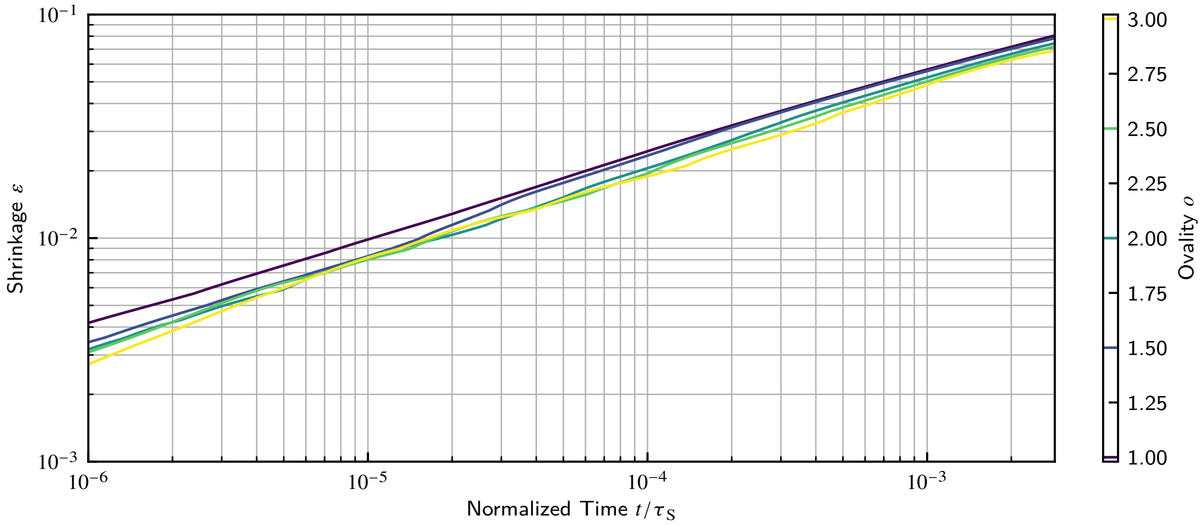


(a) Evolution Curves

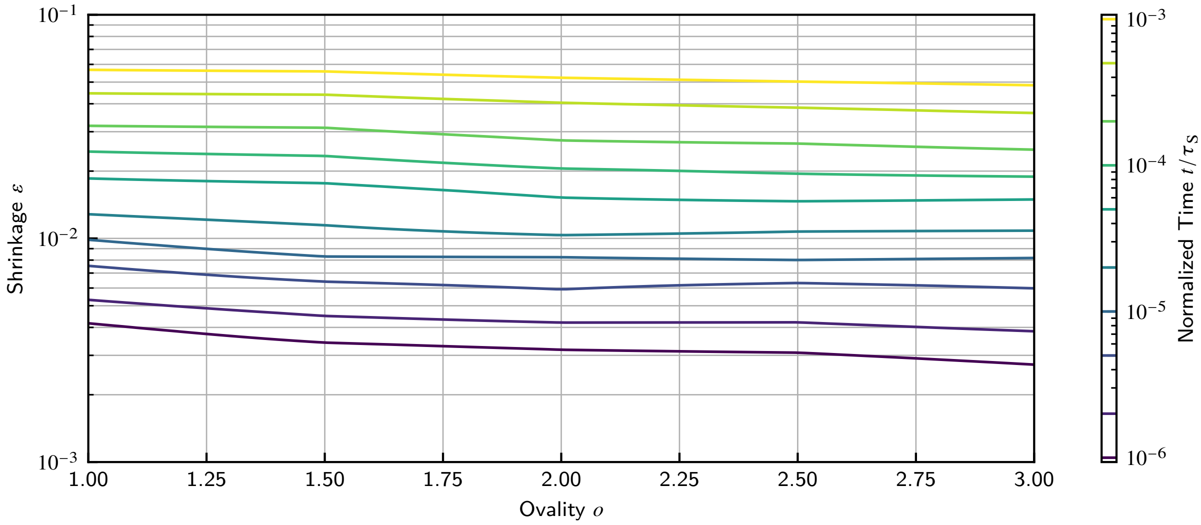


(b) Time Isolines

Figure 9.20: Neck Size in Dependence on Ovality in Flank-Flank Contacts

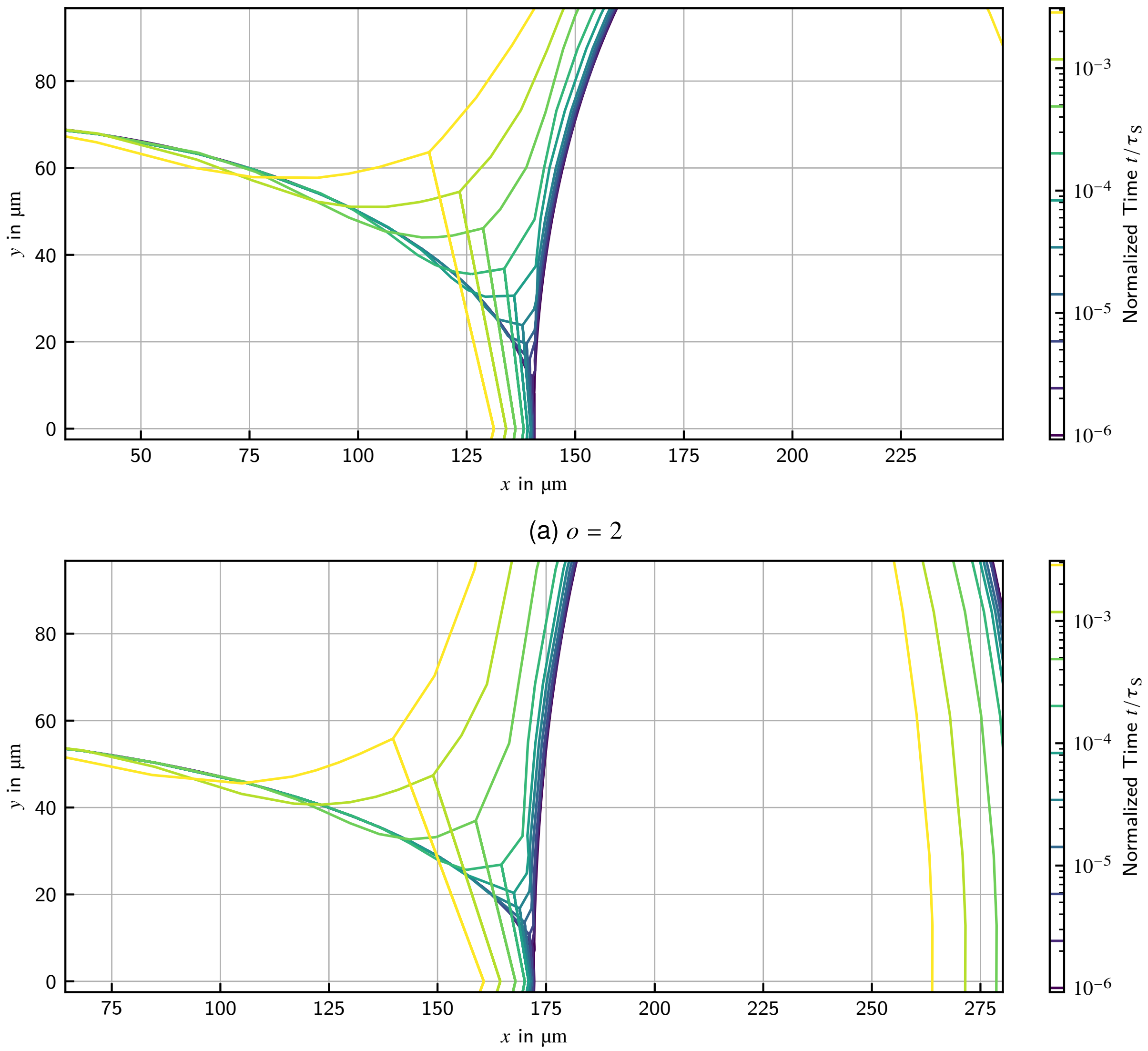


(a) Evolution Curves



(b) Time Isolines

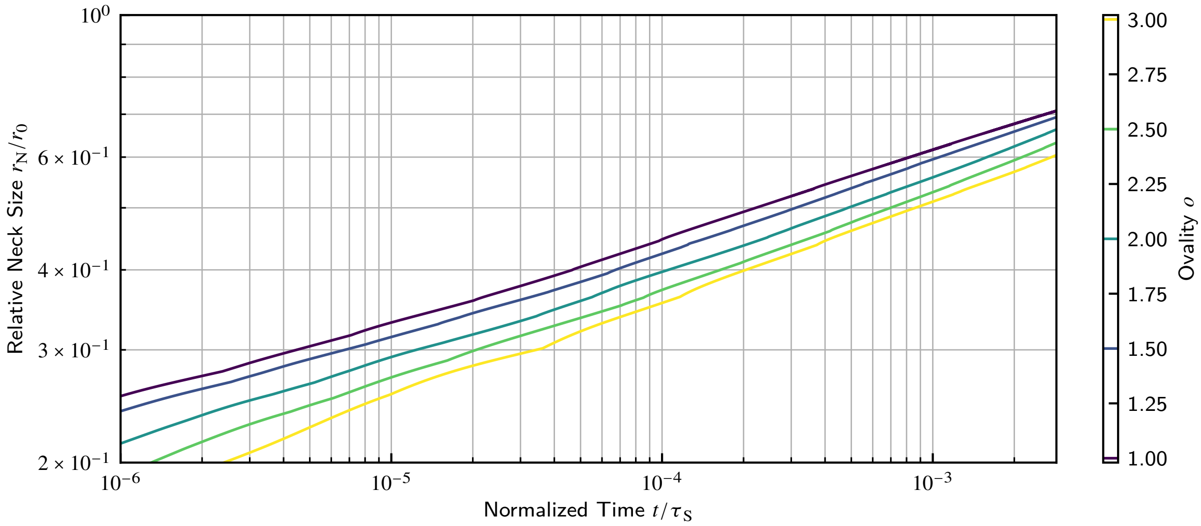
Figure 9.21: Shrinkage in Dependence on Ovality in Flank-Flank Contacts



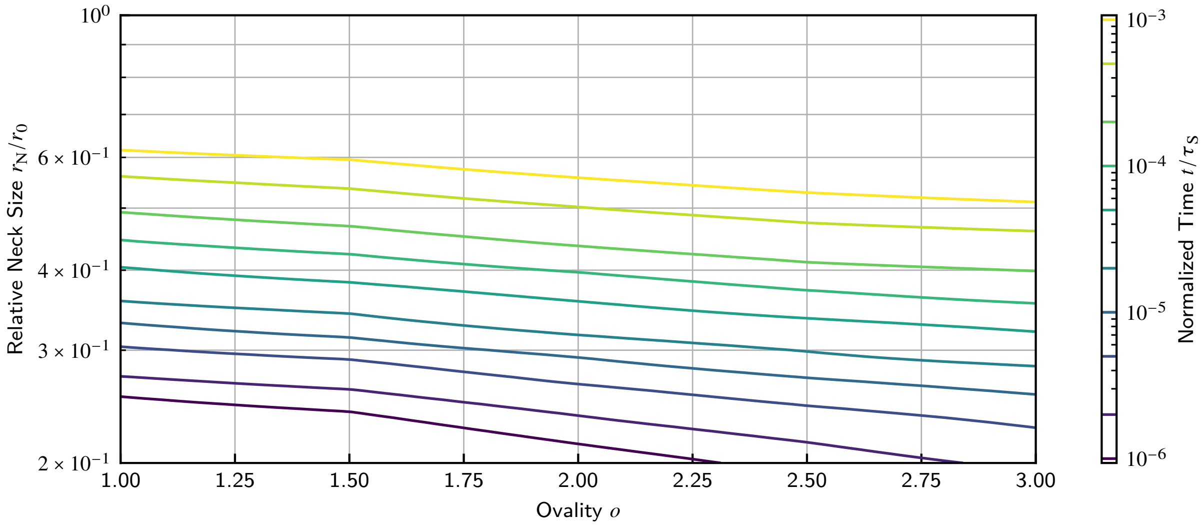
(a) $o = 2$

(b) $o = 3$

Figure 9.22: Geometry Evolutions for Selected Ovalities in Tip-Flank Contacts

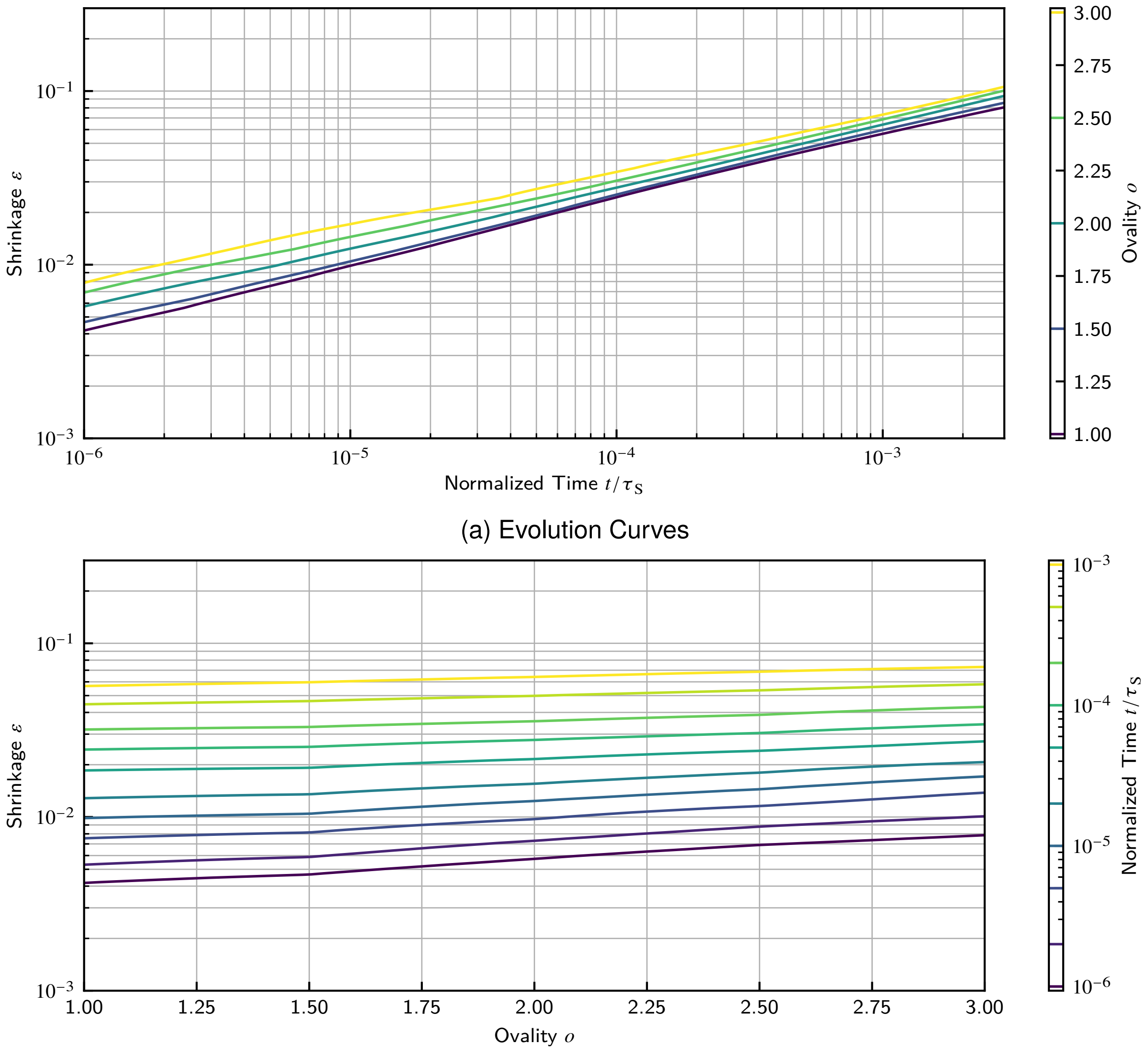


(a) Evolution Curves



(b) Time Isolines

Figure 9.23: Neck Size in Dependence on Ovality in Tip-Flank Contacts



(a) Evolution Curves

(b) Time Isolines

Figure 9.24: Shrinkage in Dependence on Ovality in Tip-Flank Contacts

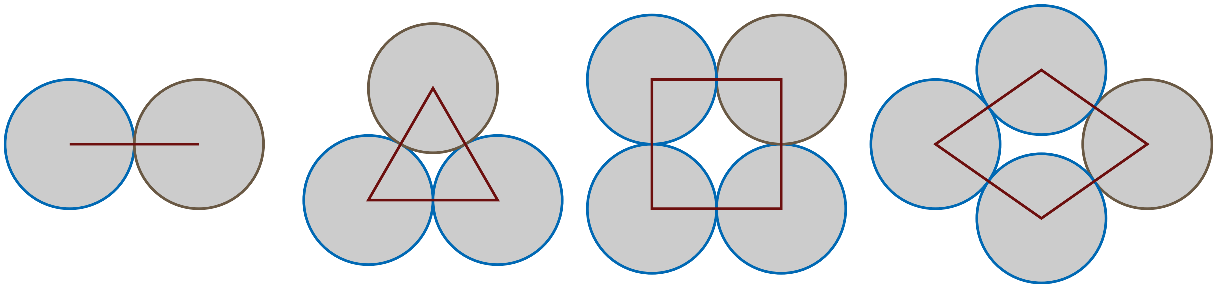


Figure 9.25: Particle Packings Regarded in this Investigation

9.2. Study on the Influence of Multiple Particle Contacts

In the previous investigations, only two-particle models have been regarded, which is a pretty common approach. However, two-particle models lack a major influence on the overall sintering response a material shows: the mutual hindering of particles with multiple contacts. In reality, particle are not able to move freely due to the requirements of a single grain boundary's evolution. However, particles form complex contacts where the conditions of all involved contacts have influence on each other. This study shall demonstrate and evaluate this issue on the example of regular sphere packings (circle packing in 2D).

9.2.1. Construction of the Study

In a regular packing of equal circles each pore in the packing shows equal evolution due to symmetry. Four different packings, displayed in figure 9.25, shall be investigated here, which resemble common configurations of sphere packings:

Pair the usual two-particle packing as a reference

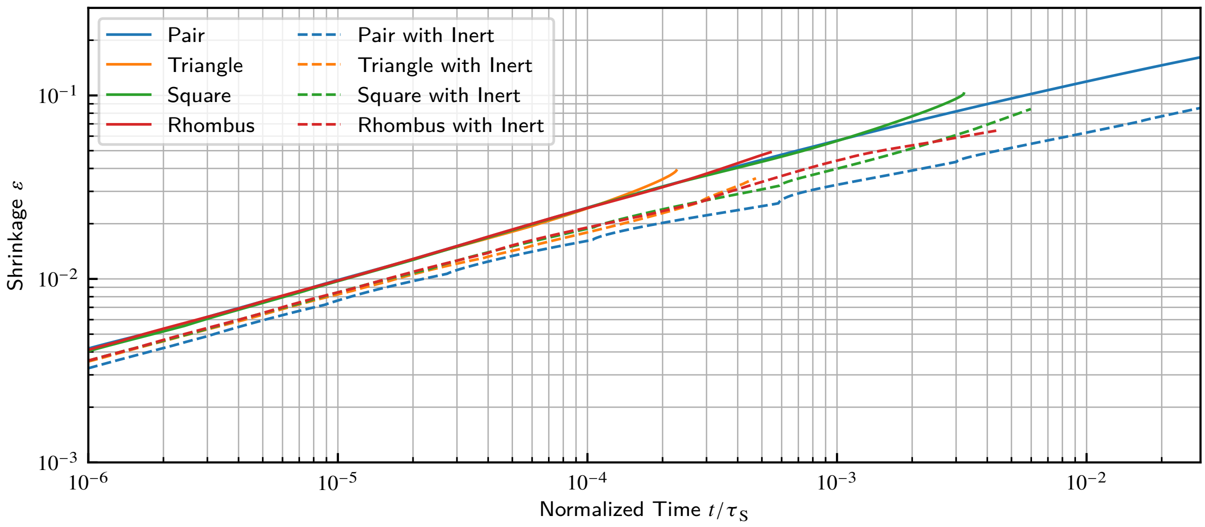
Triangle the densest possible packing of circles with particles at the corners of an equal sided triangle, resembles a $\{111\}$ cut through a face-centered cubic cell

Square the loosest regular packing with particles at the corners of a square, resembles a $\{100\}$ cut through a simple cubic cell

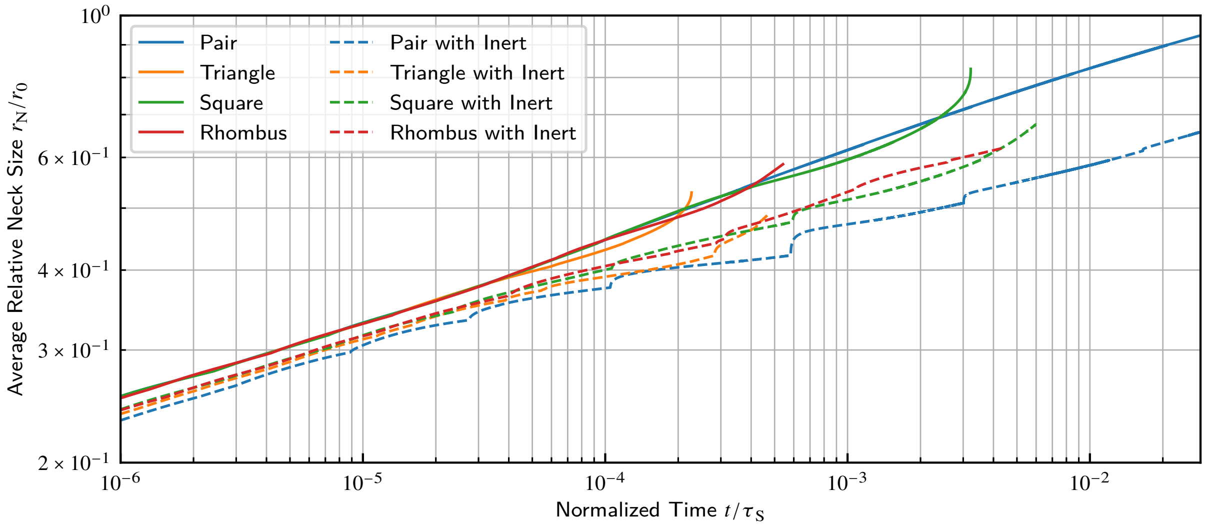
Rhombus an intermediate packing with particles at the corners of a rhombus of 70°, resembles a $\{110\}$ cut through a body centered cubic cell

The main difference between these packings is the size and the shape of the pore formed by them.

These packings are expected to show only effects of pore closing, but otherwise behave in a fully symmetric way equivalently to the pair case, as all particles are of equal size, shape and material properties. In a second step, one of the particles will be given a surface and grain boundary diffusion coefficient 1000 times smaller than the other ones to investigate hindering of sintering by the presence of an inert particle. The term inert shall be used here for a particle with remarkably lower diffusion rate on its surface as well the grain boundary. This particle is shown with brown stroke in figure 9.25. The other ones will be referred as active.



(a) Shrinkage

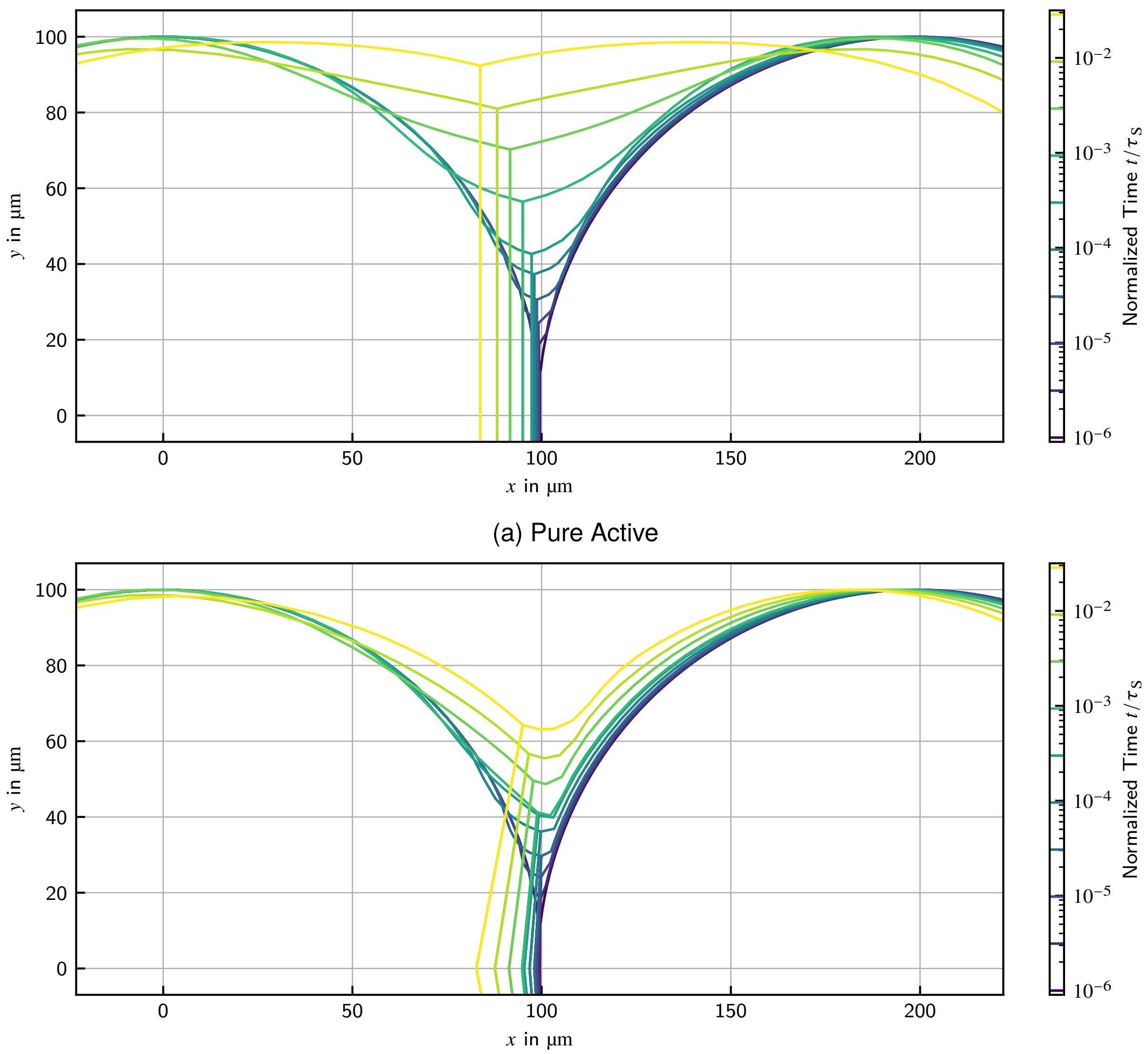


(b) Average Neck Size

Figure 9.26: Comparison of Sintering Behavior Obtained for the Investigated Packings

9.2.2. Results and Discussion

Figure 9.26a shows the shrinkage for the distinct packings obtained from the simulations. The shrinkage is calculated in all cases using the polygon approach as discussed in section 5.3, except for the pair for which the polygon approach is not applicable and so the distance approach is used. Note, that the distinct cases all show the equal behavior in early stages and the two approaches to shrinkage calculation are equivalent. At some point, specific to the respective case, the curve deviates rapidly to higher shrinkages from the reference two-particle model. This point is related to the pore closing and the change of its shape characteristic from concave to convex, as is discussed in Tikare, Braginsky, and Olevsky [100]. Therefore, the deviation occurs earlier for the more dense packed cases, in order: triangle, rhombus, square. Jing, Zhao, and He [70] reported PFM simulations of a single pore which first show the deviation to the upper side, also observed here, and then the transition into steady state when the pore was fully closed. The effect there is likely not that pronounced as here, since the PFM approach blurs the pore due to the diffuse interface which



(a) Pure Active

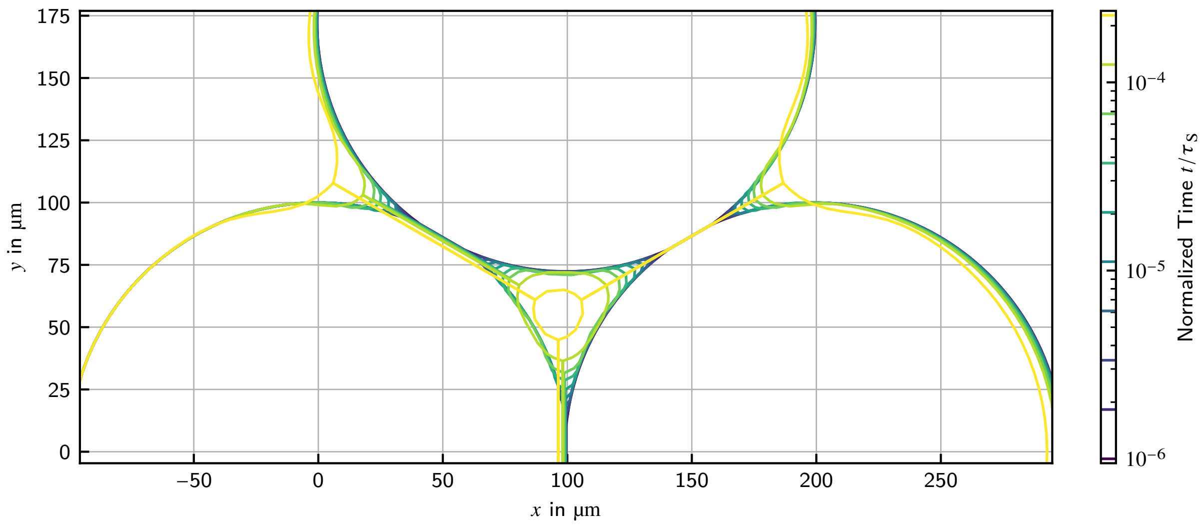
(b) With Inert Right

Figure 9.27: Geometry Evolution of the Pair Cases

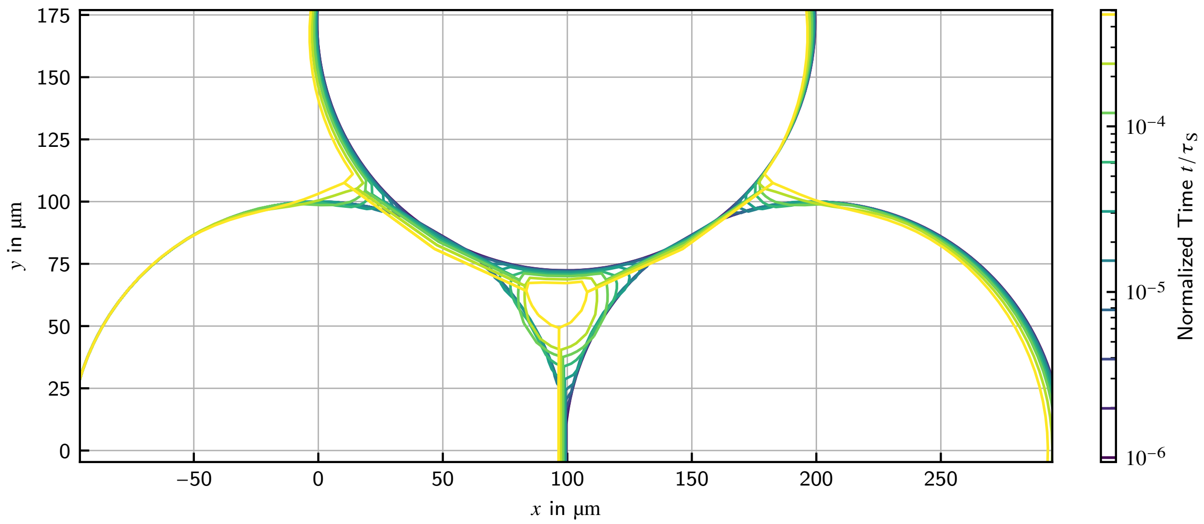
acts like a regularization when the pore size is similar to the numerical interface width.

At the point where the pore was completely closed, the simulations were cancelled, since the current model does not support triple points yet. As the model is only aimed at initial and intermediate sintering stages, this problem can be neglected. Complete closure was determined at the point, were the last surface node between two neck nodes was removed by the neck remeshing procedure. Note that equal colors represent equal times in all cases. The shrinkage response shows a drastic increase in slope prior to pore closing, as the curvature of the pore increases while its size decreases. This enhances the driving force.

Figure 9.26b shows the neck sizes for the distinct packings obtained from the simulations. The curve drawn represents the average of all necks for the multiple-particle cases. The general trends are the same as for shrinkage. The beginning of pore closing is first noticed by a deviation to the lower from the straight line. As the pore closes further, the neck growth rate increases drastically so that the neck size curve crosses the straight again.



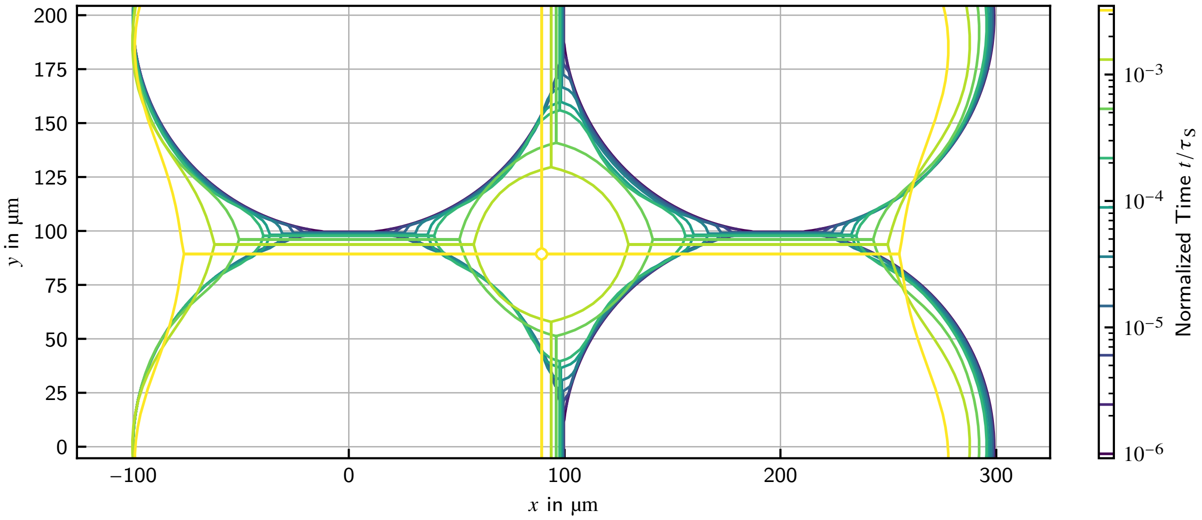
(a) Pure Active



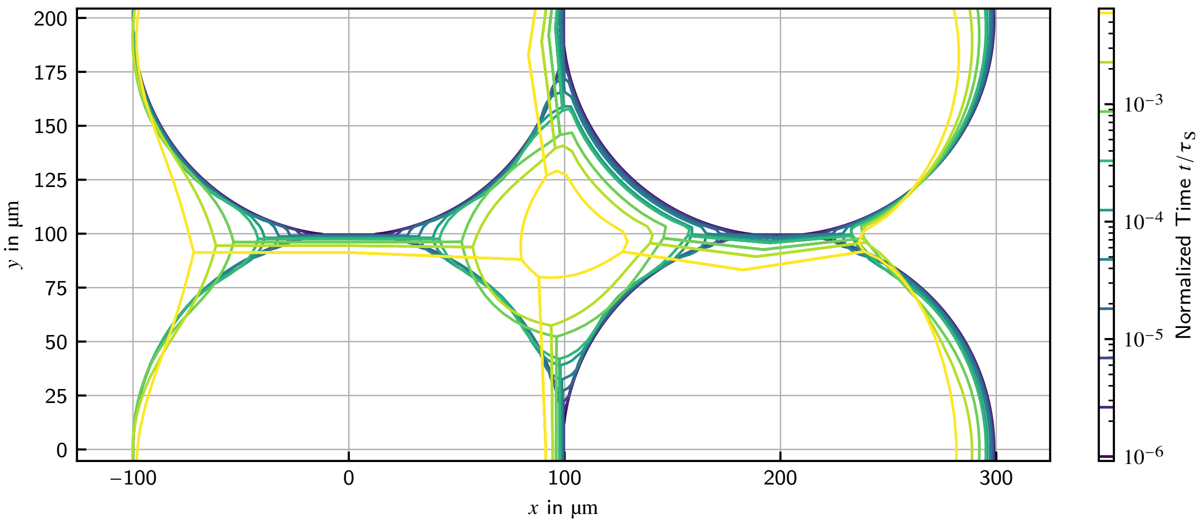
(b) With Inert Upper

Figure 9.28: Geometry Evolutions of the Triangle Cases

The cases with an inert particle show a remarkably lower shrinkage and neck size, since the contribution of volume removed from the grain boundary of the inert particle is missing. This is in accordance with experimental observations [2]. The numerical behavior of all cases with an inert particle is remarkably worse than of the pure active cases, as is apparent from the kinks in the result curves. The detailed logs of the simulation suggest the reason to the more erratic neck remeshing on the inert particle. The pore closing is delayed as the pore is kept open by the inert particle's surface. The three multiple-particle packings with an inert behave similarly, show equal behavior in early stages and pore closing occurs in the same order as for the pure active cases. The pair with an inert shows a distinct behavior to the other inert cases. Its shrinkage is the lowest. A correlation with the count of active particles present seems unlikely, since the two-particle case is the only one behaving differently. German [2] reported possible crack formation in the active-active and the active-inert interfaces due to tension stresses occurring, since the inert particle inhibits free and uniform particle movement. This did not occur in the current investigations, likely because the inert particle was not fully surrounded by active particles. Nevertheless, the simulation would currently



(a) Pure Active

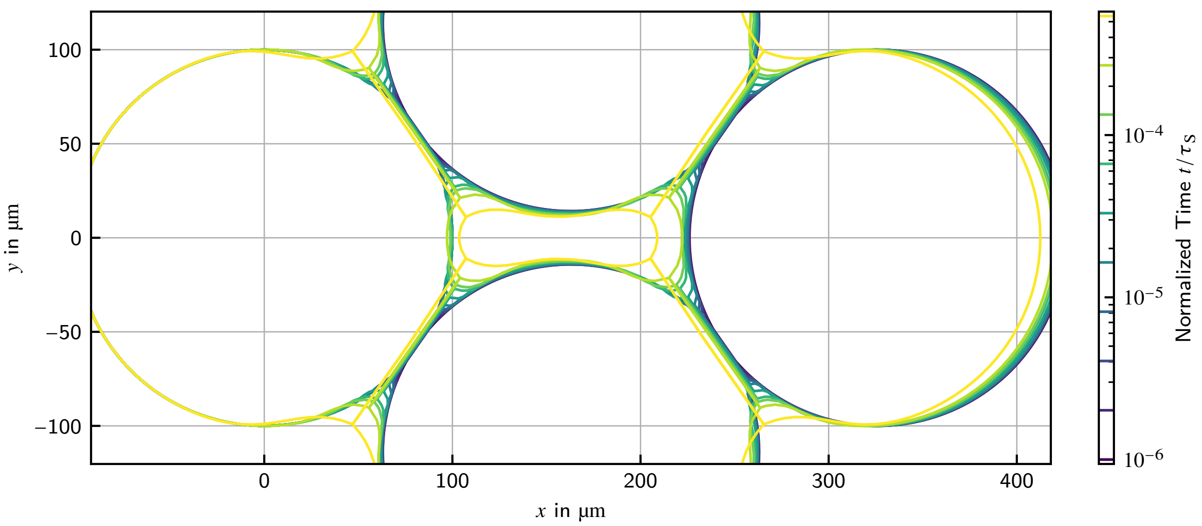


(b) With Inert Upper-Right

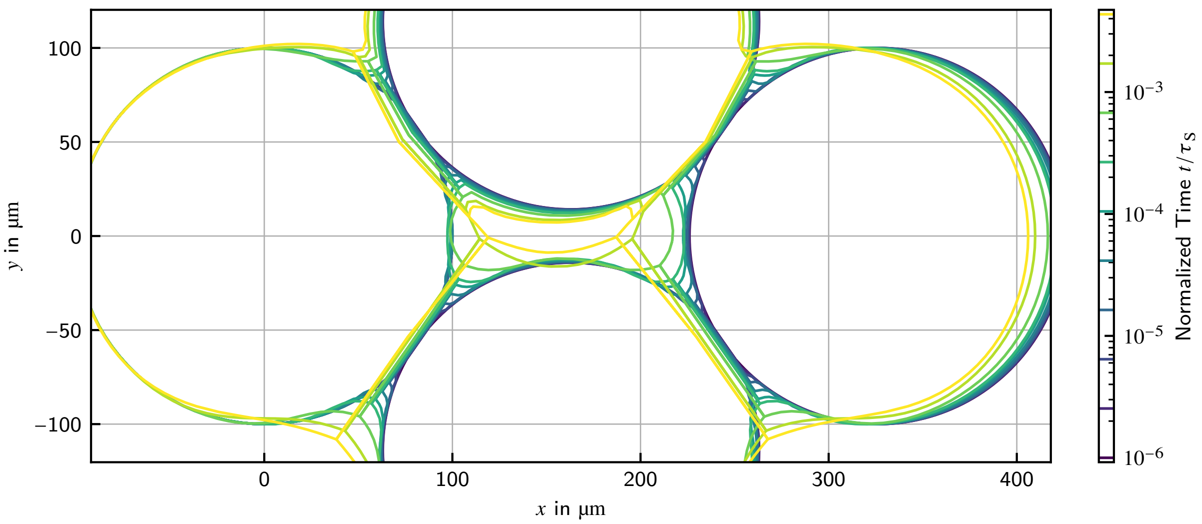
Figure 9.29: Geometry Evolutions of the Square Cases

cancel if a neck shrinks and approaches zero width, as contact release is not implemented.

Looking at the geometry evolutions in figures 9.28 to 9.30, the convexity respectively concavity of the pore can clearly be observed. The pore surface to the inert particle stays concave during the whole process. As previously observed in section 9.1.5, the grain boundaries between inert and active particles are curved, since the active particle creeps around the inert one. On the inert particle an undercut evolves near the neck. The remaining pore volume at the state of simulation cancellation is significantly higher than in the pure active cases.



(a) Pure Active



(b) With Inert Upper

Figure 9.30: Geometry Evolutions of the Square Cases

10. Randomized Simulation of Particle Packings

In the previous chapter the numerical and physical behavior of the model was analyzed. Thereby, the transition from a two-particle-model to a multi-particle-model was performed. The results of section 9.2 show, that the effects of pore closure can be incorporated by investigating a particle arrangement of at least three particles. The sintering behavior of all investigated arrangements was the same except for the region where the pore is almost closed. Also, the effects of multi-material contacts can be regarded. In the following investigation, the particle arrangement shall be randomized using statistically generated particles as developed in section 5.4.

10.1. Construction of the Study

The study will stick to three-particle arrangements to avoid major difficulties in construction of the initial state. As the size and shape of the particles will now be variable, the initial position can not be given a-priori but must be determined numerically. The study will compare four cases: the nominal case, with three circular particles of equal size, and three powders with equal particle size distribution, but one assumed to have circular particles, one where only an ovality according to equation (5.15) is considered and one where also first order waves according to equation (5.16) are considered. The differences and similarities of the cases will be investigated to rate the need for inclusion of this data into sintering simulations. The core question is if the inclusion of size distribution and shape data in the simulation leads to significantly different results and if those differences are in accordance with experimental experience and theoretical expectations. Regarding shape data, it is investigated if the first order waves have major influence or if regard on ovality is enough, since ovality is a parameter that can be measured without problems using state-of-the-art automatic particle classifiers. First order waves, however, require a high amount of manual work. The author has already published a paper describing a method to obtain these for aggregate particles via laser scanning microscopy images and least-square fitting in Weiner, Schmidtchen, and Prahl [148], but there has been used another approach of sintering model. The data obtained there will be reused in this study.

10.1.1. Statistical Description of the Example Powder

For each of the shape parameters included in the chosen shape function, a suitable distribution function has to be selected and fitted on the experimentally obtained data. The types of the distributions are selected according to their typical shape and support. The choice and fitting will be discussed in section 10.2.

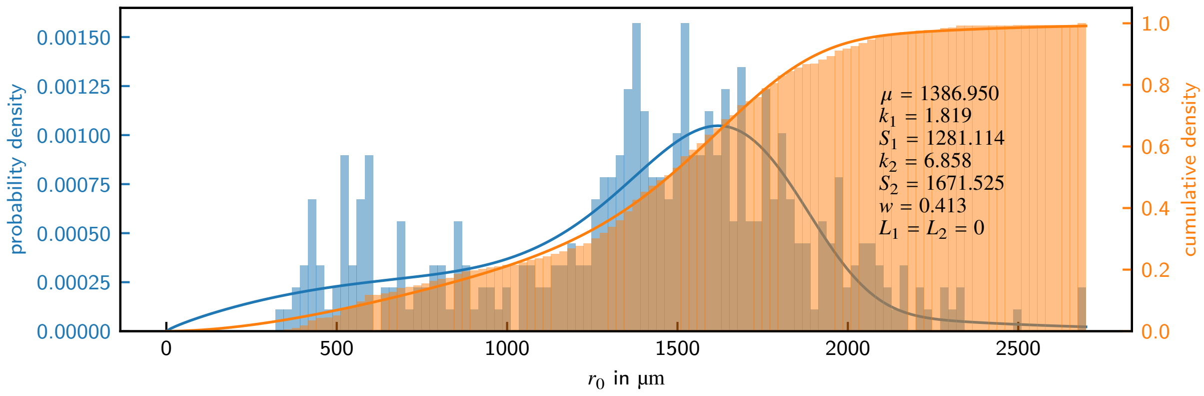


Figure 10.1: Particle Size Distribution when Assuming Circular Particles

10.1.2. Creation of the Initial State

The initial state generation starts by sampling three particles randomly. For each a radius r_0 and one of every shape parameter is sampled from the given distribution. Also, the rotation of each particle is sampled from a uniform distribution with value range $\omega \in [0, 2\pi)$. The initial positions of the particles are created in sufficient distance to each other, so that it is impossible for them to be in contact.

The first particle is kept fixed in space. Then, the second is moved along the vector towards the center of the first in small steps until contact is created by a defined minimum intersection of the two surfaces. Then the third is moved along the sum of the vectors from the thirds center to the others' centers. If contact with one is created, then the vector towards this particle is counted negative to allow sideward movement. When the third particle has contact to both, the procedure is completed.

10.2. Acquisition of Particle Shape Data

The underlying data is the same as in Weiner, Schmidtchen, and Prahl [148], but there has been used a different shape function. The fits are therefore repeated with the current shape function as discussed in section 5.4. The dataset (available in [149]) includes contours of aggregate composite particles produced by sintering, crushing and sieving as described in Zienert et al. [128, 129], which have been extracted from laser scanning micrographs.

The fit is accomplished in all cases by least squares optimization on the Euclidean distance between data points and the model function predictions. The experimental particle contour is shifted prior to fitting, so that its barycenter is in $(0, 0)$.

10.2.1. Circular Particles

In the first case, all shape terms are deactivated in the function and fitting is solely done assuming circular particle shape. So, the only parameter regarded is the particle radius r_0. Figure 10.1 shows the histogram of the fit results alongside the theoretical distribution fit. Note the bimodal distribution shape of the histogram.

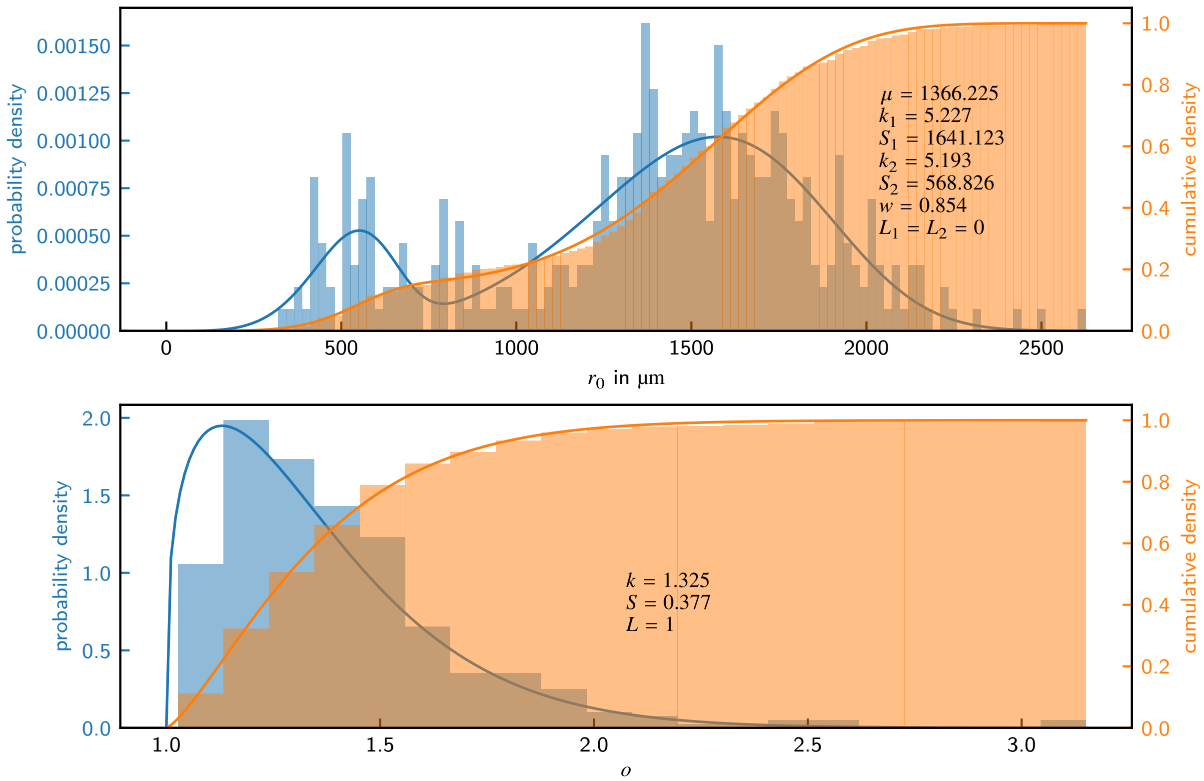


Figure 10.2: Particle Size and Ovality Distributions when Assuming Oval Particles

For particle size distributions it is common to use a Weibull distribution (equation (10.1)), with the shape parameter k, the scale parameter S and the location parameter L. The support of the distribution is $x \in [L, \infty)$. As particle sizes must always be > 0, the location $L = 0$ is used here.

$$p(x) = \frac{k}{S}\left(\frac{x-L}{S}\right)^{k-1} \exp\left[-\left(\frac{x-L}{T}\right)^{k}\right] \tag{10.1}$$

The expectation μ of the Weibull distribution is given by equation (10.2) with the gamma function Γ. The expected radius μr_0 will be used as the reference radius to calculate the characteristic time norm according to equation (2.14).

$$\mu = S \cdot \Gamma\left(1 + \frac{1}{k}\right) \tag{10.2}$$

To obtain a bimodal distribution, a mixture distribution of two Weibull distributions is used with parameters (k_1, S_1) and (k_2, S_2). The probability of sampling from the first one is the weight w, whereas of choosing the second $1 - w$, respectively.

10.2.2. Oval Particles

In the second case, the ovality term (equation (5.15)) is added to the shape function. Beneath the ovality o, also the rotation ω of the particle has to be fitted, since the shape function is no longer invariant counter rotation. As the latter is needed for correct fitting but represents no actual particle shape feature but the placement of the particle, it is not analyzed further. Fitting the rotation is numerically unstable as there are many local extrema. To get all local extrema a brute force

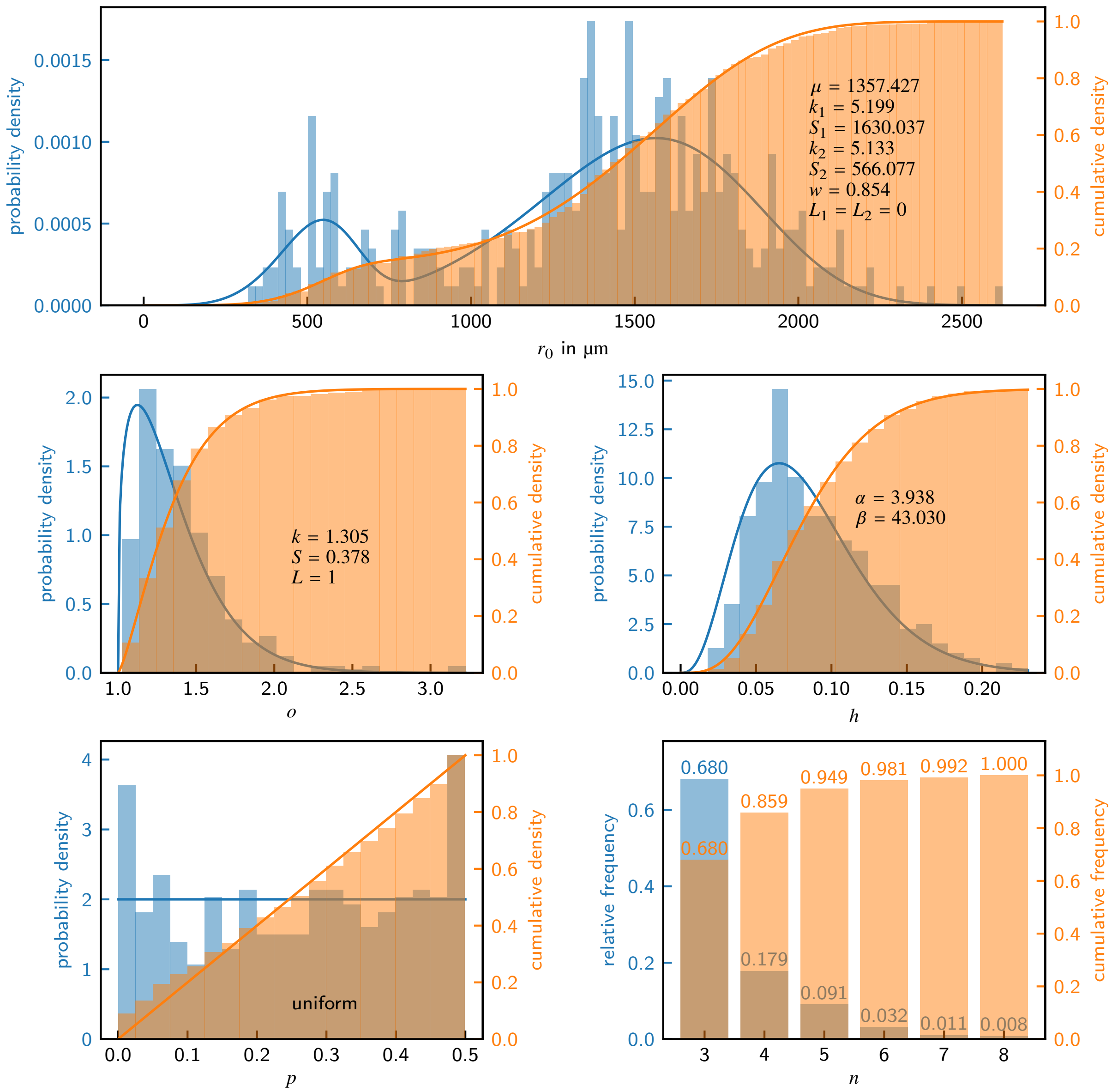


Figure 10.3: Particle Size and Shape Parameter Distributions when Assuming Oval Particles with First Order Waves

method is applied which chooses a couple of linearly spaced initial values for the rotation parameter and selects the best achieved fit afterward. This method is simple and easy to implement while the impact of computational effort hardly matters here. Figure 10.2 shows the histograms of the fit results alongside the theoretical distribution fits. The ovality o is modeled by a single Weibull distribution. In contrast to particle size, here a location $L = 1$ is used, as ovality $o \geq 1$.

10.2.3. First-Order-Waved Particles

In the third case, first-order waves according to equation (5.16) are considered, too. So, the fit additionally includes the parameters wave height h, wave count n and wave shift p. As before the rotation has to be fitted accordingly. As n is integer valued and only few distinct values have to be expected, it is treated similarly to the rotation by a brute force method. Mixed continuous and integer optimization is a generally hard problem, were several sophisticated algorithms exist. For the current case they have no computational benefit, as they have typically high effort per step and the value range of n is small. Figure 10.3 shows the histograms of the fit results alongside the theoretical distribution fits.

The wave height h is modeled using a beta distribution, which supports a variety of shapes and has a defined support for $[0, 1]$ which fits ideally to the parameter. α and β are the free parameters of the distribution.

$$p(h) = \frac{1}{B(\alpha, \beta)} h^{\alpha-1} (1-h)^{\beta-1} \tag{10.3}$$

The wave count parameter n is of discrete nature and only few distinct values were present in the data. So, the parameter is not modeled by a theoretical distribution but handled as a categorical variable with defined relative frequencies. The wave shift p is remarkably close to a uniform distribution, so no fitting of a distribution function is required here.

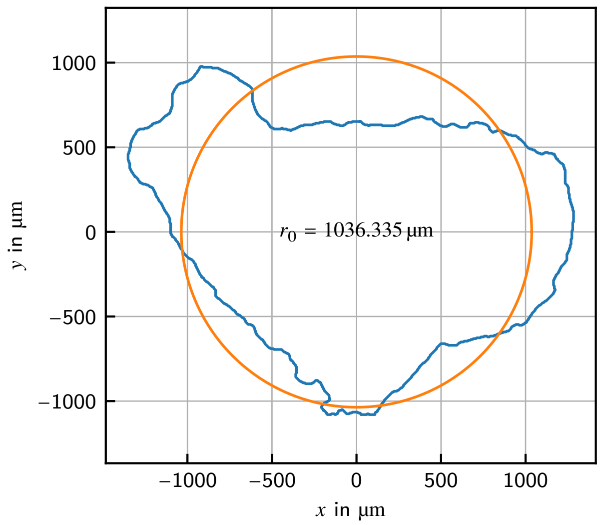
10.2.4. Comparison of the Three Approaches

The obtained particle size distributions from the three approaches are quite similar, despite the changes in the shape descriptions. Due to improved shape accuracy, the mean of particle size r_0 is slightly decreasing from 1386 µm to 1357 µm. To ensure comparability, the reference radius for normalization will be fixed to the approximate mean value of 1370 µm for the rest of this chapter. This radius is also used as the radius of the nominal case particles.

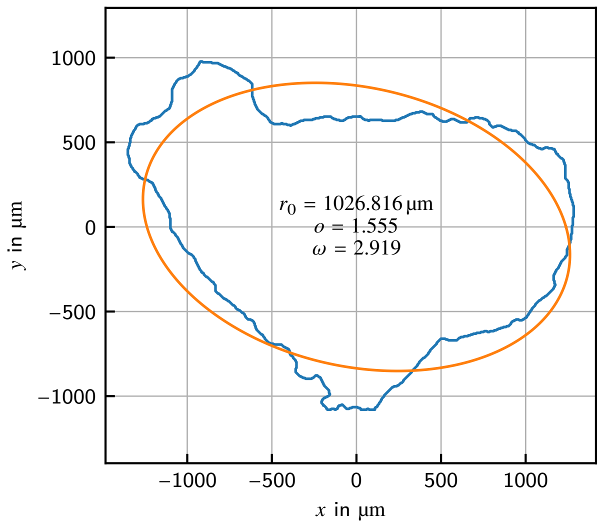
Figure 10.4 shows a typical particle from the dataset, which has been fitted by the three distinct approaches. One could clearly observe the improved shape representation when adding the additional terms. Fine surface structures obviously can not be represented by the models, but the overall shape characteristic is well modeled by the complete shape function with ovality and first-order waves.

10.3. Simulation Results

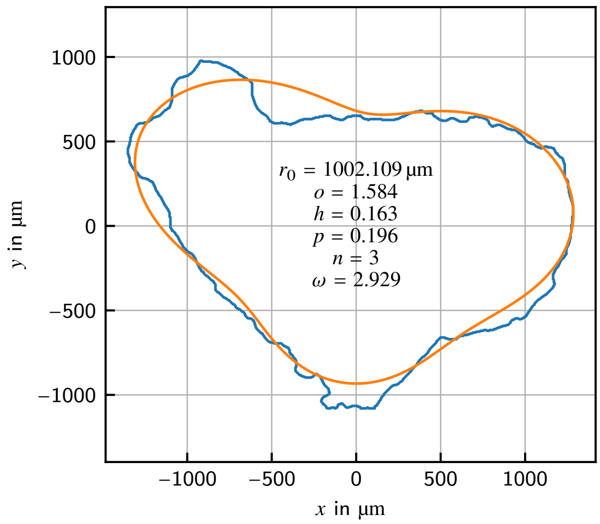
The figures 10.5 to 10.10 show the shrinkage respectively neck size evolution of the randomized simulations. The distinct samples are drawn as a partially transparent lines, so that the density of samples can be estimated visually by the intensity of the color. For comparison the arithmetic mean of the samples and the result of the nominal case are drawn alongside. The arithmetic mean curve is calculated by interpolating the sample curves at the respective time. When the respective sample curve was does not provide a value for the respective time, because the simulation was already canceled due to pore closing, than the last value of the curve is taken. This is equivalent to the assumption that the particle group is inert (does not evolve any further) after pore closing has happened. This assumption has to be imposed, as simulation after pore closing is not supported in the current state of the model (see section 9.2). Extrapolation of the curves is not feasible, as they approach a singularity (approach infinite slope) when the pore is almost closed. In each figure, also two histograms at the normalized times $t/\tau_S = 10^{-6}$ and $t/\tau_S = 10^{-6}$ are drawn, which also show the respective mean value and the standard deviation. The histograms are calculated using the same assumption on pore closing.



(a) Circular



(b) Oval



(c) First-Order Waves

Figure 10.4: Comparison of the Three Shape Fit Approaches for a Typical Particle

To measure the size of an effect, the absolute effect e (difference of means) is often not sufficient, as it neglects the spread of the data within each group. Cohen [150] defined a relative effect size e_r, also known after the author as Cohen's effect size. It is given as the absolute effect size e divided by the standard deviation σ as in equation (10.4). Which standard deviation to take is an issue, especially when sample sizes are different. For equal sample sizes Cohen [150] gave the expression in equation (10.5). Here, the case is simpler, as the comparison is done between each randomized and the nominal case, which is fully certain. So, the reference standard deviation is just that of the randomized case. Table 10.1 lists the obtained means and standard deviation for the respective cases alongside with the absolute and relative effect sizes. For rating the value of Cohen's effect size, Sawilowsky [151] gave rules of thumb, according to which an effect of $e_r \approx 0.2$ is to be classified as small, $e_r \approx 0.5$ as medium, $e_r \approx 0.8$ as large and $e_r \approx 1.2$ as very large.

$$e_r = \frac{e}{\sigma} = \frac{\hat{\mu}_1 - \hat{\mu}_2}{\sigma} \tag{10.4}$$

Table 10.1: Comparison of Means and Standard Deviations and their Effect Sizes for the Distinct Cases of Randomized Simulations

	Nominal		Circular		Oval		Waves	
t/τ_S	10^{-6}	10^{-4}	10^{-6}	10^{-4}	10^{-6}	10^{-4}	10^{-6}	10^{-4}
Shrinkage								
$\hat{\mu}$ in %	1.189	5.502	1.156	4.843	1.139	4.778	1.185	4.377
$\hat{\sigma}$ in %	–	–	0.428	1.038	0.364	1.063	0.468	1.559
e in %	–	–	–0.033	–0.659	–0.050	–0.724	–0.004	–1.125
e_r in %	–	–	–0.077	–0.635	–0.137	–0.681	–0.009	–0.722
Neck Size								
$\hat{\mu}$ in %	24.861	45.734	23.664	41.792	23.267	41.782	21.502	38.138
$\hat{\sigma}$ in %	–	–	3.856	8.670	3.215	7.427	4.017	9.369
e in %	–	–	–1.197	–3.942	–1.594	–3.952	–3.359	–7.596
e_r in %	–	–	–0.310	–0.458	–0.496	–0.532	–0.836	–0.811

$$\bar{\sigma} = \sqrt{\frac{1}{2}\left(\sigma_1^2 + \sigma_2^2\right)} \tag{10.5}$$

Below, the results of shrinkage and neck size will be described with respect to the plots and the calculated effect sizes under consideration of the effect size rules of thumb.

10.3.1. Observations on Shrinkage

The figures 10.5 to 10.7 show the shrinkage results of the randomized simulations.

Regarding the early stages, the following observations are to be noted:

1. The deviations between the nominal and the randomized cases is very small, and both curves are visually in well agreement.
2. The relative effect sizes can be classified as small or less. The effect size of the waved case is even much lower than of the other ones, which is related to the fact that in this case the time section at $t/\tau_S = 10^{-6}$ is accidentally near an intersection of the mean and nominal curves.
3. The standard deviation of the oval case is slightly smaller compared to the circular case, whereas that of the waved case is slightly higher.
4. The histograms show an approximately symmetric shape when plotted in logarithmic scale.

Regarding the late stages, the following observations are to be noted:

1. The deviations between the nominal and the randomized cases is much larger than in early stages.
2. The relative effect size has to be classified between medium and large, where the effect slightly increases from circular, over oval, to waved.

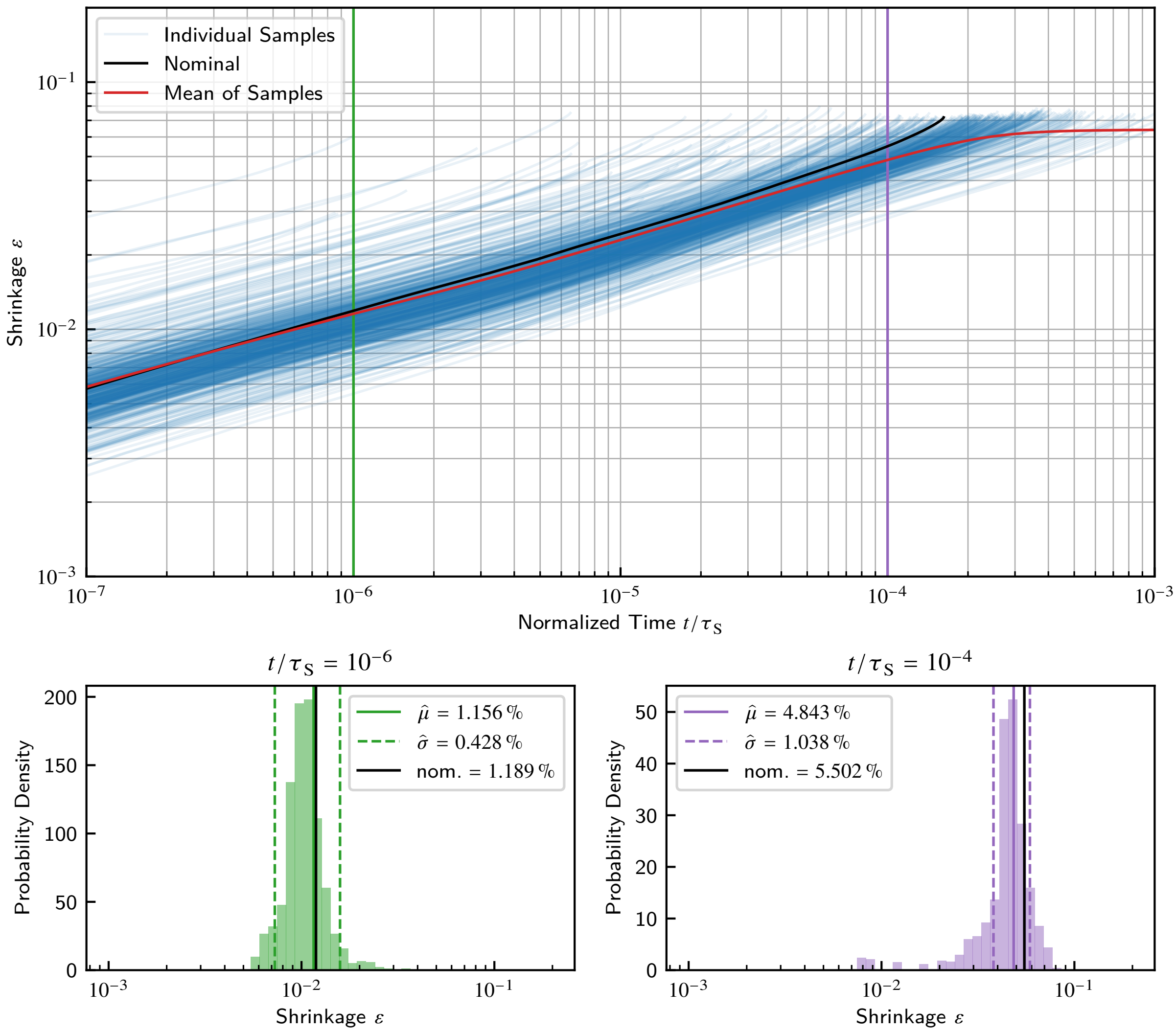


Figure 10.5: Shrinkage of the Circular Case

3. The circular and oval case have comparable standard deviation, whereas that of the waved case is remarkably higher.
4. The waved case has a drastically higher absolute effect, but the relative effect, although higher than that of the oval case, is not that pronounced due to the higher standard deviation.
5. The histograms show a heavy tail on the left side, formed by samples with early pore closing. The heavy tail is especially pronounced in the waved case. This is in accordance with the smoother transition into the stationary state.

Regarding the overall trends with time, the following observations are to be noted:

1. The nominal curve shows the typical behavior of a closing pore with increasing slope as already observed in section 9.2 and in the individual samples.
2. The mean curve shows a flattening characteristic with transition into a stationary state. The transition into the stationary state is more sudden in the circular case over the oval to the waved case.
3. In all cases the standard deviation increases with time.

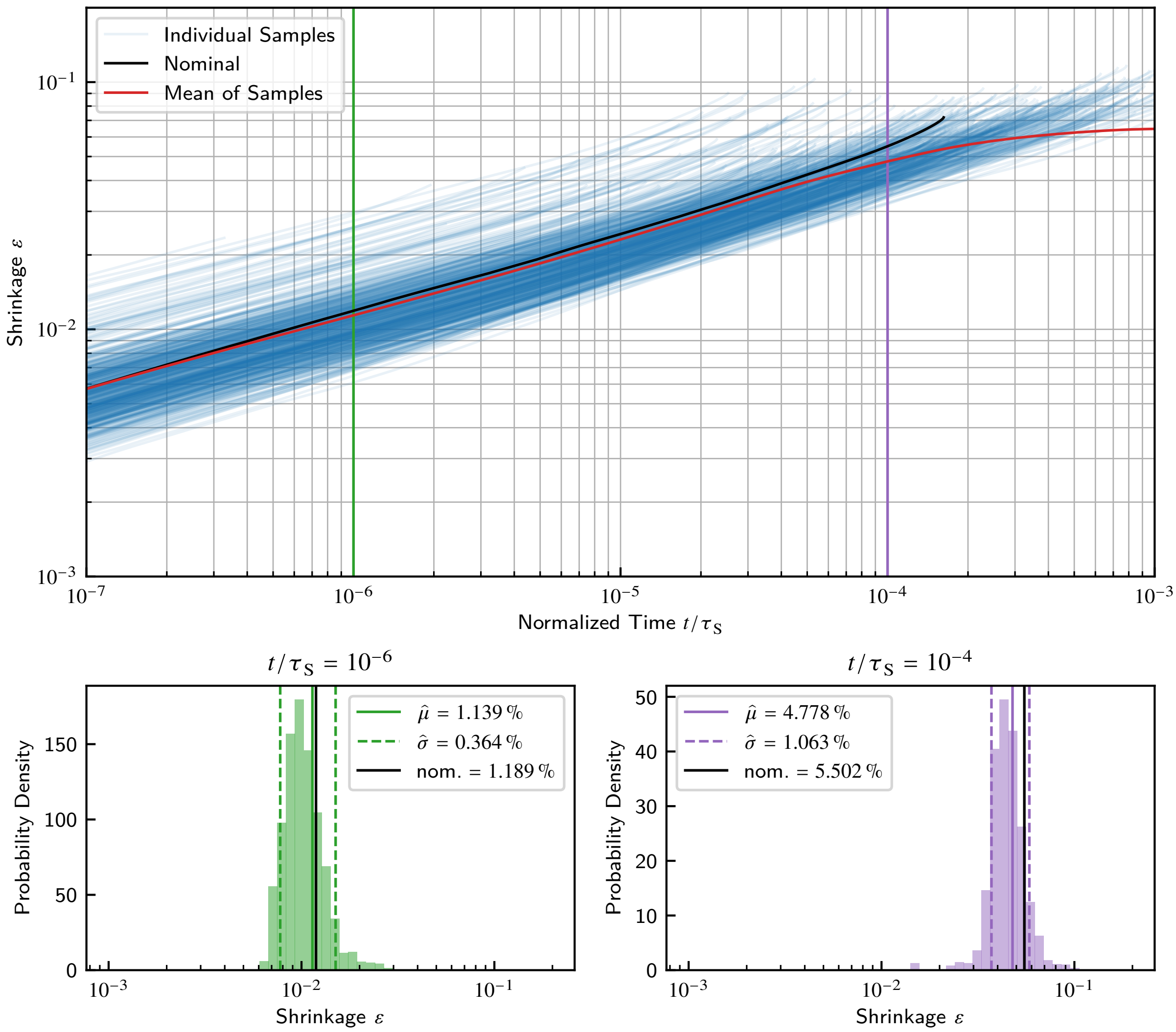


Figure 10.6: Shrinkage of the Oval Case

10.3.2. Observations on Neck Size

The figures 10.8 to 10.10 show the neck size results of the randomized simulations.

Regarding the early stages, the following observations are to be noted:

1. In contrast to shrinkage, neck size is also in early stages significantly affected by the randomized consideration of size and shape features.
2. The absolute effect heavily increases from the circular, over the oval to the waved case.
3. The standard deviation of the oval case is smaller compared to the circular case, whereas this of the waved case is higher.
4. The relative effect sizes are to be classified as small to medium for the circular, medium for the oval case and large for the waved case.
5. The histogram of the circular case (figure 10.8) has a heavily right skewed shape when plotted in logarithmic scale, which corresponds to a symmetric shape in linear scale. This skew vanishes over the oval case to the waved case.

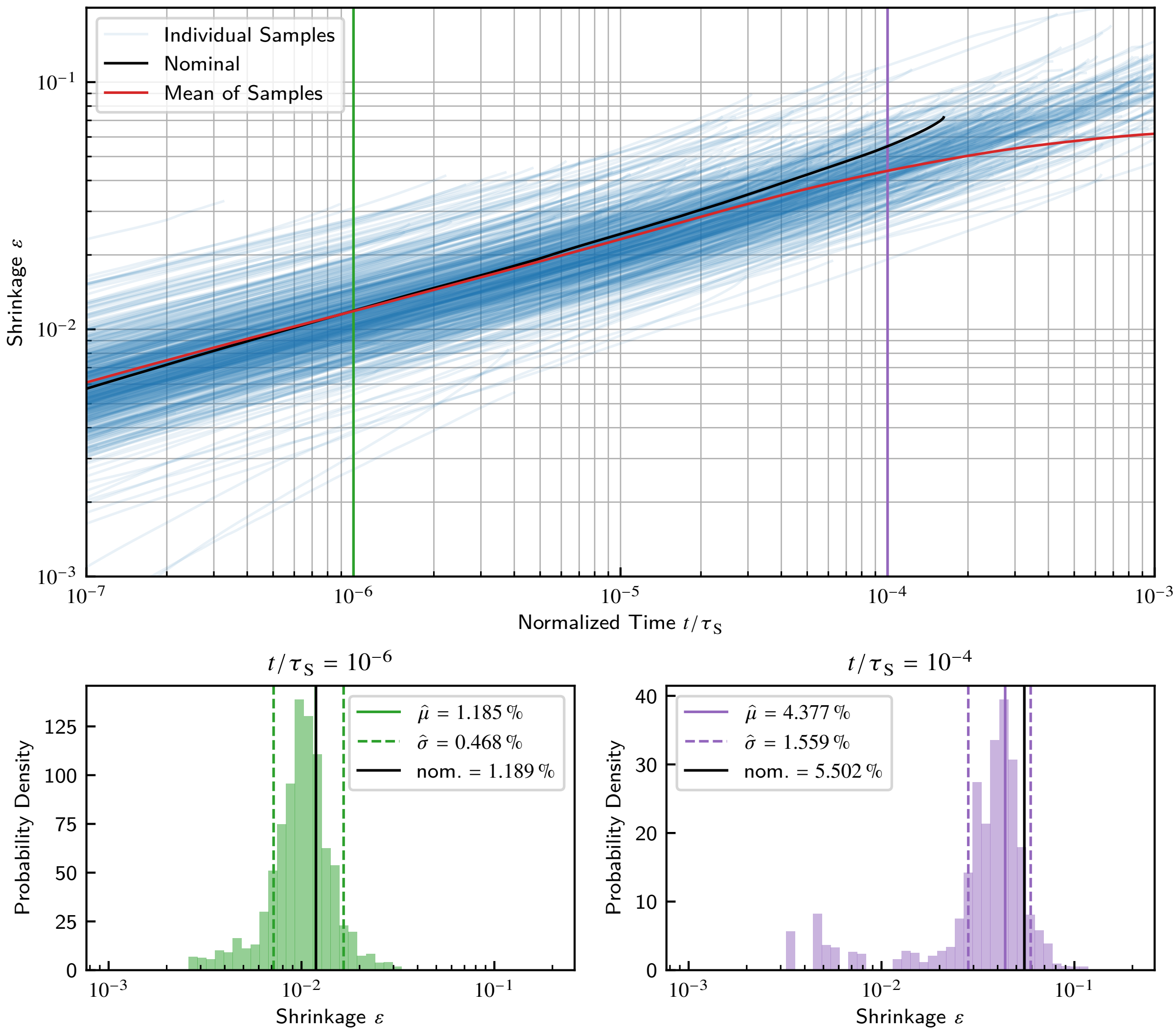


Figure 10.7: Shrinkage of the First-Order Waves Case

Regarding the late stages, the following observations are to be noted:

1. The absolute effect is similar for the circular and the oval case. The waved case has a remarkably higher effect.
2. The standard deviation follows the same trend as in early stages.
3. The relative effect sizes are similar for the circular and oval cases, both to be classified as medium. The waved case has a higher effect, to be classified as large.
4. The histograms show a heavily right skewed shape when plotted in logarithmic scale. The tail gets heavier from the circular, over the oval, to the waved case.

Regarding the overall trends with time, the following observations are to be noted:

1. Similarly to shrinkage, the nominal curve shows the typical pore closing behavior, whereas the mean curve transits into a stationary state with the same trends observed before.
2. As in shrinkage, the standard deviation increases with time.

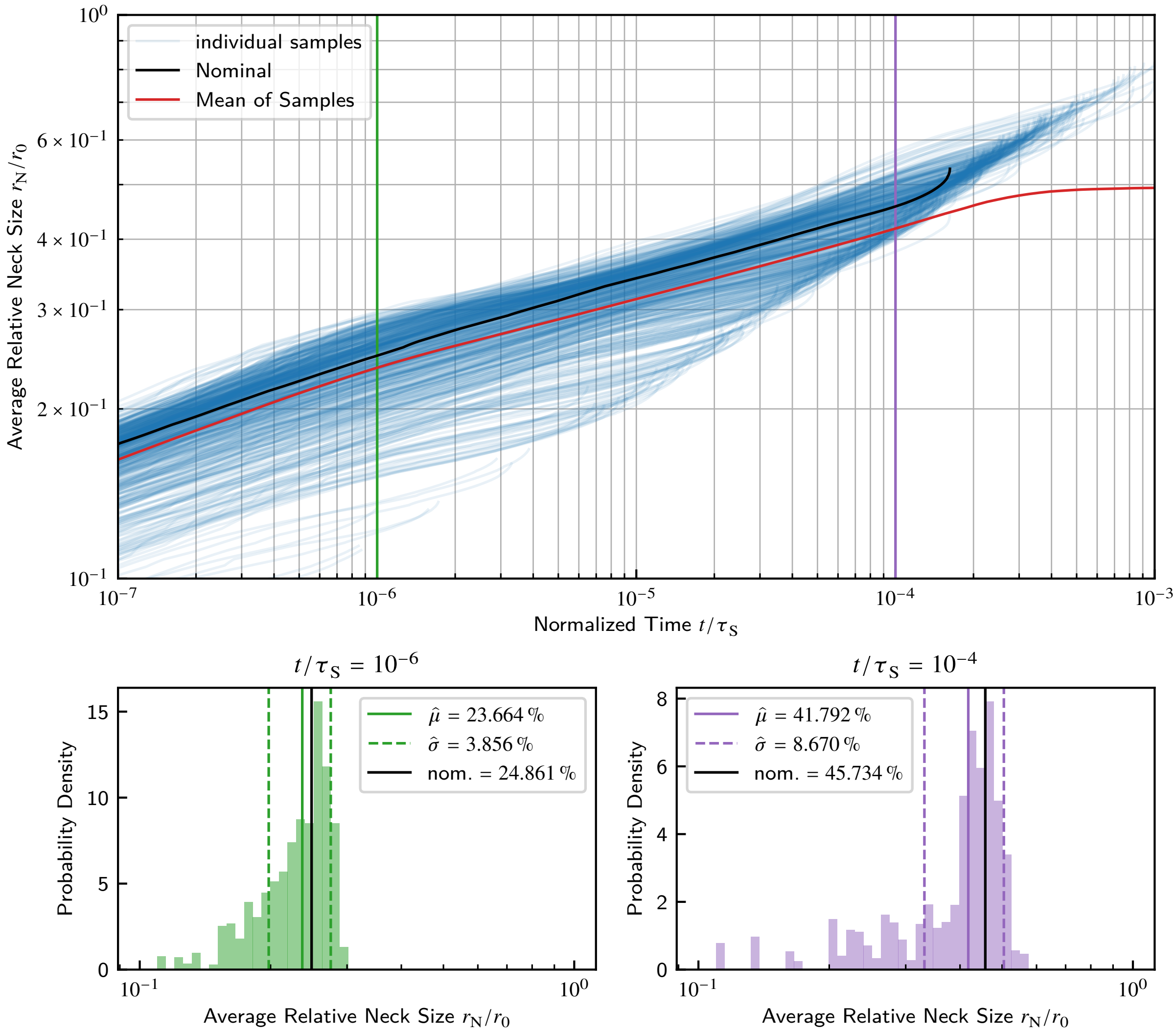


Figure 10.8: Neck Size of the Circular Case

3. In contrast to the shrinkage, the relative effect sizes are not remarkably increasing with time, the larger absolute effect is canceled out by the increase in standard deviation.

10.4. Discussion and Conclusions

The main reason for the observations is suspected in the effects of pore closing, where the sintering behavior significantly deviates from the straight linear one (in double logarithmic scale). As denoted before in section 9.2, the shrinkage and neck size evolution of a particle group, forming a single pore, features a singularity with the slope tending to infinity when the pore volume approaches zero. Macroscopical sintering bodies in contrast, exhibit a transition into a stationary state, in accordance to the flattening behavior shown by the mean curve. This flattening behavior is created numerically due to the increasing proportion of closed pores at later times, which do not contribute to sintering anymore as they are considered to be fully inert by assumption (see above). Principally, if the model would support regard on a triple junction of grain boundaries, further neck growth and shrinkage may occur due to volume displacement from the inner to the free surfaces on the outside of the

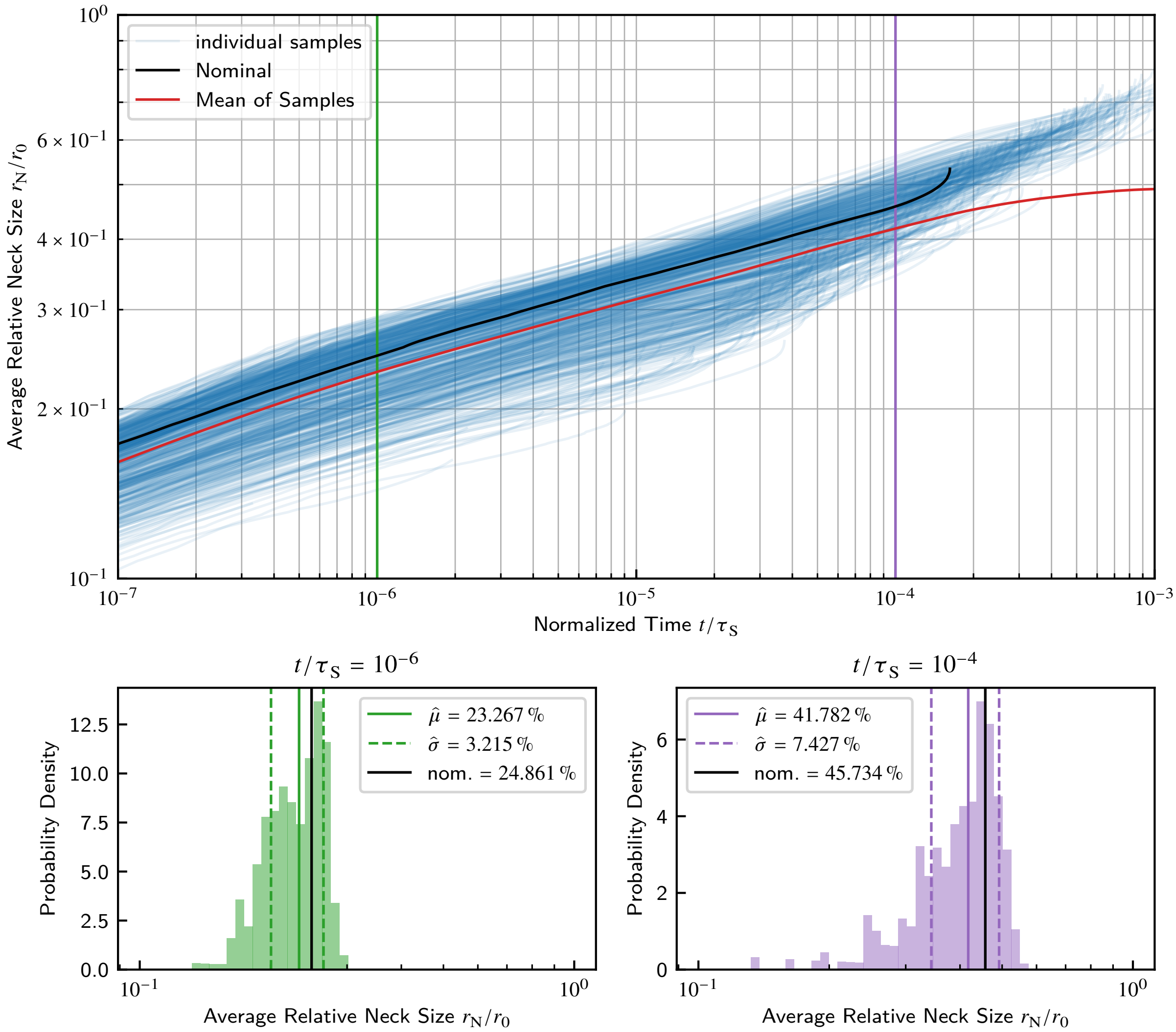


Figure 10.9: Neck Size of the Oval Case

particle group. It is questionable, if this would be a major improvement to the predictions of the model, as in a real powder compact this outer surface does not really exist (except on the body surface), instead there is another pore. The quantitative validity of the prediction by this numerical effect is a pending task, but at least the qualitative behavior appears to be valid.

A similar effect is reported in simulations of the kinetic Monte-Carlo-Method (kMCM) type, which are able to work with large counts of particles in complex microstructures. Braginsky, Tikare, and Olevsky [98] compared the two cases of a regular circle packing and a random packaging of unequally sized circular particles. In the first case they obtained a sharp transition into stationary state, since pore closing happens (approximately) at the same time for all pores. In the second case the transition was much smoother, due to distributed pore closing. Tikare et al. [104] obtained similar results for a 3D model.

Many models, except for the simplest ones, show a stationary state in later stages. The reasons and mechanisms behind this are manifold. Classic two-particle [38] or single-pore models [152] do show a flattening of the sintering curves, but at drastically later times, when a stationary state is reached due to vanishing driving force. This is also the case for the current model in a two-particle arrangement for very large sintering times up to $t/\tau_S = 10^{-1}$ to 1 (not shown). Two-particle models

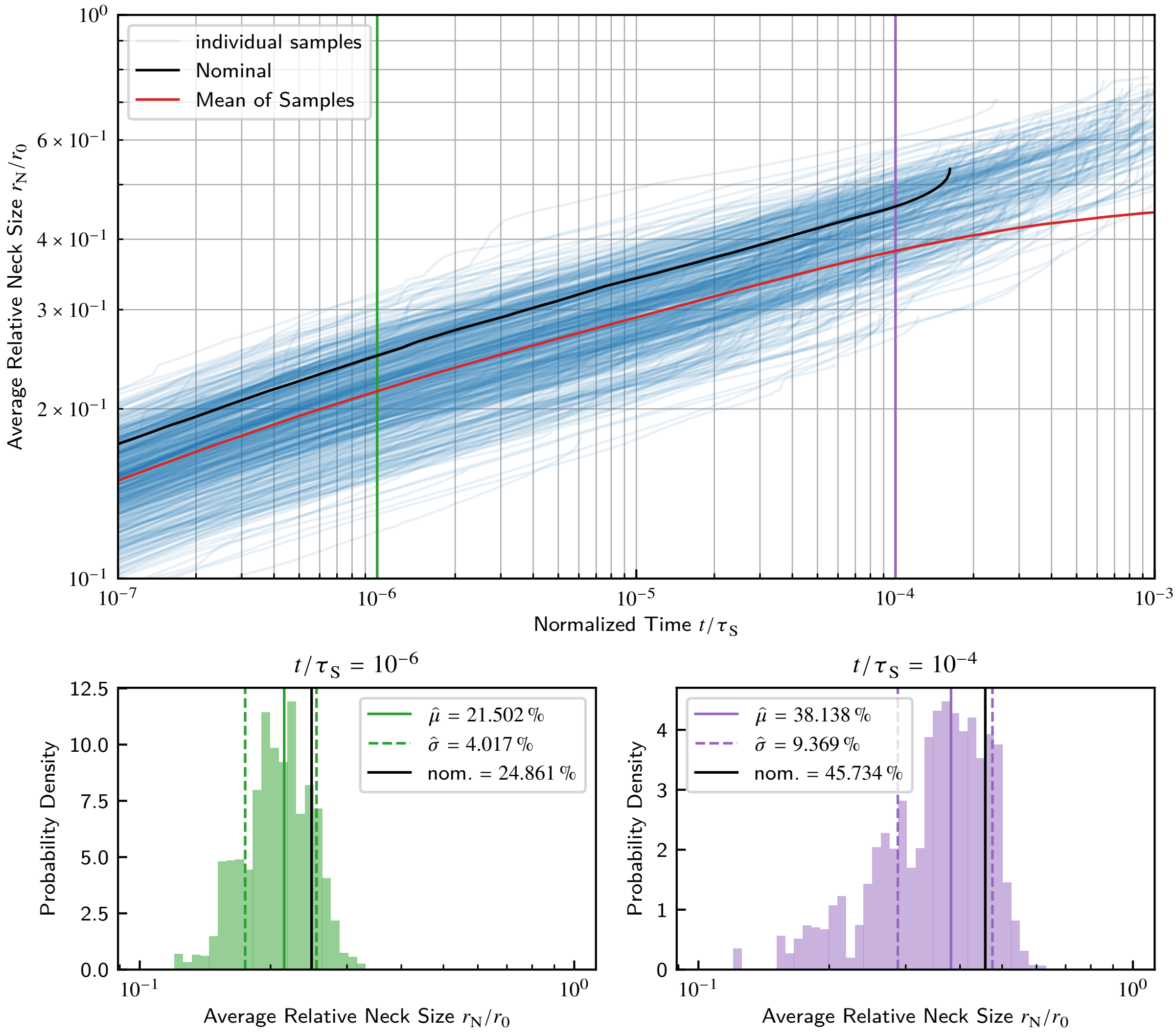


Figure 10.10: Neck Size of the First-Order Waves Case

with unequal sized particles feature first a steady state and then a sudden drop in neck size due to the consumption of the smaller particle [79, 74, 75]. Some heavily simplified models of two particles seem to show this behavior, too, but this is due to an accompanying grain growth model which increases the mean grain size and thus reduces the predicted sintering rate [31, 32].

The circular particles behave more uniformly in their pore closing compared to the waved ones. The behavior of the oval particles lies in between, but more similar to the circular ones. Pores formed initially between circular or oval particles are always of concave shape, as the particle surfaces between the necks are always convex. When considering first-order waves, the initial pore shape may be complex: being partially or fully concave and featuring narrow channels of varying width. Therefore, much more possible varieties of behavior are present in this case. This is suspected to be the reason of the less sudden transition into stationary state compared to the circular case, where the pore closing happens more concentrated in time.

In accordance with the results of section 9.1, the shape features' influences on shrinkage average out except for pore closing. Therefore, the shrinkage behavior is well described by the nominal case in the early stages. The validity of circular two-particle models was stated up to a neck size of $r_N/r_0 \approx 0.2$ [31]. This is supported by the current results.

Bjørk et al. [103] reported from kMCM results a decrease in densification when assuming normal and log-normal distributed particle sizes over monosized ones. The densification decreased further when increasing the standard deviation. In contrast, Termuhlen et al. [82] reported an increase in densification from 3D PFM simulation results, with a volume cell filled with randomly sized spherical particles in comparison to one with equally sized particles. The reasons for this contradiction are unclear. Both authors used linear timescale plots, so the early stages are not visually distinguishable. But for late stages, the results of Bjørk et al. [103] support the current findings.

The picture is different for the neck size results. The influence of circular particle size distribution is remarkable, and even higher under consideration of shape features. With regard on the results on the particle size ratio of section 9.1.3, the reason is suspected in the asymmetry of the particle size influence. The neck growth is affected mainly by the size of the smaller particle in a contact, not by the involved particles' average size. This introduces a bias in averaging, because small particles have higher weight in the average influence. So when taking the particle size distribution in consideration, the mean neck size achieved will be lower than that predicted by a model regarding particles of equal mean size. In accordance with the results on ovality of section 9.1.6, particle shape features behave like smaller or larger particles in regard on neck evolution, depending on the contact configuration. The therein introduced term of an apparent radius can be applied to first-order waves to.

In contrast to their shrinkage results, the neck size prediction of Termuhlen et al. [82] supports the current findings. They found a higher neck size for the monosized particles, although only at later stages. In early stages monosized and multisized particles were approximately equivalent. The reported microstructural state images suggest that the final state of their simulation is at far earlier time then in the current case, as only small necks and still connected pore networks are present.

A classic study of Johnson [153] states that for the usual diffusion mechanisms of surface, grain boundary and volume diffusion, the respective diffusion coefficients can be determined from experimental data of shrinkage and neck size when an appropriate sintering model is used. The present results on neck size show that one should pay special attention on the particle size and morphology of the used powder when conducting a procedure like that. One should either use near-spherical powders or use a model that correctly regards morphological influences. Shrinkage data, however, appears to be uncritical.

11. Conclusions and Outlook

A novel approach to model sintering behavior of two or multiple particles in 2D space has been developed which supports asymmetric geometry and substance conditions. The approach uses sharp interfaces and a finite difference approximation of the particle surfaces. The model was based on the TEP to obtain a concise and flexible formulation of the evolution equations. The model allows direct identification of physical interface properties due to the sharp interface formulation. Additional requirements such as contact conditions can be formulated as equality constraints and are implicitly ensured by the solution procedure.

The numerical behavior of the model has been examined to define feasible parameters for numerical routines such as time step control and remeshing procedures. The computational effort for time integration of the model equation can be heavily lowered by feasible time step control and rigorous remeshing procedures. In early stages of the simulation remeshing has to be done with care to keep the fine details of geometry, in later stages however remeshing can be aggressive to reduce the system size and accelerate the solution procedure. The latter is an significant improvement over state-of-the-art PFM approaches, which usually have larger system sizes and may not reduce the system size during the simulation. Although remeshing procedures are applied there, too, the discretization sizes are determined by the phase field gradient requirements, rather than the interface geometry as in the current case. So in the current case the requirements on discretization fineness decrease during the simulation, where in PFM they do not.

The model has then be used for extensive parameter studies on two-particle configurations. The varied parameters include classic ones like the ratio of surface energy to grain boundary energy or the ratio of surface and grain boundary diffusion rate, for which previous results have been confirmed and the new model has been validated. Other parameters, for which no previous results could be found in literature, have been investigated regarding the influence of asymmetric substance pairings and particle geometries. Explanations of the observed behavior have been attempted.

Some special crafted packings of more than two particles have been investigated regarding the characteristics of pore closing, without and with inert particles present. The distinct packings without inert particles were found to behave similarly until pore closing happens. The point of pore closing is dependent on the type of packing, respectively the size of the initial pore. Inert particles have direct retarding influence on shrinkage and neck evolution, but also keep the pore open so that the previously discussed effects happen later or are prevented at all.

The dimensionless parameter studies, as well as the investigations of crafted packings have lead to several general statements about the influence on shrinkage and neck size evolution during sintering. The results are summarized graphically in figure 11.1 for the shrinkage and figure 11.2 for the neck size.

In a fourth step, the behavior of powder mixtures has been investigated by simulating the sintering of random arrangements of three particles. The randomized simulations were found to incorporate the effects of subsequent pore closing and to be in accordance with models of large volume cells with high counts of particles. Shape features of the particles were found to have especially impact

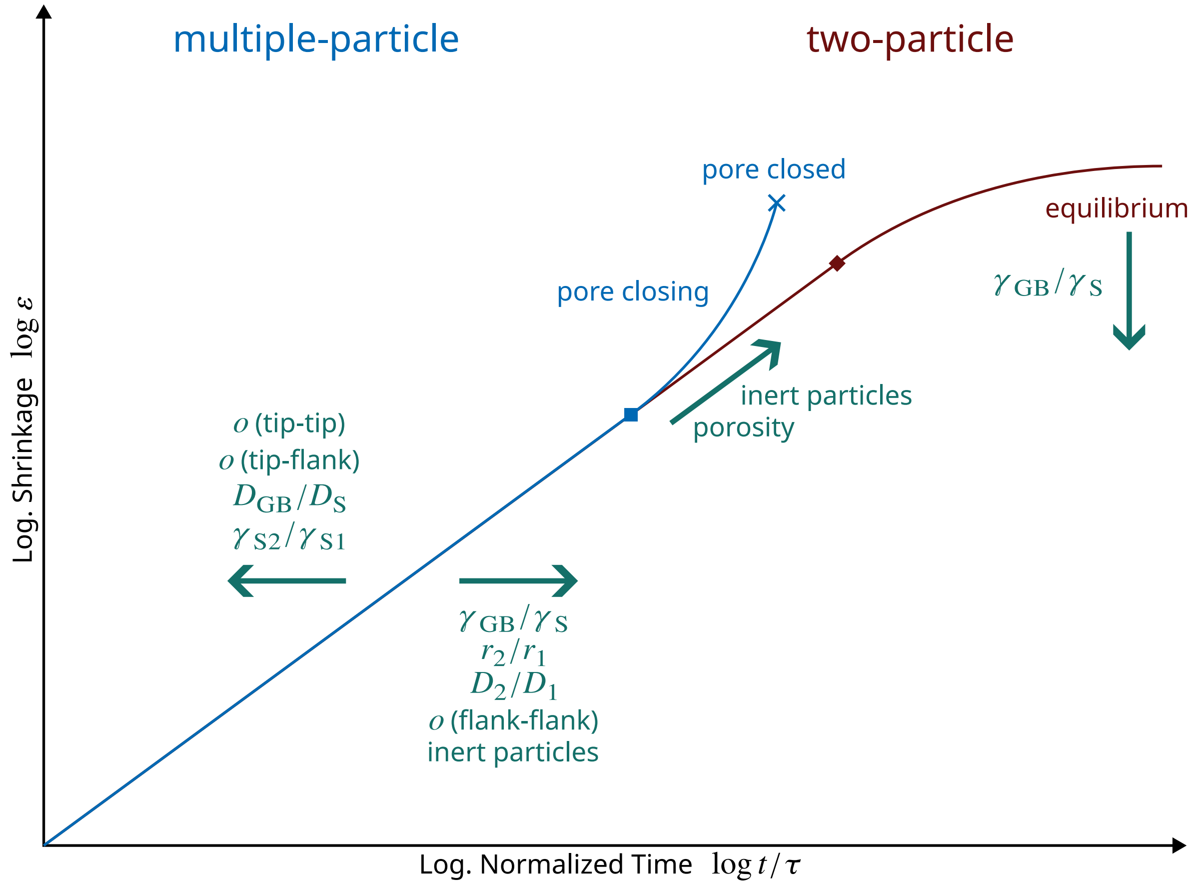


Figure 11.1: Influence of Dimensionless Parameters and Packing Properties on Shrinkage Summarized from Investigations in This Work

on the pore closing and therefore on the formation of the final stationary state. Findings from the parameter studies have been used to explain the observations in the randomized simulations. So, classic particle size averaging was found to introduce a bias regarding neck size prediction, as the influence of particle size on neck size is determined mainly by the smaller of the particles in contact.

In the current state of the model, several limitations have been observed during the investigations conducted:

1. Triple points where three grain boundaries meet are currently not supported. This limits the applicability till the closing of the first pore present in the system.
2. In contacts between particles with high difference in surface energy, a drastic undercut near the neck occurs as discussed in section 9.1.4. This eventually leads to a crossing between grain boundary and surface, which is obviously not physical. This is a numerical effect due to the removal of volume from a node, where in fact no volume is left between the respective surface and the grain boundary.
3. The grain boundary can currently only consist of a single node (plus the two neck nodes) as discussed in section 6.4, since otherwise the contact conditions create collinear rows in the system matrix.
4. Transfer between particles was neglected, which disallowed the occurrence of grain growth in the parameter studies of asymmetric geometry (sections 9.1.3 and 9.1.6) and the randomized

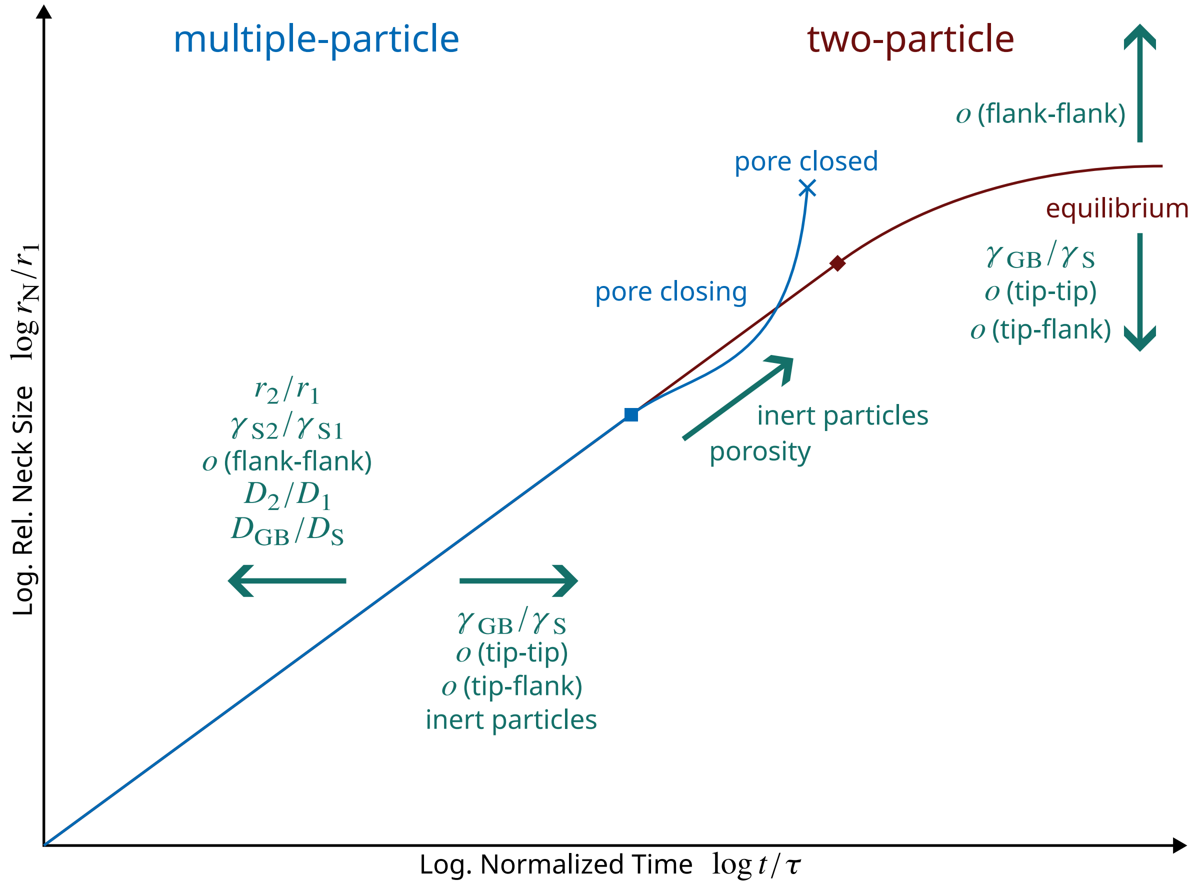


Figure 11.2: Influence of Dimensionless Parameters and Packing Properties on Neck Size Evolution Summarized from Investigations in This Work

simulations (chapter 10), but allowed to disregard concentration gradients within the particle volume for the case of unequal substances.

Beside resolving these, further development of the model can include:

1. Introduction of a multiscale approach that models fine-grained powder fractions as a solid continuum of viscoelastic behavior filling the pores with ability to apply a pressure on the large particles' surfaces and transfer mass to them.
2. Extension of the approach to 3D space.
3. Application of the randomized approach to development of powder mixtures with differing size, shape and material properties.
4. Transfer the approach to simulation of microstructure evolution of compact polycrystals regarding recrystallization, possibly also with phase transformations.

References

List of Abbreviations

CA cellular automaton 20, 22

CDF cumulative density function 25

CIT classical irreversible thermodynamics 15, 17

DEM discrete element method 24

EVS external variable of state 16, 27

FDM finite differences method 19, 20, 24, 33

FEM finite elements method 19, 24, 33

IDF internal degree of freedom 17, 44

IVS internal variable of state 15–17, 27, 44

kMCM kinetic Monte-Carlo-Method 22, 109, 111

LSM level-set method 20, 21, 33

MCM Monte-Carlo-Method 24–26, 33

PDF probability density function 25

PFM phase-field method 20, 22, 23, 33, 93, 111, 112

PRNG pseudo random number generator 25

RVE representative volume element 19, 34

TEP thermodynamic extremal principle 19, 27, 33, 53, 112

TIP theory of irreversible processes 15

List of Symbols

Symbol	Meaning
$\mathscr{L}$	Lagrangian functional
λ	Lagrangian multiplicator
$\bullet^{\mathrm{T}}$	transposed matrix
$\mathrm{\delta}$	step difference operator
Δ	difference operator
d	differential operator
∇	gradient operator
$\bar{\ }$	projected value
r	radius coordinate
φ	angle coordinate
x	horizontal coordinate
y	vertical coordinate
t	time
τ	characteristic sintering time
κ	curvature of a surface
V	volume
A	area
a	surface distance between nodes
α	angle between surface line and radius
β	angle between surface line and surface normal resp. tangent
s	shift (displacement) of a node or particle
ω	rotation
d	diameter
δ	thickness of an interface
d	arbitrary distance
T	absolute temperature
ϑ	celsius temperature
p	pressure
R	specific gas constant

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(Continued)

k_B	Boltzmann's constant
V_m	molar volume
Ω	atom volume
ϱ	mass density
j	flux
c	concentration
c_Va	vacancy concentration
D	diffusion coefficient
M	mobility coefficient
x	internal variable of state
$\dot{x}$	internal velocity of state
X	external variable of state
g	constraint to TEP
ξ	internal degree of freedom in TEP
De	Deborah number
n	amount of substance
G	Gibbs energy (free enthalpy)
g	specific Gibbs energy (free enthalpy)
G_m	molar Gibbs energy (free enthalpy)
F	Helmholtz energy (free energy)
f	specific Helmholtz energy (free energy)
F_m	molar Helmholtz energy (free energy)
S	entropy
s	specific entropy
S_m	molar entropy
γ	interface energy
$\mathscr{D}$	dissipation
$\mathscr{Q}$	dissipation function
μ	chemical potential
Q	activation energy
S	index for surface
GB	index for grain boundary
N	index for neck
u	index for upper neighbor node
l	index for lower neighbor node
M	index for matrix material

Continued on next page

(Continued)

C	index for contact
$\perp$	index for normal direction
$\parallel$	index for tangential direction
P	index for particle
$\circ$	index for standard conditions
$\vert^{\bullet}$	index specifying the coordinate system
eq	index for thermodynamic equilibrium
$\mathbb{N}$	set of nodes
$\mathbb{P}$	set of particles
$\mathbb{C}$	set of contacts
$\mathbb{Z}$	set of integer numbers
$\mathbb{Z}^+$	set of integer numbers
$\mathbb{R}$	set of integer numbers
σ	normal stress
μ	stochastic expectation
σ	standard deviation
σ^2	variance
$\hat{}$	estimation of the respected value
p	probability
P	cumulative probability
Γ	gamma function
o	ovality of particle
h	relative height of first order waves
n	count of first order waves
p	phase shift of first order waves
f	shape function
x	fractional amount in a powder mixture
N	particle count

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Bibliography

[1] R. M. German. *Sintering Theory and Practice*. New York: Wiley, 1996. 550 pp. ISBN: 978-0-471-05786-4.

[2] R. M. German. *Sintering: From Empirical Observations to Scientific Principles*. Amsterdam: Elsevier/Butterworth-Heinemann, 2014. 536 pp. ISBN: 978-0-12-401682-8.

[3] J. E. Geguzin. *Physik des Sinterns*. Leipzig: VEB Deutscher Verlag für Grundstoffindustrie, 1973.

[4] S. J. L. Kang. *Sintering*. Elsevier, 2005. ISBN: 978-0-7506-6385-4. DOI: 10.1016/b978-0-7506-6385-4.x5000-6.

[5] H. E. Exner. *Grundlagen von Sintervorgängen*. Ed. by G. Petzow. Materialkundlich-Technische Reihe. Stuttgart: Gebrüder Borntraeger, 1978. ISBN: 3-443-23005-9.

[6] Z. Z. Fang. *Sintering of Advanced Materials: Fundamentals and Processes*. Oxford: Woodhead, 2010. ISBN: 978-1-84569-562-0.

[7] M. N. Rahaman. *Ceramic Processing*. Taylor & Francis, 2017. ISBN: 978-1-4987-1641-3. DOI: 10.1201/9781315157160.

[8] G. Routschka and H. Wuthnow, eds. *Praxishandbuch feuerfeste Werkstoffe: Aufbau - Eigenschaften - Prüfung*. 6th ed. Prozesswärme. Essen: Vulkan, 2017. 383 pp. ISBN: 978-3-8027-3168-6.

[9] J. C. Fisher. "Calculation of Diffusion Penetration Curves for Surface and Grain Boundary Diffusion". In: *Journal of Applied Physics* 22.1 (1951), pp. 74–77. DOI: 10.1063/1.1699825.

[10] A. Fick. "Ueber Diffusion". In: *Annalen der Physik* 170.1 (1855), pp. 59–86. DOI: 10.1002/andp.18551700105.

[11] J. B. J. Fourier. *Théorie analytique de la chaleur*. Oeuvres de Fourier. Paris: Gauthier-Villars, 1888. URL: https://books.google.de/books?id=Ez-dvn6anL8C.

[12] D. W. Hoffman and J. W. Cahn. "A Vector Thermodynamics for Anisotropic Surfaces: I. Fundamentals and Application to Plane Surface Junctions". In: *Surface Science* 31 (1972), pp. 368–388. DOI: 10.1016/0039-6028(72)90268-3.

[13] A. H. King. "The Geometric and Thermodynamic Properties of Grain Boundary Junctions". In: *Interface Science* 7.3 (1999), pp. 251–271. DOI: 10.1023/A:1008769209265.

[14] S. R. de Groot and P. Mazur. *Non-Equilibrium Thermodynamics*. Reprint Dover Pub. 1984. North-Holland, 1962.

[15] G. A. Maugin. *The Thermomechanics of Nonlinear Irreversible Behaviors*. Singapore: World Scientific, 1999. 375 pp. ISBN: 978-981-02-3375-4.

[16] G. Lebon, D. Jou, and J. Casas-Vázquez. *Understanding Non-Equilibrium Thermodynamics: Foundations, Applications, Frontiers*. Berlin: Springer, 2008. 325 pp. ISBN: 978-3-540-74251-7.

[17] G. A. Maugin and W. Muschik. “Thermodynamics with Internal Variables. Part I. General Concepts”. In: *Journal of Non-Equilibrium Thermodynamics* 19.3 (1994). DOI: 10.1515/jnet.1994.19.3.217.

[18] J. Bataille and J. Kestin. “Irreversible Processes and Physical Interpretation of Rational Thermodynamics”. In: *Journal of Non-Equilibrium Thermodynamics* 4.4 (1979), pp. 229–258. DOI: 10.1515/jnet.1979.4.4.229.

[19] J. Kestin. “A Note on the Relation Between the Hypothesis of Local Equilibrium and the Clausius-Duhem Inequality”. In: *Journal of Non-Equilibrium Thermodynamics* 15.3 (1990), pp. 193–212. DOI: 10.1515/jnet.1990.15.3.193.

[20] j. Kestin. “Local-Equilibrium Formalism Applied to Mechanics of Solids”. In: *International Journal of Solids and Structures* 29.14 (1992), pp. 1827–1836. DOI: 10.1016/0020-7683(92)90174-R.

[21] W. Muschik. “Internal Variables in Non-Equilibrium Thermodynamics”. In: *Journal of Non-Equilibrium Thermodynamics* 15.2 (1990), pp. 127–137. DOI: 10.1515/jnet.1990.15.2.127.

[22] G. A. Maugin and W. Muschik. “Thermodynamics with Internal Variables. Part II. Applications”. In: 19.3 (1994), pp. 250–289. DOI: 10.1515/jnet.1994.19.3.250.

[23] G. A. Maugin. “The Saga of Internal Variables of State in Continuum Thermo-Mechanics (1893–2013)”. In: *Mechanics Research Communications* 69 (2015), pp. 79–86. DOI: 10.1016/j.mechrescom.2015.06.009.

[24] G. C. Kuczynski. “Self-Diffusion in Sintering of Metallic Particles”. In: *Transactions of the AIME* 85 (1949), pp. 169–178. DOI: 10.1007/978-94-009-0741-6_33.

[25] D. L. Johnson and I. B. Cutler. “Diffusion Sintering: I, Initial Stage Sintering Models and Their Application to Shrinkage of Powder Compacts”. In: *Journal of the American Ceramic Society* 46.11 (1963), pp. 541–545. DOI: 10.1111/j.1151-2916.1963.tb14606.x.

[26] W. D. Kingery and M. Berg. “Study of the Initial Stages of Sintering Solids by Viscous Flow, Evaporation-Condensation, and Self-Diffusion”. In: *Journal of Applied Physics* 26.10 (1955), pp. 1205–1212. DOI: 10.1063/1.1721874.

[27] W. S. Coblenz et al. “Initial Stage Solid State Sintering Models - A Critical Analysis and Assessment”. In: *Materials Science Research* 13 (1980), pp. 141–157. DOI: 10.1007/978-1-4899-5301-8_12.

[28] R. M. German. “The Emergence of Quantitative Sintering Theory from 1945 to 1955”. In: *Journal of Metals* 69.4 (2017), pp. 630–634. DOI: 10.1007/s11837-016-2242-1.

[29] R. L. Coble. “Sintering Crystalline Solids. I. Intermediate and Final State Diffusion Models”. In: *Journal of Applied Physics* 32.5 (1961), pp. 787–792. DOI: 10.1063/1.1736107.

[30] F. B. Swinkels and M. F. Ashby. “A Second Report on Sintering Diagrams”. In: *Acta Metallurgica* 29.2 (1981), pp. 259–281. DOI: 10.1016/0001-6160(81)90154-1.

[31] S.-J. L. Kang and Y.-Il Jung. “Sintering Kinetics at Final Stage Sintering: Model Calculation and Map Construction”. In: *Acta Materialia* 52.15 (2004), pp. 4573–4578. DOI: 10.1016/j.actamat.2004.06.015.

[32] D. Gouvêa. “Thermodynamic of Solid-State Sintering: Contributions of Grain Boundary Energy”. In: *Journal of the European Ceramic Society* 44.14 (2024), p. 116677. DOI: 10.1016/j.jeurceramsoc.2024.116677.

[33] E. A. Olevsky. “Theory of Sintering: From Discrete to Continuum”. In: *Materials Science and Engineering R: Reports* 23.2 (1998), pp. 41–100. DOI: 10.1016/s0927-796x(98)00009-6.

[34] A. Safonov et al. “Numerical Simulation of Sintering for 3D-printed Ceramics via SOVS Model”. In: *Ceramics International* 45.15 (2019), pp. 19027–19035. DOI: 10.1016/j.ceramint.2019.06.144.

[35] T. Kraft and H. Riedel. “Numerical Simulation of Solid State Sintering – Model and Application”. In: *Journal of the European Ceramic Society* 24.2 (2004), pp. 345–361. DOI: 10.1016/s0955-2219(03)00222-x.

[36] A. Al-Qudsi et al. “Comparison between Different Numerical Models of Densification during Solid-State Sintering of Pure Aluminium Powder”. In: *Production Engineering* (2014). DOI: 10.1007/s11740-014-0574-7.

[37] H. Shi et al. “Efficient Modelling of Ceramic Sintering Processes: Application to Bilayers and Membranes”. In: *Journal of the European Ceramic Society* 43.11 (2023), pp. 4939–4949. DOI: 10.1016/j.jeurceramsoc.2023.03.053.

[38] J. Svoboda and H. Riedel. “New Solutions Describing the Formation of Interparticle Necks in Solid-State Sintering”. In: *Acta Metallurgica et Materialia* 43.1 (1995), pp. 1–10. DOI: 10.1016/0956-7151(95)90255-4.

[39] D. Bouvard and R. M. McMeeking. “Deformation of Interparticle Necks by Diffusion-Controlled Creep”. In: *Journal of the American Ceramic Society* 79.3 (1996), pp. 666–672. DOI: 10.1111/j.1151-2916.1996.tb07927.x.

[40] Z. S. Nikolic. “Computer Simulation of Grain Growth by Grain Boundary Migration during Liquid Phase Sintering”. In: *Journal of Materials Science* 34.4 (1999), pp. 783–794. DOI: 10.1023/a:1004577014266.

[41] A. Jagota and P. R. Dawson. “Micromechanical Modeling of Powder Compacts—I. Unit Problems for Sintering and Traction Induced Deformation”. In: *Acta Metallurgica* 36.9 (1988), pp. 2551–2561. DOI: 10.1016/0001-6160(88)90200-3.

[42] J. Léchelle et al. “A Sub-Granular Scale Model for Solid State Free Sintering: Results on the Evolution of Two Grains”. In: *Journal of Chemical Technology and Metallurgy* 49.3 (2014), pp. 263–274. URL: https://hal.science/hal-01265013v1.

[43] J. Svoboda and H. Riedel. “Pore-Boundary Interactions and Evolution Equations for the Porosity and the Grain Size during Sintering”. In: *Acta Metallurgica et Materialia* 40.11 (1992), pp. 2829–2840. DOI: 10.1016/0956-7151(92)90448-n.

[44] F. Wakai et al. “Sintering Forces Acting among Particles during Sintering by Grain-Boundary/Surface Diffusion”. In: *Journal of the American Ceramic Society* (2018), pp. 1–10. DOI: 10.1111/jace.15716.

[45] L. Klinger and E. Rabkin. “Sintering of Spherical Particles of Two Immiscible Phases Controlled by Surface and Interphase Boundary Diffusion”. In: *Acta Materialia* 61.7 (2013), pp. 2607–2616. DOI: 10.1016/j.actamat.2013.01.040.

[46] L. Klinger and E. Rabkin. “Capillary-Driven Motion of Nanoparticles Attached to Curved Rigid Substrates”. In: *Acta Materialia* 60.17 (2012), pp. 6065–6075. DOI: 10.1016/j.actamat.2012.07.043.

[47] L. Klinger and E. Rabkin. “Shape Evolution by Surface and Interface Diffusion with Rigid Body Rotations”. In: *Acta Materialia* 59.17 (2011), pp. 6691–6699. DOI: 10.1016/j.actamat.2011.07.026.

[48] K. A. Brakke. “The Surface Evolver”. In: *Experimental Mathematics* 1.2 (1992), pp. 141–165. DOI: 10.1080/10586458.1992.10504253.

[49] F. Wakai and G. Okuma. “Rigid Body Motion of Multiple Particles in Solid-State Sintering”. In: *Acta Materialia* 235 (2022), p. 118092. DOI: 10.1016/j.actamat.2022.118092.

[50] F. Wakai and K. A. Brakke. “Mechanics of Sintering for Coupled Grain Boundary and Surface Diffusion”. In: *Acta Materialia* 59.14 (2011), pp. 5379–5387. DOI: 10.1016/j.actamat.2011.05.006.

[51] S. Osher and R. P. Fedkiw. “Level Set Methods: An Overview and Some Recent Results”. In: *Journal of Computational Physics* 169.2 (200), pp. 463–502. DOI: 10.1006/jcph.2000.6636.

[52] F. Gibou, R. Fedkiw, and S. Osher. “A Review of Level-Set Methods and Some Recent Applications”. In: *Journal of Computational Physics* 353 (2018), pp. 82–109. DOI: 10.1016/j.jcp.2017.10.006.

[53] N. P. van Dijk et al. “Level-Set Methods for Structural Topology Optimization: A Review”. In: *Structural and Multidisciplinary Optimization* 48.3 (2013), pp. 437–472. DOI: 10.1007/s00158-013-0912-y.

[54] J. Bruchon et al. “3D Simulation of the Matter Transport by Surface Diffusion within a Level-Set Context”. In: *European Journal of Computational Mechanics* 19.1–3 (2010), pp. 281–292. DOI: 10.3166/ejcm.19.281-292.

[55] J. Bruchon et al. “Finite Element Simulation of Mass Transport During Sintering of a Granular Packing. Part I. Surface and Lattice Diffusions”. In: *Journal of the American Ceramic Society* 95.8 (2012). Ed. by D. Bouvard, pp. 2398–2405. DOI: 10.1111/j.1551-2916.2012.05073.x.

[56] D. Pino Muñoz et al. “Direct 3D Simulation of Powder Sintering by Surface and Volume Diffusion”. In: *Key Engineering Materials* 554–557 (2013), pp. 714–723. DOI: 10.4028/www.scientific.net/KEM.554-557.714.

[57] L.-Q. Chen and Y. Zhao. “From Classical Thermodynamics to Phase-Field Method”. In: *Progress in Materials Science* 124 (2022), p. 100868. DOI: 10.1016/j.pmatsci.2021.100868.

[58] R. S. Qin and H. K. Bhadeshia. “Phase Field Method”. In: *Materials Science and Technology* 26.7 (2010), pp. 803–811. DOI: 10.1179/174328409X453190.

[59] L.-Q. Chen. “Phase-Field Models for Microstructure Evolution”. In: *Annual Review of Materials Science* 32 (2002), pp. 113–140. DOI: 10.1146/annurev.matsci.32.112001.132041.

[60] Ingo Steinbach. “Phase-Field Models in Materials Science”. In: *Modelling and Simulation in Materials Science and Engineering* 17 (2009), p. 073001. DOI: 10.1088/0965-0393/17/7/073001.

[61] I. Steinbach. “Phase-Field Model for Microstructure Evolution at the Mesoscopic Scale”. In: *Annual Review of Materials Research* 43 (2013), pp. 89–107. DOI: 10.1146/annurev-matsci-071312-121703.

[62] Y. Wang and J. Li. “Phase Field Modeling of Defects and Deformation”. In: *Acta Materialia* 58.4 (2010), pp. 1212–1235. DOI: 10.1016/j.actamat.2009.10.041.

[63] M. R. Tonks and L. K. Aagesen. “The Phase Field Method: Mesoscale Simulation Aiding Material Discovery”. In: *Annual Review of Materials Research* 49 (2019), pp. 79–102. DOI: 10.1146/annurev-matsci-070218-010151.

[64] S. M. Allen and J. W. Cahn. “A Microscopic Theory for Antiphase Boundary Motion and Its Application to Antiphase Domain Coarsening”. In: *Acta Metallurgica* 27.6 (1979), pp. 1085–1095. DOI: 10.1016/0001-6160(79)90196-2.

[65] J. W. Cahn. “On Spinodal Decomposition”. In: *Acta Metallurgica* 9.9 (1961), pp. 795–801. DOI: 10.1016/0001-6160(61)90182-1.

[66] J. Deng. “A Phase Field Model of Sintering with Direction-Dependent Diffusion”. In: *Materials Transactions* 53.2 (2012), pp. 385–389. DOI: 10.2320/matertrans.m2011317.

[67] Y. Yang et al. “Investigation on Temperature-Gradient-Driven Effects in Unconventional Sintering via Non-Isothermal Phase-Field Simulation”. In: *Scripta Materialia* 186 (2020), pp. 152–157. DOI: 10.1016/j.scriptamat.2020.05.016.

[68] Y. U. Wang. “Computer Modeling and Simulation of Solid-State Sintering: A Phase Field Approach”. In: *Acta Materialia* 54.4 (2006), pp. 953–961. DOI: 10.1016/j.actamat.2005.10.032.

[69] K. Asp and J. Ågren. “Phase-Field Simulation of Sintering and Related Phenomena – A Vacancy Diffusion Approach”. In: *Acta Materialia* 54.5 (2006), pp. 1241–1248. DOI: 10.1016/j.actamat.2005.11.005.

[70] X. N. Jing, J. H. Zhao, and L. H. He. “2D Aggregate Evolution in Sintering Due to Multiple Diffusion Approaches”. In: *Materials Chemistry and Physics* 80.3 (2003), pp. 595–598. DOI: 10.1016/s0254-0584(03)00102-0.

[71] V. Kumar, Z. Z. Fang, and P. C. Fife. “Phase Field Simulations of Grain Growth during Sintering of Two Unequal-Sized Particles”. In: *Materials Science and Engineering A* 528.1 (2010), pp. 254–259. DOI: 10.1016/j.msea.2010.08.061.

[72] V. Kumar. “Simulations and Modeling of Unequal Sized Particles Sintering”. PhD thesis. University of Utah, 2011. URL: https://www.proquest.com/dissertations-theses/simulations-modeling-unequal-sized-particles/docview/864832939/se-2?accountid=17183.

[73] K. Shinagawa. “Simulation of Grain Growth and Sintering Process by Combined Phase-Field / Discrete-Element Method”. In: *Acta Materialia* 66 (2014), pp. 360–369. DOI: 10.1016/j.actamat.2013.11.023.

[74] S. Biswas et al. “A Study of the Evolution of Microstructure and Consolidation Kinetics during Sintering Using a Phase Field Modeling Based Approach”. In: *Extreme Mechanics Letters* 7 (2016), pp. 78–89. DOI: 10.1016/j.eml.2016.02.017.

[75] S. Biswas, D. Schwen, and V. Tomar. “Implementation of a Phase Field Model for Simulating Evolution of Two Powder Particles Representing Microstructural Changes during Sintering”. In: *Journal of Materials Science* 53.8 (2017), pp. 5799–5825. DOI: 10.1007/s10853-017-1846-3.

[76] N. Moelans, B. Blanpain, and P. Wollants. “An Introduction to Phase-Field Modeling of Microstructure Evolution”. In: *Calphad* 32.2 (2008), pp. 268–294. DOI: 10.1016/j.calphad.2007.11.003.

[77] K. Chockalingam et al. “2D Phase Field Modeling of Sintering of Silver Nanoparticles”. In: *Computer Methods in Applied Mechanics and Engineering* 312 (2016), pp. 492–508. DOI: 10.1016/j.cma.2016.07.002.

[78] D. Choudhuri and L. Blake. “Particle Curvature Effects on Microstructural Evolution during Solid-State Sintering: Phenomenological Insights from Phase-Field Simulations”. In: *Journal of Materials Science* 56.12 (2021), pp. 7474–7493. DOI: 10.1007/s10853-021-05802-8.

[79] K. Ahmed et al. “Phase Field Modeling of the Effect of Porosity on Grain Growth Kinetics in Polycrystalline Ceramics”. In: *Modelling and Simulation in Materials Science and Engineering* 21.6 (2013), p. 065005. DOI: 10.1088/0965-0393/21/6/065005.

[80] J. Hötzer et al. “Phase-Field Simulation of Solid State Sintering”. In: *Acta Materialia* 164 (2019), pp. 184–195. DOI: 10.1016/j.actamat.2018.10.021.

[81] R. Zhang et al. “Thermodynamic Consistent Phase Field Model for Sintering Process with Multiphase Powders”. In: *Transactions of Nonferrous Metals Society of China* 24.3 (2014), pp. 783–789. DOI: 10.1016/s1003-6326(14)63126-5.

[82] R. Termuhlen et al. “Three-Dimensional Phase Field Sintering Simulations Accounting for the Rigid-Body Motion of Individual Grains”. In: *Computational Materials Science* 186 (2021), p. 109963. DOI: 10.1016/j.commatsci.2020.109963.

[83] V. Ivannikov et al. “Capturing Shrinkage and Neck Growth with Phase Field Simulations of the Solid State Sintering”. In: *Modelling and Simulation in Materials Science and Engineering* 29.7 (2021), p. 075008. DOI: 10.1088/1361-651X/ac1f87.

[84] I. Greenquist et al. “Development of a Microstructural Grand Potential-Based Sintering Model”. In: *Computational Materials Science* 172 (2020), p. 109288. DOI: 10.1016/j.commatsci.2019.109288.

[85] M. R. Tonks et al. “An Object-Oriented Finite Element Framework for Multiphysics Phase Field Simulations”. In: *Computational Materials Science* 51.1 (2012), pp. 20–29. DOI: 10.1016/j.commatsci.2011.07.028.

[86] W. M. Feng et al. “Spectral Implementation of an Adaptive Moving Mesh Method for Phase-Field Equations”. In: *Journal of Computational Physics* 220.1 (2006), pp. 498–510. DOI: 10.1016/j.jcp.2006.07.013.

[87] S. Burke, C. Ortner, and E. Süli. “An Adaptive Finite Element Approximation of a Variational Model of Brittle Fracture”. In: *SIAM Journal on Numerical Analysis* 48.3 (2010), pp. 980–1012. DOI: 10.1137/080741033.

[88] M. Artina et al. “Anisotropic Mesh Adaptation for Crack Detection In Brittle Materials”. In: *SIAM Journal on Scientific Computing* 37.4 (2015), B633–B659. DOI: 10.1137/140970495.

[89] N. Ferro, S. Micheletti, and S. Perotto. “Anisotropic Mesh Adaptation for Crack Propagation Induced by a Thermal Shock in 2D”. In: *Computer Methods in Applied Mechanics and Engineering* 331 (2018), pp. 138–158. DOI: 10.1016/j.cma.2017.11.024.

[90] S. Micheletti, S. Perotto, and M. Signorini. “Anisotropic Mesh Adaptation for the Generalized Ambrosio–Tortorelli Functional with Application to Brittle Fracture”. In: *Computers & Mathematics with Applications* 75.6 (2018), pp. 2134–2152. DOI: 10.1016/j.camwa.2017.08.009.

[91] Hirshikesh et al. “Adaptive Phase Field Method for Quasi-Static Brittle Fracture Using a Recovery Based Error Indicator and Quadtree Decomposition”. In: *Engineering Fracture Mechanics* 220 (2019). DOI: 10.1016/j.engfracmech.2019.106599.

[92] S. Goswami, C. Anitescu, and T. Rabczuk. “Adaptive Phase Field Analysis with Dual Hierarchical Meshes for Brittle Fracture”. In: *Engineering Fracture Mechanics* 218 (2019). DOI: 10.1016/j.engfracmech.2019.106608.

[93] T. Heister, M. F. Wheeler, and T. Wick. “A Primal-Dual Active Set Method and Predictor-Corrector Mesh Adaptivity for Computing Fracture Propagation Using a Phase-Field Approach”. In: *Computer Methods in Applied Mechanics and Engineering* 290 (2015), pp. 466–495. DOI: 10.1016/j.cma.2015.03.009.

[94] H. Badnava et al. "An H-Adaptive Thermo-Mechanical Phase Field Model for Fracture". In: *Finite Elements in Analysis and Design* 138 (2018), pp. 31–47. DOI: 10.1016/j.finel.2017.09.003.

[95] F. Tian et al. "A Hybrid Adaptive Finite Element Phase-Field Method for Quasi-Static and Dynamic Brittle Fracture". In: *International Journal for Numerical Methods in Engineering* 120.9 (2019), pp. 1108–1125. DOI: 10.1002/nme.6172.

[96] M. Klinsmann et al. "An Assessment of the Phase Field Formulation for Crack Growth". In: *Computer Methods in Applied Mechanics and Engineering* 294 (2015), pp. 313–330. DOI: 10.1016/j.cma.2015.06.009.

[97] C. Lemieux. *Monte Carlo and Quasi-Monte Carlo Sampling*. New York: Springer, 2009. ISBN: 978-0-387-78165-5. DOI: 10.1007/978-0-387-78165-5.

[98] M. Braginsky, V. Tikare, and E. Olevsky. "Numerical Simulation of Solid State Sintering". In: *International Journal of Solids and Structures* 42.2 (2005), pp. 621–636. DOI: 10.1016/j.ijsolstr.2004.06.022.

[99] A. Luque et al. "Geometrical Monte Carlo Model of Liquid-Phase Sintering". In: *Mathematics and Computers in Simulation* 80.7 (2010), pp. 1469–1486. DOI: 10.1016/j.matcom.2009.09.014.

[100] V. Tikare, M. Braginsky, and E. A. Olevsky. "Numerical Simulation of Solid-State Sintering: I, Sintering of Three Particles". In: *Journal of the American Ceramic Society* 86.1 (2003), pp. 49–53. DOI: 10.1111/j.1151-2916.2003.tb03276.x.

[101] X. Wang and A. Atkinson. "Combining Densification and Coarsening in a Cellular Automata-Monte-Carlo Simulation of Sintering: Methodology and Calibration". In: *Computational Materials Science* 143 (2018), pp. 338–349. DOI: 10.1016/j.commatsci.2017.11.023.

[102] M. Matsumoto and T. Nishimura. "Mersenne Twister: A 623-Dimensionally Equidistributed Uniform Pseudo-Random Number Generator". In: *ACM Transactions on Modeling and Computer Simulation* 8.1 (1998), pp. 3–30. DOI: 10.1145/272991.272995.

[103] R. Bjørk et al. "The Effect of Particle Size Distributions on the Microstructural Evolution During Sintering". In: *Journal of the American Ceramic Society* 96.1 (2012). Ed. by J. Blendell, pp. 103–110. DOI: 10.1111/jace.12100.

[104] V. Tikare et al. "Numerical Simulation of Microstructural Evolution during Sintering at the Mesoscale in a 3D Powder Compact". In: *Computational Materials Science* 48.2 (2010), pp. 317–325. DOI: 10.1016/j.commatsci.2010.01.013.

[105] A. Luque et al. "Simulation of the Microstructural Evolution during Liquid Phase Sintering Using a Geometrical Monte Carlo Model". In: *Modelling and Simulation in Materials Science and Engineering* 13.7 (2005), pp. 1057–1070. DOI: 10.1088/0965-0393/13/7/004.

[106] J. Svoboda and I. Turek. "On Diffusion-Controlled Evolution of Closed Solid-State Thermodynamic Systems at Constant Temperature and Pressure". In: *Philosophical Magazine B* 64.6 (1991), pp. 749–759. DOI: 10.1080/13642819108207635.

[107] H. Ziegler. *An Introduction to Thermomechanics*. New York: North Holland, 1983.

[108] H. Ziegler. "Some Extremum Principles in Irreversible Thermodynamics, with Application to Continuum Mechanics". In: *Progress in Solid Mechanics* 4 (1963), p. 91.

[109] L. Onsager. "Reciprocal Relations in Irreversible Processes. I." In: *Physical Review* 37.4 (1931), pp. 405–426. DOI: 10.1103/PhysRev.37.405.

[110] F. D. Fischer, J. Svoboda, and H. Petryk. “Thermodynamic Extremal Principles for Irreversible Processes in Materials Science”. In: *Acta Materialia* 67 (2014), pp. 1–20. DOI: 10.1016/j.actamat.2013.11.050.

[111] K. Hackl, F. D. Fischer, and J. Svoboda. “Constraints in Thermodynamic Extremal Principles for Non-Local Dissipative Processes”. In: *Continuum Mechanics and Thermodynamics* 32.5 (2020), pp. 1337–1345. DOI: 10.1007/s00161-019-00846-3.

[112] W. D. Kingery. “Metal-Ceramic Interactions:IV, Absolute Measurement of Metal-Ceramic Interfacial Energy and the Interfacial Adsorption of Silicon from Iron-Silicon Alloys”. In: *Journal of the American Ceramic Society* 37.2 (1954), pp. 42–45. DOI: 10.1111/j.1151-2916.1954.tb14002.x.

[113] P. Nikolopoulos. “Surface, Grain-Boundary and Interfacial Energies in Al2O3 and Al2O3-Sn, Al2O3-Co Systems”. In: *Journal of Materials Science* 20.11 (1985), pp. 3993–4000. DOI: 10.1007/BF00552390.

[114] S. V. Radcliffe. “The Surface Energy of Solid Niobium”. In: *Journal of the Less Common Metals* 3.5 (1961), pp. 360–366. DOI: 10.1016/0022-5088(61)90011-X.

[115] E. N. Hodkin, M. G. Nicholas, and D. M. Poole. “The Surface Energies of Solid Molybdenum, Niobium, Tantalum and Tungsten”. In: *Journal of the Less Common Metals* 20.2 (1970), pp. 93–103. DOI: 10.1016/0022-5088(70)90093-7.

[116] W. M. Robertson and R. Chang. “The Kinetics of Grain-Boundary Groove Growth on Alumina Surfaces”. In: *The Role of Grain Boundaries and Surfaces in Ceramics*. Ed. by W. W. Kriegel and H. Palmour. Boston: Springer, 1966, pp. 49–60. ISBN: 978-1-4899-6162-4 978-1-4899-6311-6. DOI: 10.1007/978-1-4899-6311-6_4.

[117] T. Maruyama and W. Komatsu. “Surface Diffusion of Single-Crystal Al2O3 by Scratch-Smoothing Method”. In: *Journal of the American Ceramic Society* 58.7–8 (1975), pp. 338–339. DOI: 10.1111/j.1151-2916.1975.tb11494.x.

[118] J. F. Shackelford and W. D. Scott. “Relative Energies of Tilt Boundaries in Aluminum Oxide”. In: *Journal of the American Ceramic Society* 51.12 (1968), pp. 688–692. DOI: 10.1111/j.1151-2916.1968.tb15929.x.

[119] R. L. Coble. “Initial Sintering of Alumina and Hematite”. In: *Journal of the American Ceramic Society* 41.2 (1958), pp. 55–62. DOI: 10.1111/j.1151-2916.1958.tb13519.x.

[120] G. C. Kuczynski, L. Abernethy, and J. Allan. “Sintering Mechanisms of Aluminum Oxide”. In: *Kinetics of High-Temperature Processes*. Kinetics of High-Temperature Processes. Ed. by W. D. Kingery. MIT Press, 1959, pp. 163–172. ISBN: 978-0-262-31081-9. DOI: 10.7551/mitpress/4061.003.0029.

[121] A. R. Gaddipati and W. D. Scott. “Surface Mass Transport of Alumina”. In: *Journal of Materials Science* 21.2 (1986), pp. 419–423. DOI: 10.1007/bf01145503.

[122] T. K. Gupta. “Instability of Cylindrical Voids in Alumina”. In: *Journal of the American Ceramic Society* 61.5–6 (1978), pp. 191–195. DOI: 10.1111/j.1151-2916.1978.tb09276.x.

[123] R. M. Cannon and R. L. Coble. “Review of Diffusional Creep of Al2O3”. In: *Deformation of Ceramic Materials*. Deformation of Ceramic Materials. Ed. by R. C. Bradt and R. E. Tressler. Boston: Springer, 1975, pp. 61–100. ISBN: 978-1-4613-4433-9 978-1-4613-4431-5. DOI: 10.1007/978-1-4613-4431-5_3.

[124] B. C. Allen. “The Surface Self-Diffusion of Mo, Cb (Nb), and Re”. In: *Metallurgical Transactions* 3.9 (1972), pp. 2544–2547. DOI: 10.1007/BF02647068.

[125] G. A. Odishariya. “Investigation of Surface Self-Diffusion of Niobium in a Strong Electric Field by Means of Field-Emission Microscope”. In: *Sov. Phys.-Solid State (Engl. Transl.)* (1967). URL: https://www.osti.gov/biblio/4826859.

[126] S. Hok and M. Drechsler. “A Measurement of the Surface Self-Diffusion of Tantalum”. In: *Surface Science* 107.2–3 (1981), pp. L362–L366. DOI: 10.1016/0039-6028(81)90527-6.

[127] P. C. Bettler, D. H. Bennum, and C. M. Case. “Effect of Impurities on Surface Self-Diffusion and Surface Structure”. In: *Surface Science* 44.2 (1974), pp. 360–376. DOI: 10.1016/0039-6028(74)90123-X.

[128] T. Zienert et al. “Characterisation of Sintered Niobium-Alumina Refractory Composite Granules Synthesised by Castable Technology”. In: *Advanced Engineering Materials* (2022), p. 2200407. DOI: 10.1002/adem.202200407.

[129] T. Zienert et al. “Coarse-Grained Refractory Composite Castables Based on Alumina and Niobium”. In: *Advanced Engineering Materials* (2022), p. 2200296. DOI: 10.1002/adem.202200296.

[130] E. Storti et al. “Metal-Ceramic Beads Based on Niobium and Alumina Produced by Alginate Gelation”. In: *Materials* 14.19 (19 2021), p. 5483. DOI: 10.3390/ma14195483.

[131] E. Storti et al. “Full and Hollow Metal–Ceramic Beads Based on Tantalum and Alumina Produced by Alginate Gelation”. In: *Advanced Engineering Materials* 24.8 (2022), p. 2200381. DOI: 10.1002/adem.202200381.

[132] M. K. Eusterholz et al. “High-temperature Ternary Oxide Phases in Tantalum/Niobium–Alumina Composite Materials”. In: *Advanced Engineering Materials* 24.8 (2022), p. 2200161. DOI: 10.1002/adem.202200161.

[133] M. K. Eusterholz et al. “Nanoscale Oxide Formation at α-Al2O3-Nb Interfaces”. In: *Advanced Engineering Materials* 25.14 (2023), p. 2201441. DOI: 10.1002/adem.202201441.

[134] M. K. Eusterholz. “Grenzflächenreaktionen in refraktären Metall-Keramik-Verbundwerkstoffen”. PhD thesis. Karlsruher Institut für Technologie (KIT), 2024. URL: https://publikationen.bibliothek.kit.edu/1000172647.

[135] D. Prot and C. Monty. “Self-Diffusion in α-Al2O3. II. Oxygen Diffusion in ’undoped’ Single Crystals”. In: *Philosophical Magazine A* 73.4 (1996), pp. 899–917. DOI: 10.1080/01418619608243695.

[136] R. H. Doremus. “Diffusion in Alumina”. In: *Journal of Applied Physics* 100.10 (2006), p. 101301. DOI: 10.1063/1.2393012.

[137] A. E. Paladino and R. L. Coble. “Effect Of Grain Boundaries on Diffusion-Controlled Processes in Aluminum Oxide”. In: *Journal of the American Ceramic Society* 46.3 (1963), pp. 133–136. DOI: 10.1111/j.1151-2916.1963.tb11696.x.

[138] M. Weiner. *RefraSin*. Version 1.0. Institute of Metal Forming, TU Bergakademie Freiberg, 2025. URL: https://github.com/institute-of-metal-forming/refrasin.

[139] T. A. Davis. *Direct Methods for Sparse Linear Systems*. Fundamentals of Algorithms 2. Philadelphia: Society for Industrial and Applied Mathematics, 2006. ISBN: 978-0-89871-613-9 978-0-89871-888-1. DOI: 10.1137/1.9780898718881.

[140] C. Woltering. *CSparse.NET*. Version 4.2.0. 2024. URL: https://github.com/wo80/CSparse.NET.

[141] “Thermodynamic Aspects of Solid State Sintering”. In: J. A. Pask and C. E. Hoge. *Sintering and Catalysis*. Boston: Springer, 1975, pp. 229–238. ISBN: 978-1-4684-0936-9 978-1-4684-0934-5. DOI: 10.1007/978-1-4684-0934-5_15.

[142] R. M. Cannon and W. C. Carter. “Interplay of Sintering Microstructures, Driving Forces, and Mass Transport Mechanisms”. In: *Journal of the American Ceramic Society* 72.8 (1989), pp. 1550–1555. DOI: 10.1111/j.1151-2916.1989.tb07705.x.

[143] B. J. Kellett and F. F. Lange. “Thermodynamics of Densification: I, Sintering of Simple Particle Arrays, Equilibrium Configurations, Pore Stability, and Shrinkage”. In: *Journal of the American Ceramic Society* 72.5 (1989), pp. 725–734. DOI: 10.1111/j.1151-2916.1989.tb06208.x.

[144] F. F. Lange and Bruce J. Kellett. “Thermodynamics of Densification: II, Grain Growth in Porous Compacts and Relation to Densification”. In: *Journal of the American Ceramic Society* 72.5 (1989), pp. 735–741. DOI: 10.1111/j.1151-2916.1989.tb06209.x.

[145] F. Delannay. “Influence of Dihedral Angle and Grain Coordination on Densification Rate During Intermediate and Final Sintering Stages”. In: *Journal of the American Ceramic Society* 98.11 (2015), pp. 3469–3475. DOI: 10.1111/jace.13577.

[146] J. Pan et al. “A Model for the Sintering of Spherical Particles of Different Sizes by Solid State Diffusion”. In: *Acta Materialia* 46.13 (1998), pp. 4671–4690. DOI: 10.1016/S1359-6454(98)00144-X.

[147] V. Kumar et al. “Geometric Model of Sintering of Unequal-Sized Particles”. In: *Powder Metallurgy* 59.3 (2016), pp. 170–180. DOI: 10.1080/00325899.2016.1148915.

[148] M. Weiner, M. Schmidtchen, and U. Prahl. “A New Approach for Sintering Simulation of Irregularly Shaped Powder Particles - Part I: Model Development and Case Studies”. In: *Advanced Engineering Materials* 24 (2022), p. 2101513. DOI: 10.1002/adem.202101513.

[149] M. Weiner. *Data and Software Used to Write the Thesis – Supplementary Material*. GitHub, 2025. URL: https://github.com/axtimhaus/dissertation.

[150] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 1988. ISBN: 978-1-134-74270-7. DOI: 10.4324/9780203771587.

[151] S. S. Sawilowsky. “New Effect Size Rules of Thumb”. In: *Journal of Modern Applied Statistical Methods* 8.2 (2009), pp. 597–599. DOI: 10.22237/jmasm/1257035100.

[152] J. Pan and A. C. F. Cocks. “A Numerical Technique for the Analysis of Coupled Surface and Grain-Boundary Diffusion”. In: *Acta Metallurgica et Materialia* 43.4 (1995), pp. 1395–1406. DOI: 10.1016/0956-7151(94)00365-O.

[153] D. L. Johnson. “New Method of Obtaining Volume, Grain-Boundary, and Surface Diffusion Coefficients from Sintering Data”. In: *Journal of Applied Physics* 40.1 (1969), pp. 192–200. DOI: 10.1063/1.1657030.